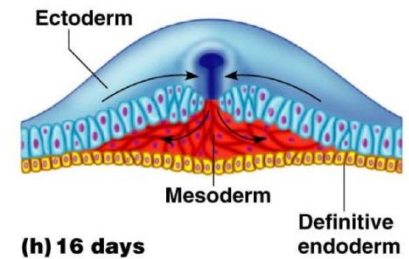
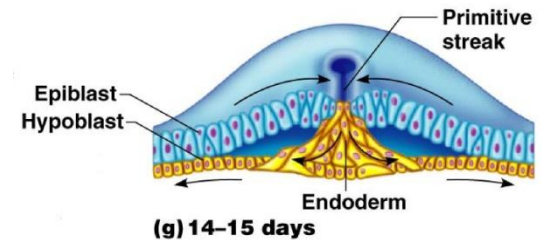
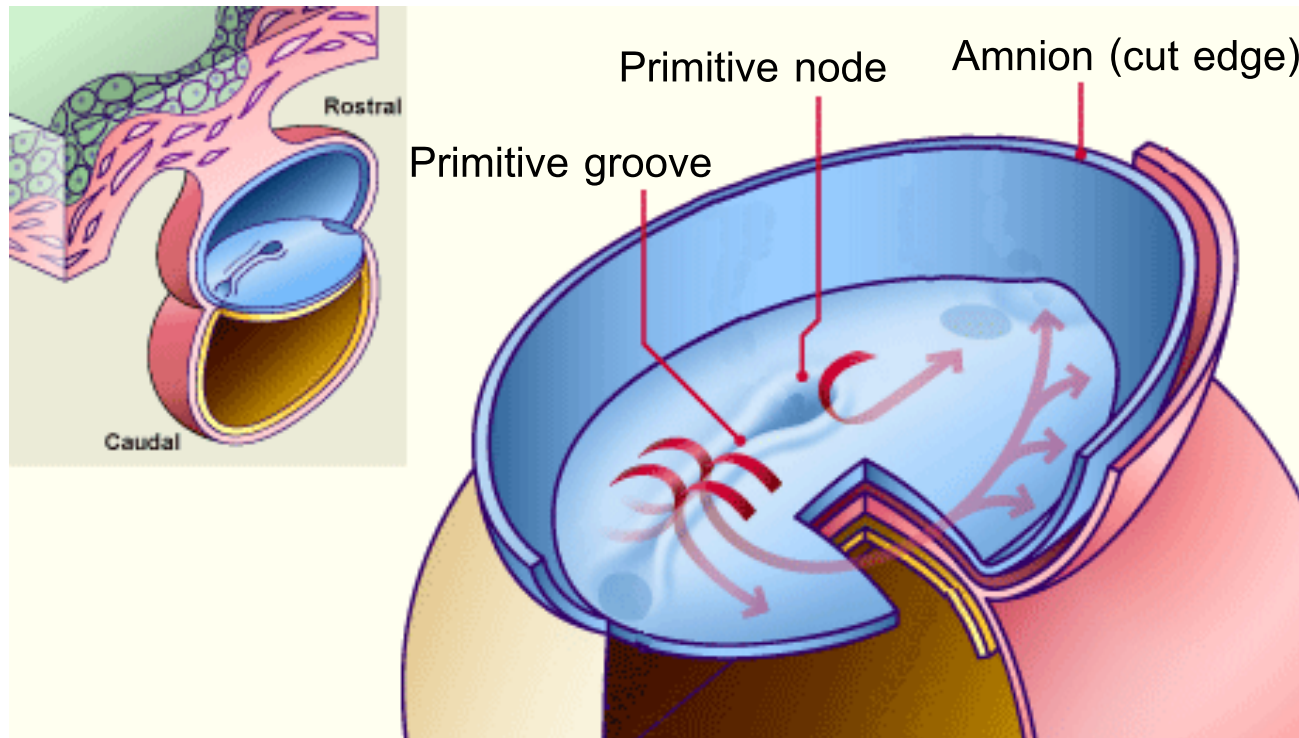




Development of the Cardiovascular System

Assoc.Prof.Dr.Pornsawan Duangsuwan

3rd week of development: Trilaminar embryonic disc

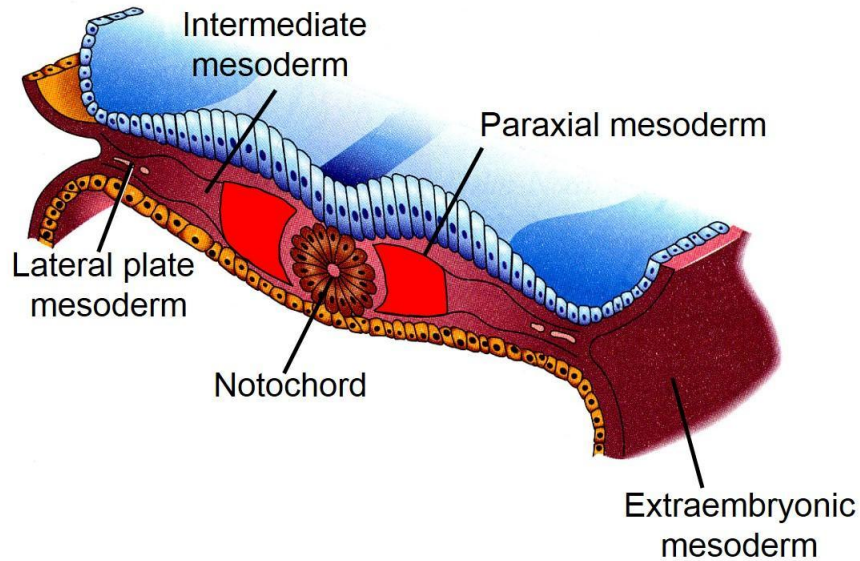


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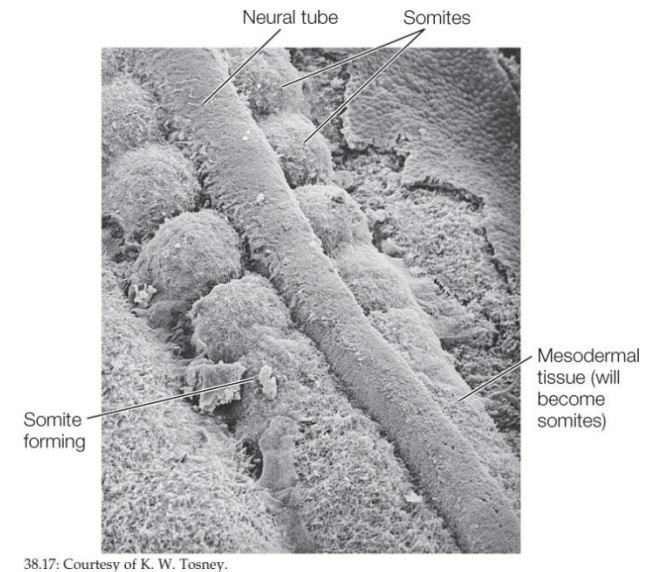
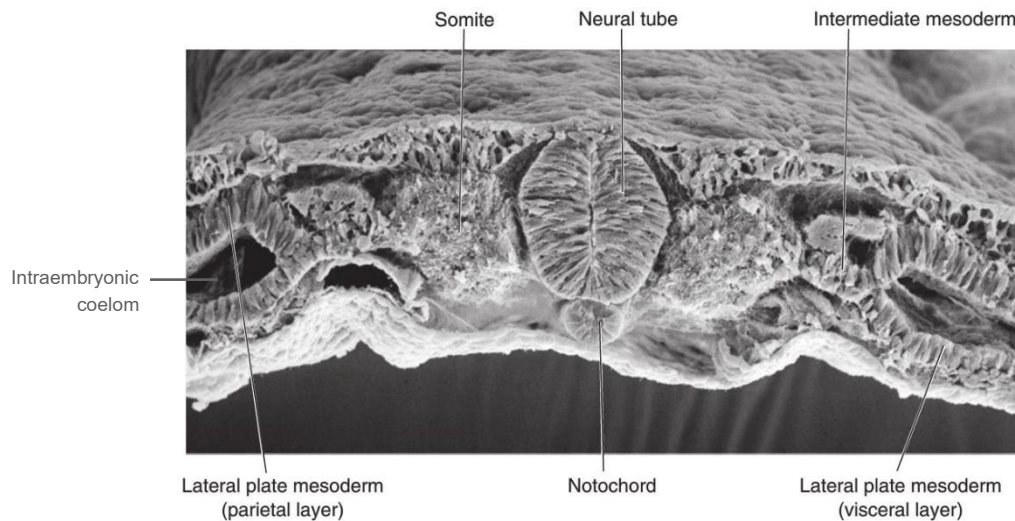
- Day 16: Epiblast cells migrate through the primitive streak between epiblast layers and definitive endoderm to form intraembryonic mesoderm

Mesoderm migration



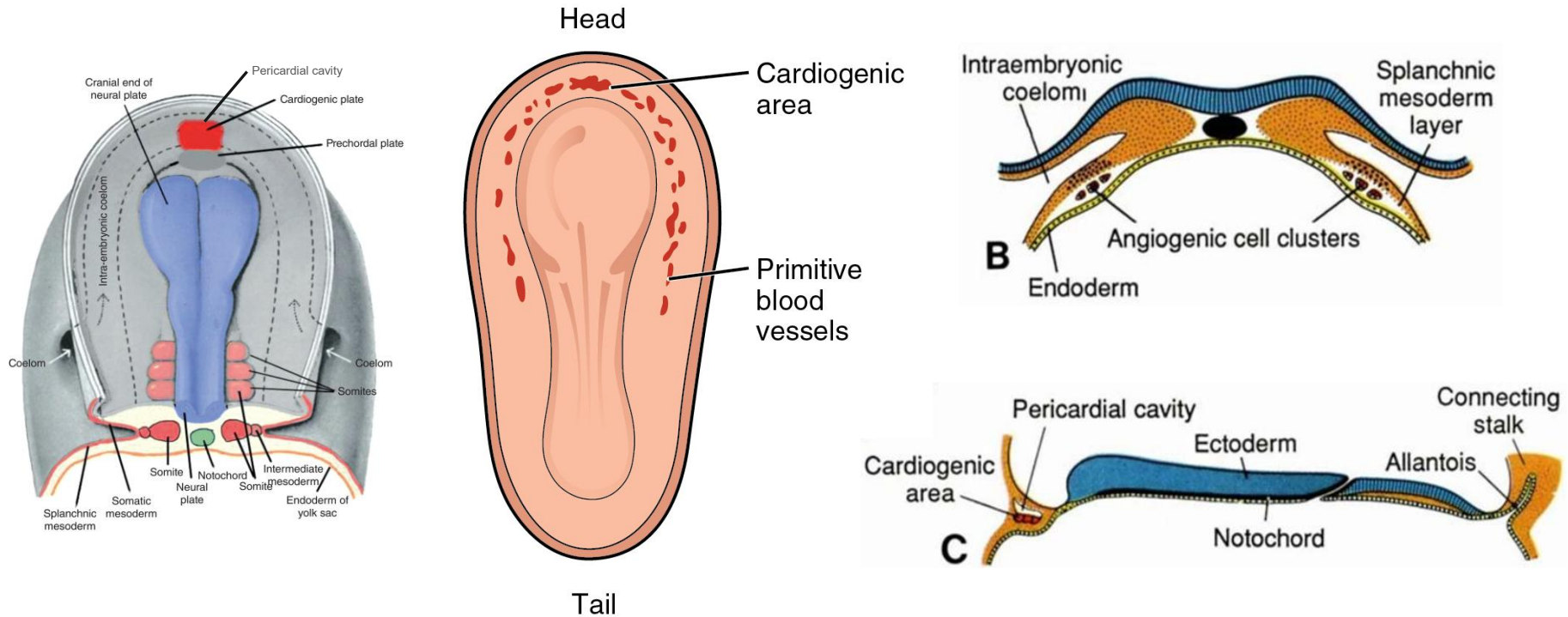
Mesoderm migrates laterally to form

- Paraxial mesoderm
- Intermediate mesoderm
- Lateral plate mesoderm



38.17: Courtesy of K. W. Tosney.

Early development of the heart

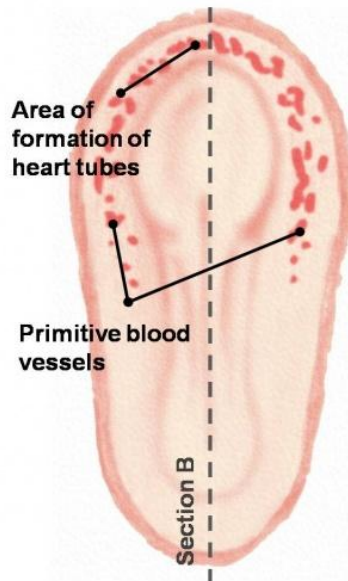


- Cardiovascular system begins to develop in the 3rd week of gestation
- Mesenchymal cells in the splanchnic mesoderm proliferate and form isolated cell clusters known as **angiogenic cell clusters** toward cephalic end in cardiogenic region of the trilaminar embryo

Early development of the heart

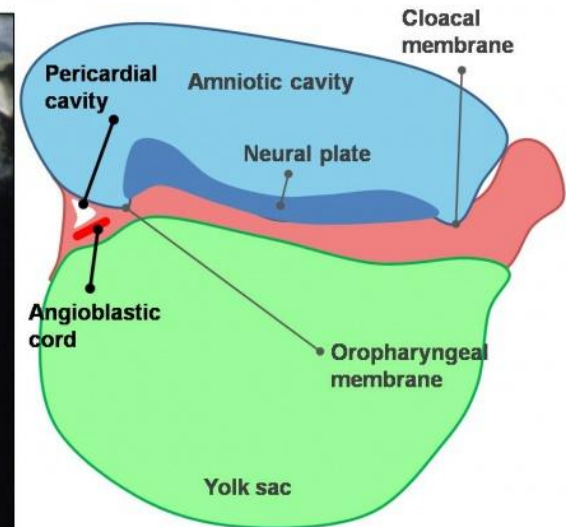
Embryo approximately 18 Days

A Dorsal view



Embryo approximately 18 Days

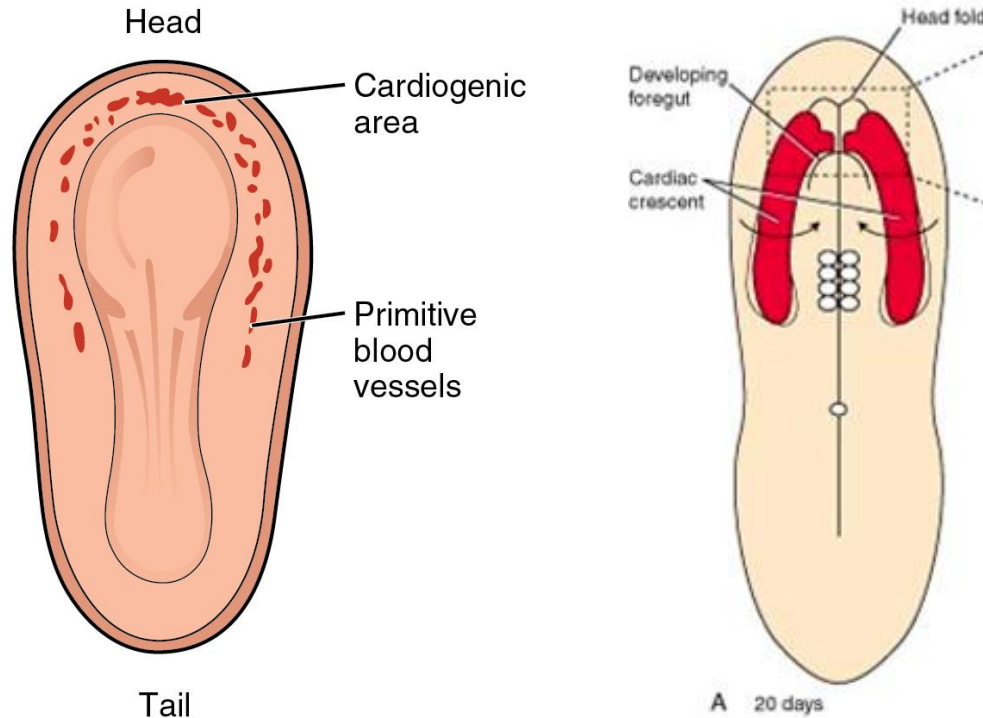
B Lateral view



https://embryology.med.unsw.edu.au/embryology/index.php?title=Intermediate_-_Primordial_Heart_Tube

- Angiogenic cell cluster (Angioblastic cord): horseshoe-shaped pattern
- Cardiogenic region : cranial end of embryonic disc
- Pericardial cavity : cranial end of intraembryonic coelom

Formation of endocardial heart tube: day 19

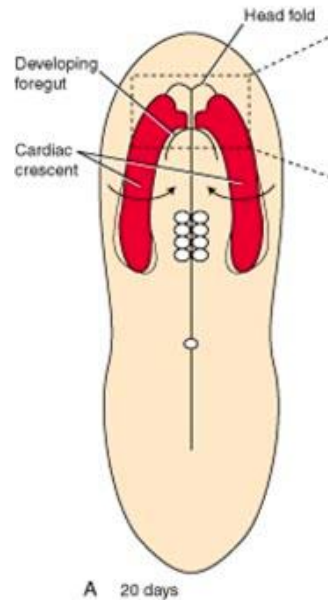


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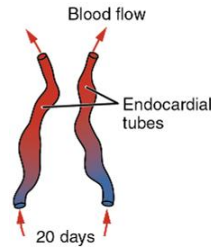
- Angioblastic cords develop in cardiogenic mesoderm and canalizes to form bilateral **endocardial heart tubes** (very thin-walled tubes)



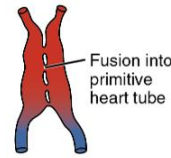
Lateral folding and fusion of heart tubes



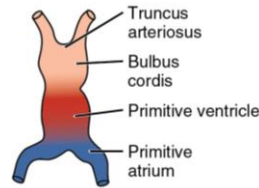
A 20 days



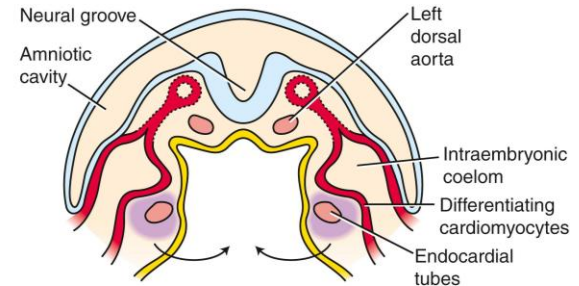
20 days



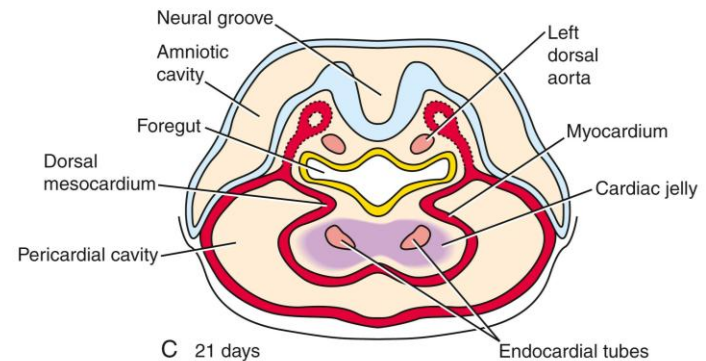
21 days



22 days



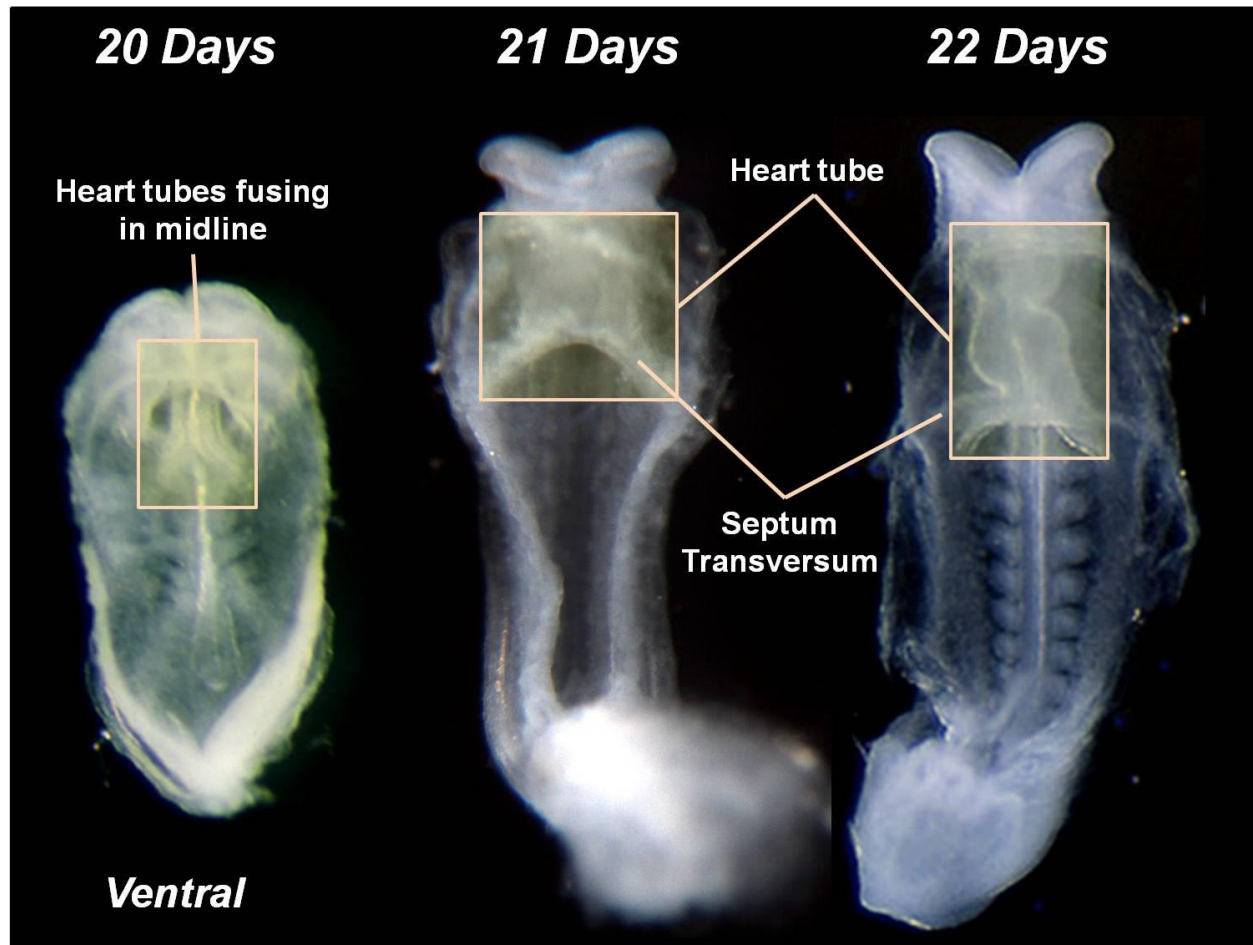
B 20 days



C 21 days

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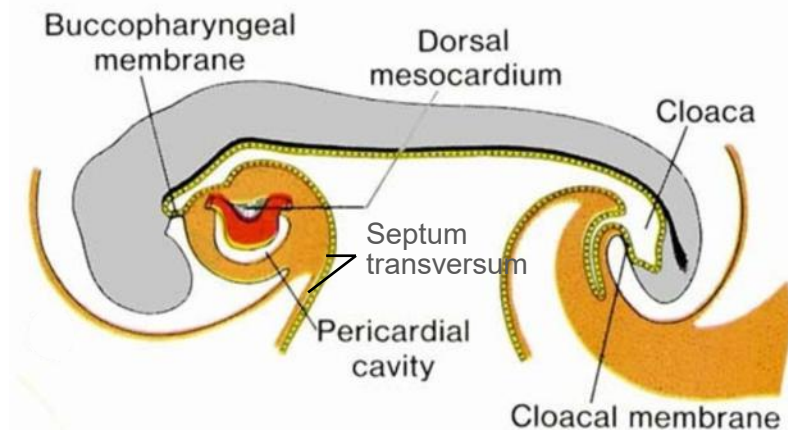
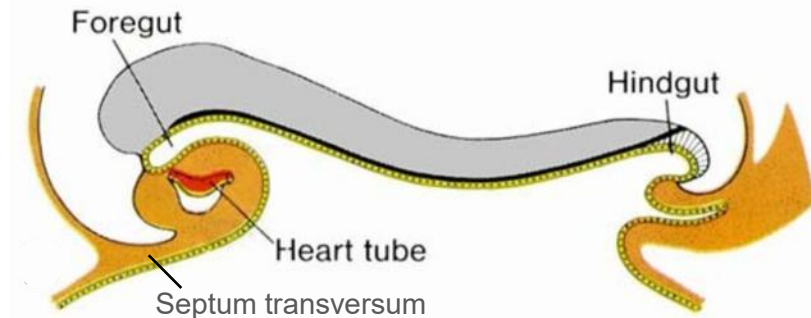
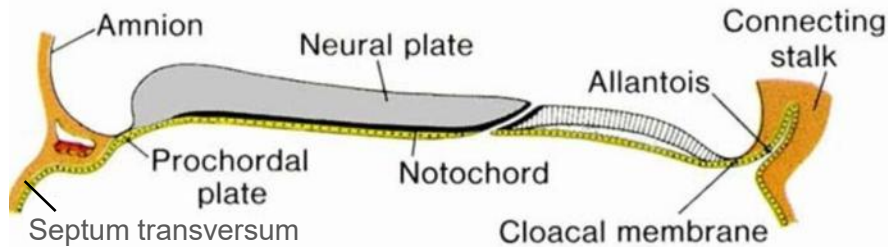
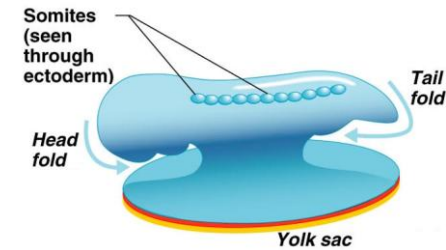
- Trilaminar embryonic disc folds around flat shape to form 3D sphere
- Primordial heart tubes fuse in midline to form single ventral heart tube
- Apoptosis of cells along two edges cause forming one continuous tube



https://embryology.med.unsw.edu.au/embryology/index.php?title=File:Heart_Tube_Fusion.jpg

- The primordial heart tubes fuse in the midline to form a single ventral heart tube.
- Fusion begins cranially and extends caudally.

Head folding

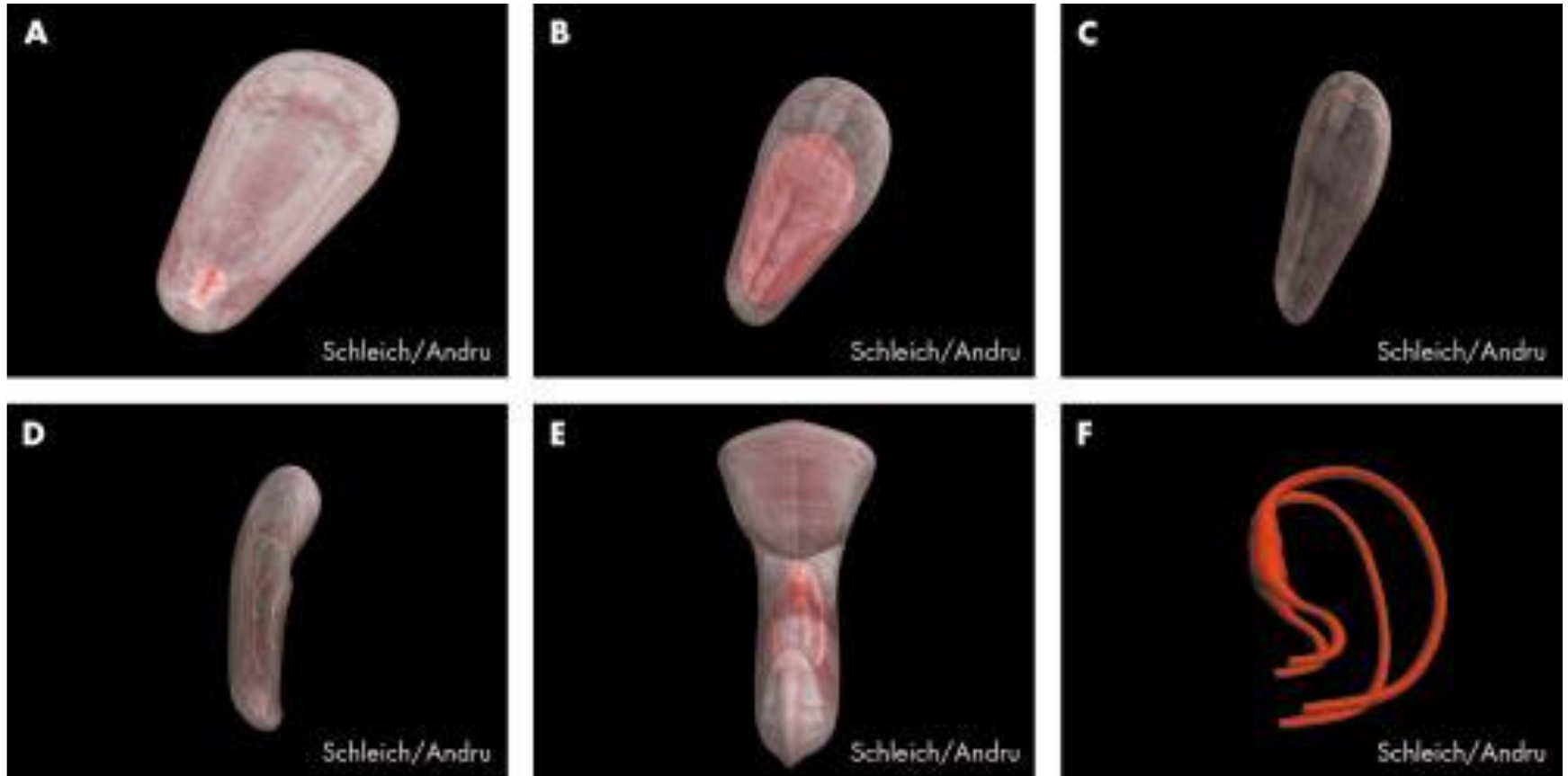


Brain growth: The rapidly growing forebrain grows cranially, causing the head region to fold ventrally

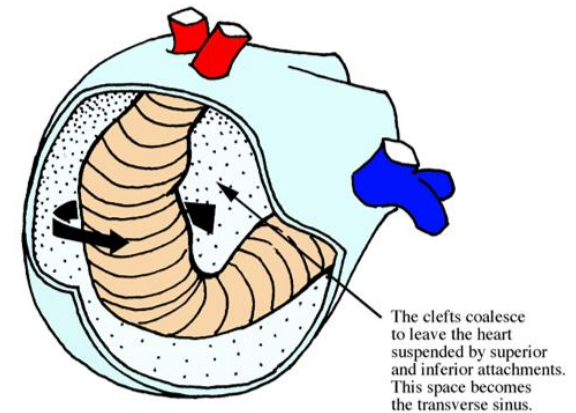
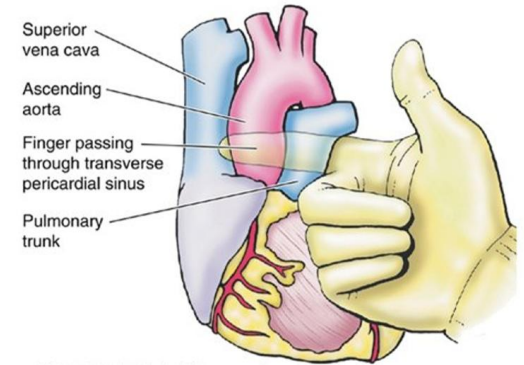
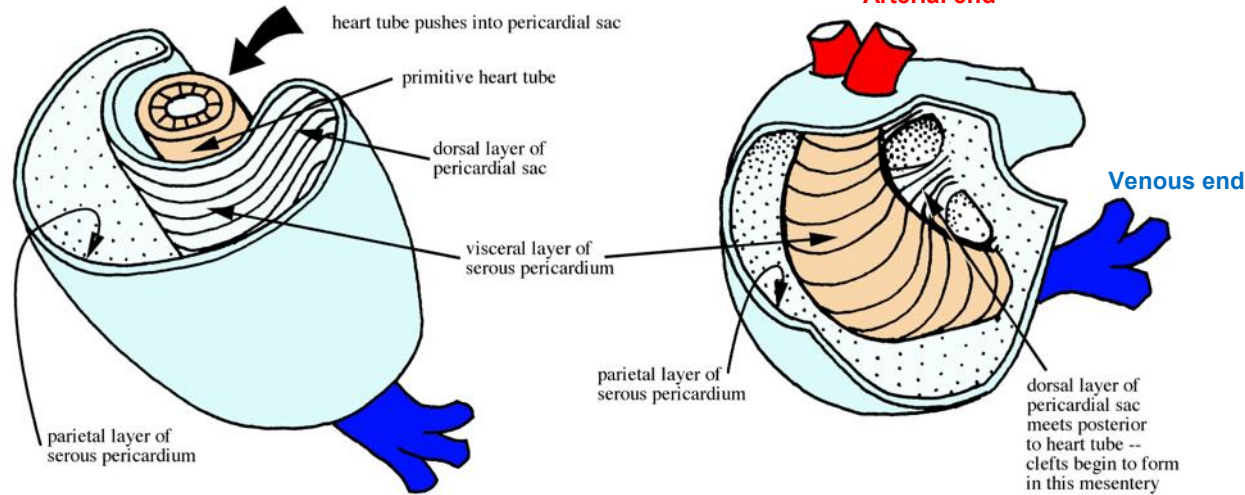
Position shift: The septum transversum, primordial heart and pericardial cavity swings 180, move from a position in front of the brain to the ventral aspect of the embryo

As the head fold completed: The heart tube lie dorsal to the pericardial cavity

Development of the human heart: Day 15-21

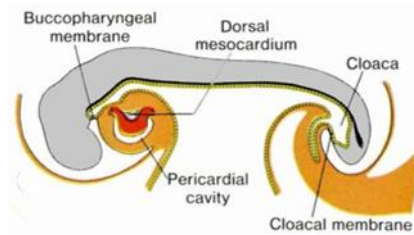


Formation of transverse pericardial sinus



- Heart tube begins as a tube surrounded by cylinder of pericardium
Heart tube is suspended by the dorsal mesocardium
- As the heart tube elongates and loops, dorsal mesocardium breakdown and disappear, leaves the heart suspended in the pericardial space by the vasculature
- Primordial arterial and venous ends are brought together to form transverse sinus

Components of the primitive heart tube



1. Endocardium

- Endothelium of heart tube

2. Cardiac jelly

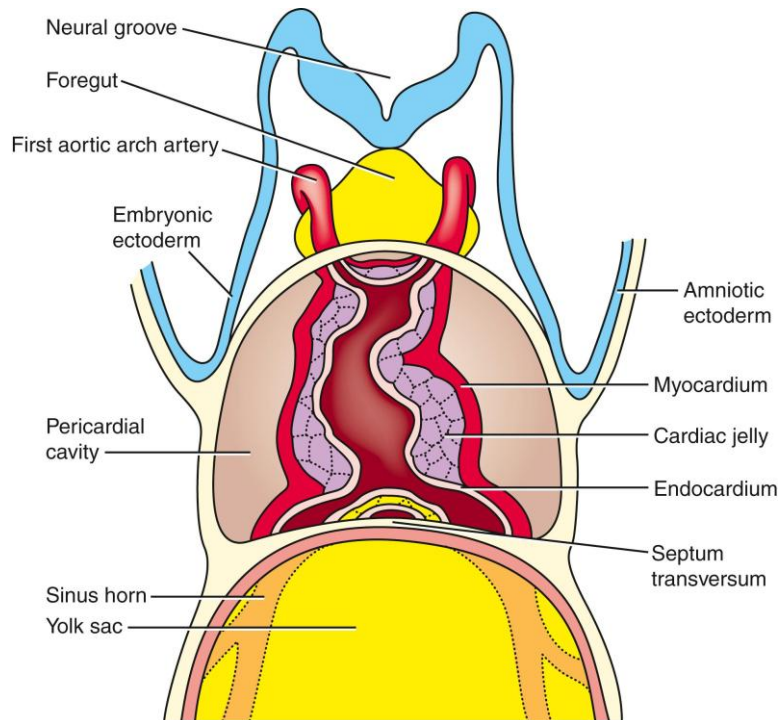
- Acellular matrix separate heart tube and myocardium
- Further subendocardial layer

3. Myocardium

- Differentiate from splanchnic mesoderm surround coelom

4. Epicardium (visceral pericardium)

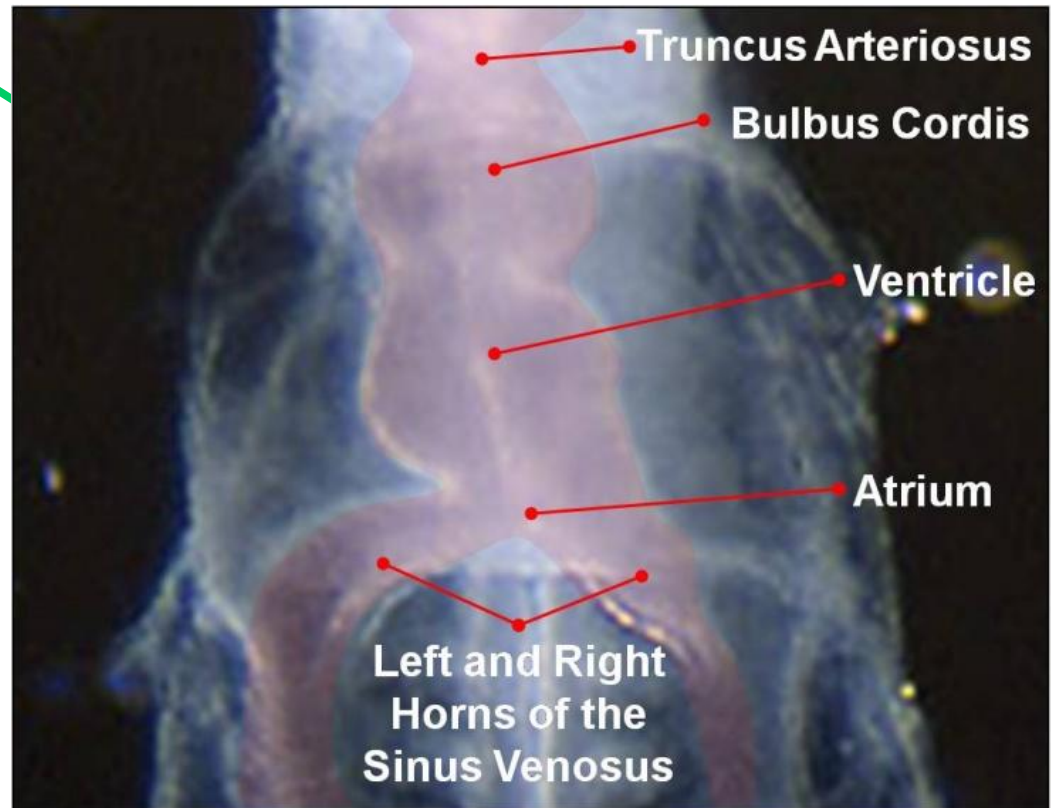
- Develops from mesothelial cells arising from the sinus venosus, spread cranially over myocardium



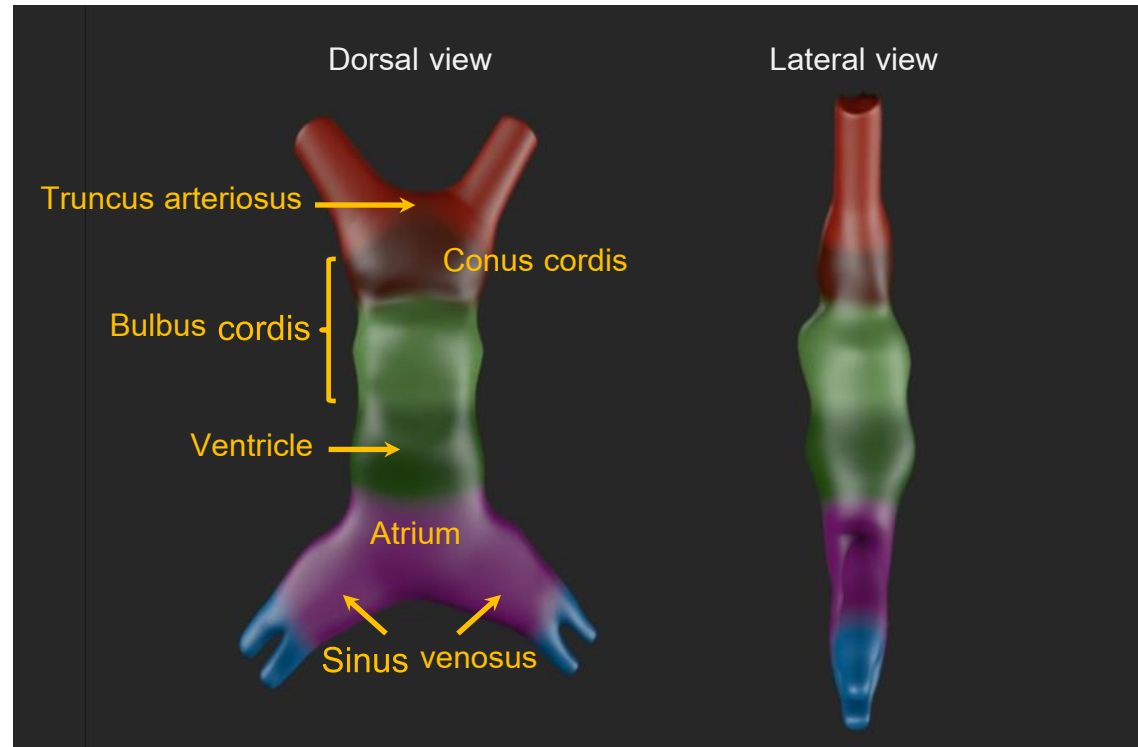
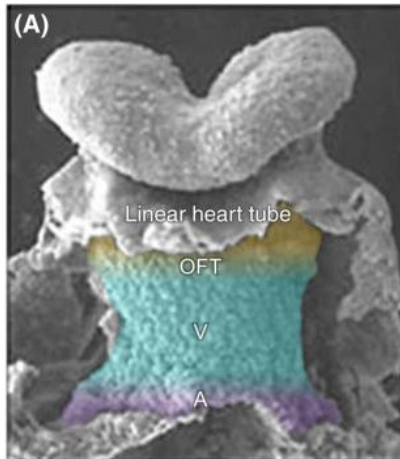
22 days

Primitive heart tube

As the tubular heart grows, it develops dilations and constrictions which form 5 basic regions



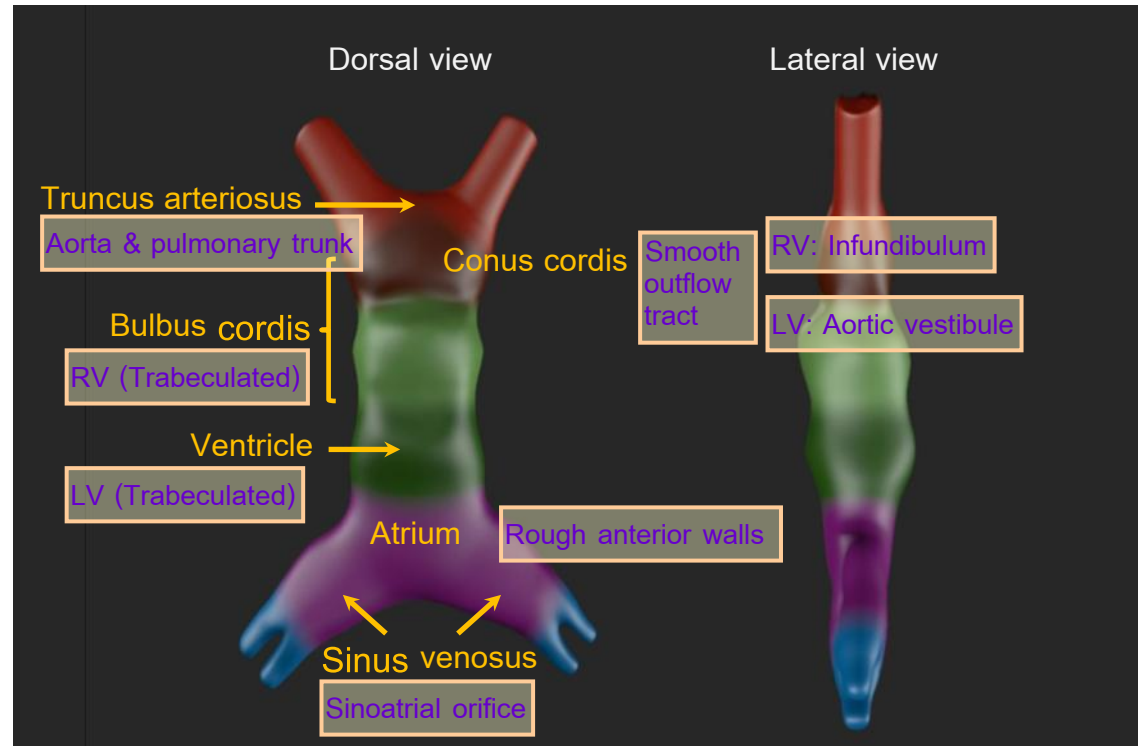
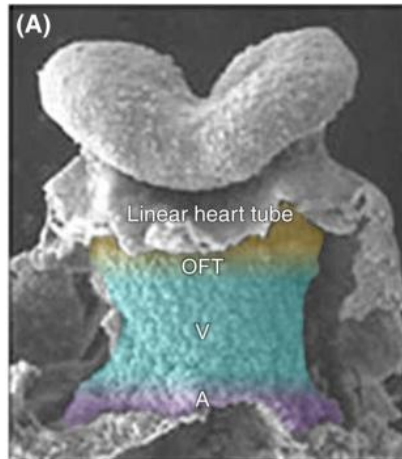
Cardiac looping: Elongate linear heart tube



<https://www.youtube.com/watch?v=a0qyaglgBPw>

- Begins around day 22 to 23 of human gestation (during the fourth week) as the heart tube rapidly elongates within a fixed pericardial cavity.

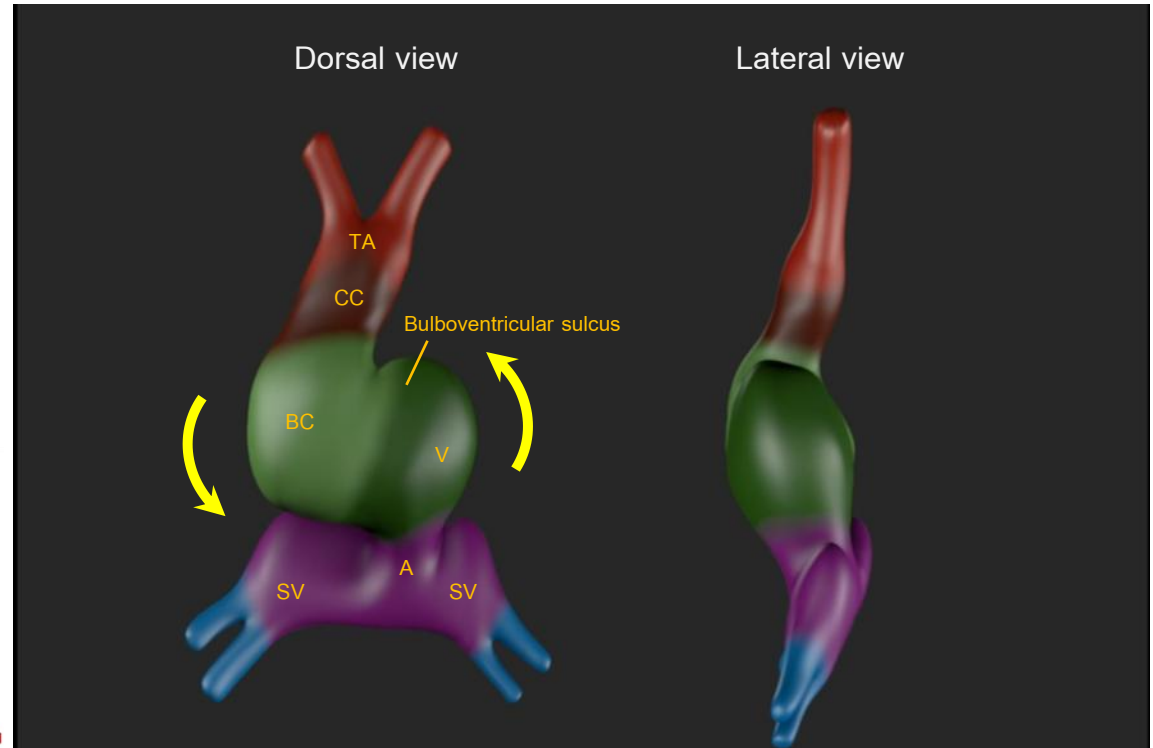
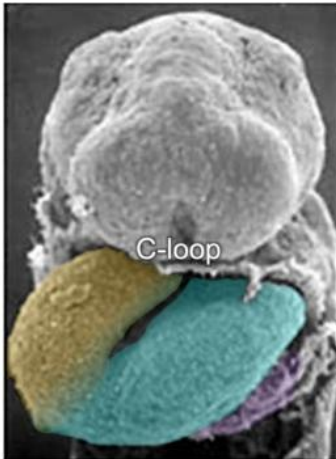
Cardiac looping: Elongate linear heart tube



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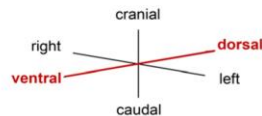
Cardiac looping: C-shaped looping



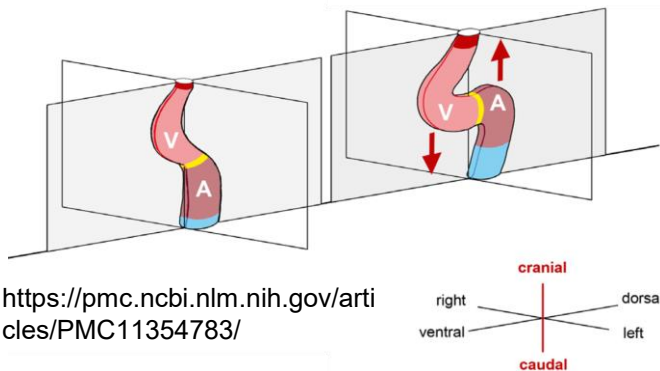
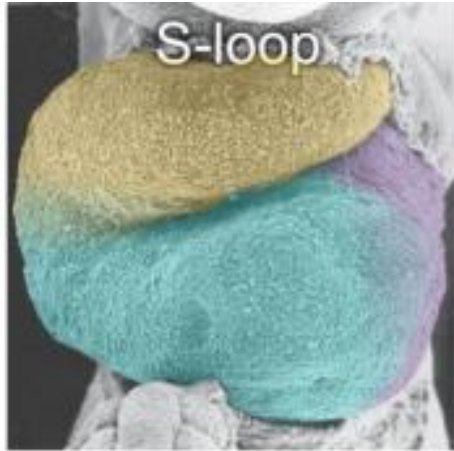
<https://www.youtube.com/watch?v=a0qyaglgBPw>

- Heart tube grows longer within a fixed pericardium space, forcing the middle part (bulbus cordis and primitive ventricle) bends ventrally
- Bulbus cordis moves down and right, while the primitive ventricle shifts to the left, forming a C-shape, Bulboventricular sulcus

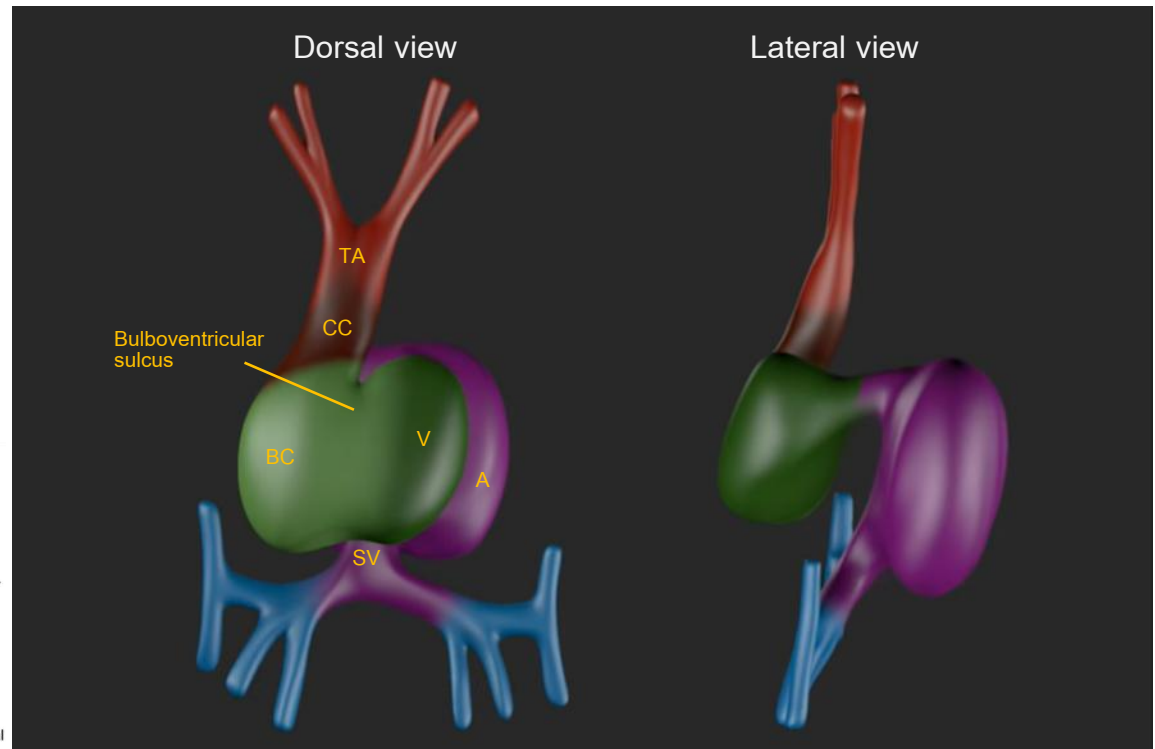
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11354783/>



Cardiac looping: S-shaped looping



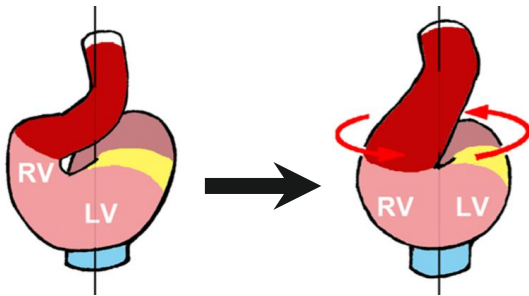
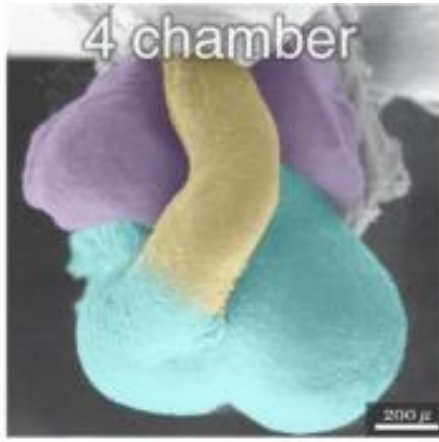
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11354783/>



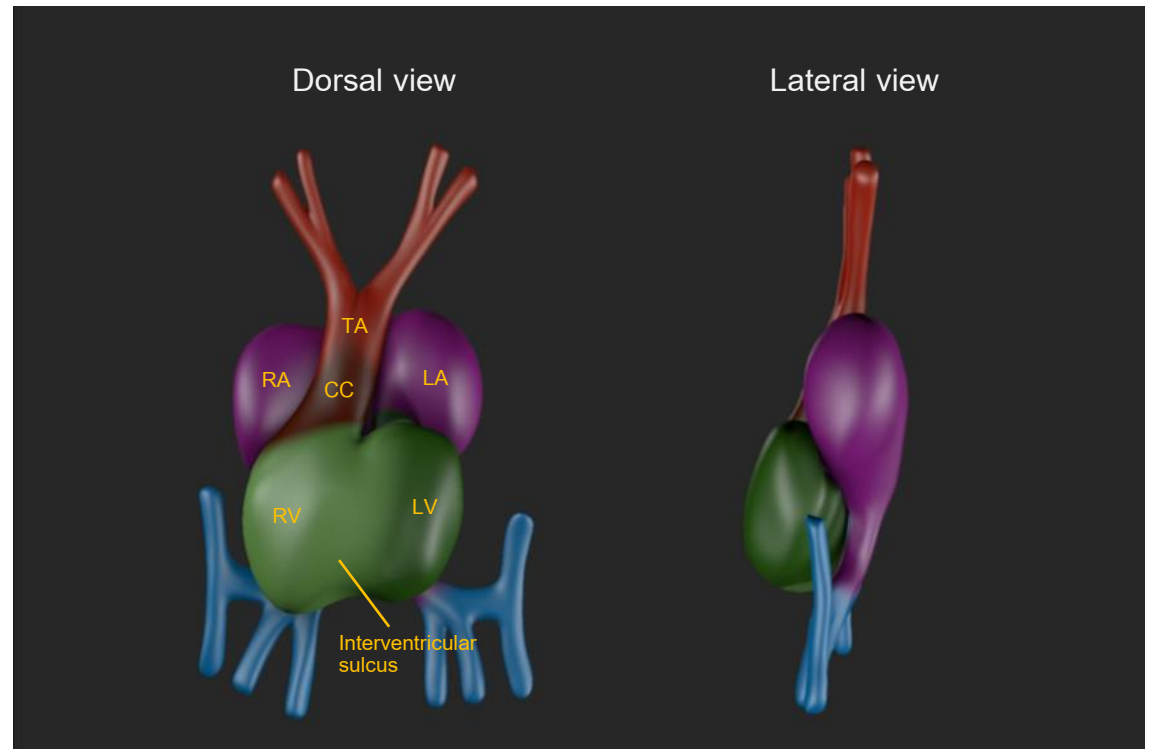
<https://www.youtube.com/watch?v=a0qyaglgBPw>

- The caudal/primitive atrium move upward and backward (dorsocranially) behind the ventricles, completing an S-shaped loop by approximately day 28.
- Bulbus cordis : ventricle / Atrium : sinus venosus

Cardiac looping: Chamber Alignment



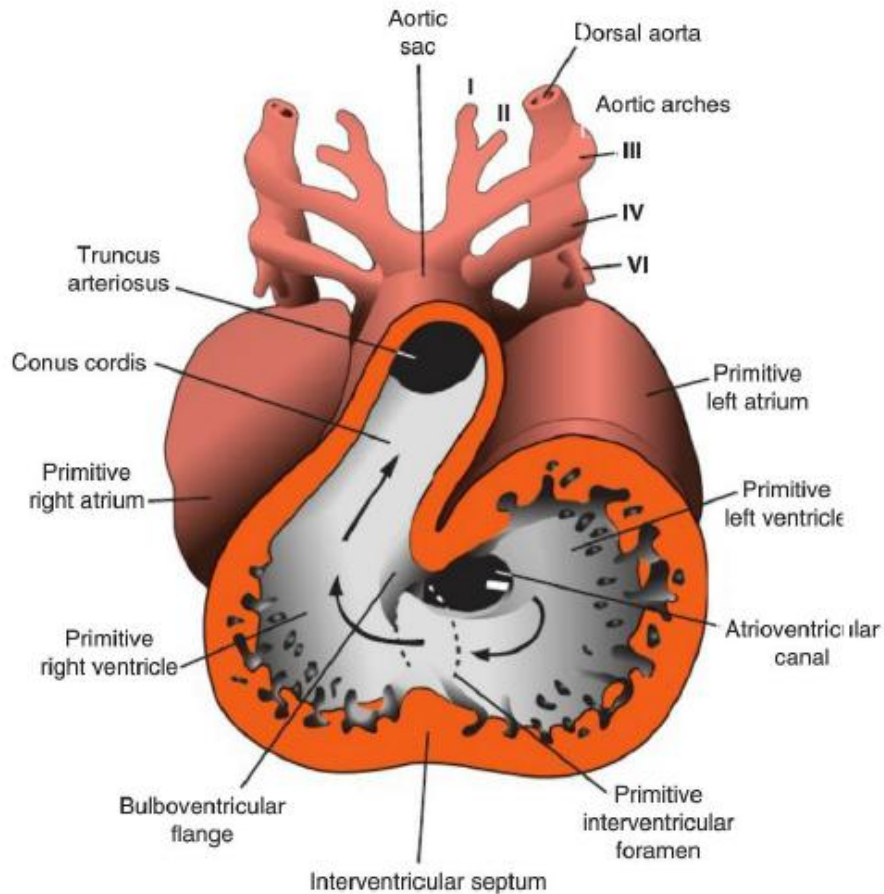
<https://www.mdpi.com/2308-3425/11/8/252>



<https://www.youtube.com/watch?v=a0qyaglgBPw>

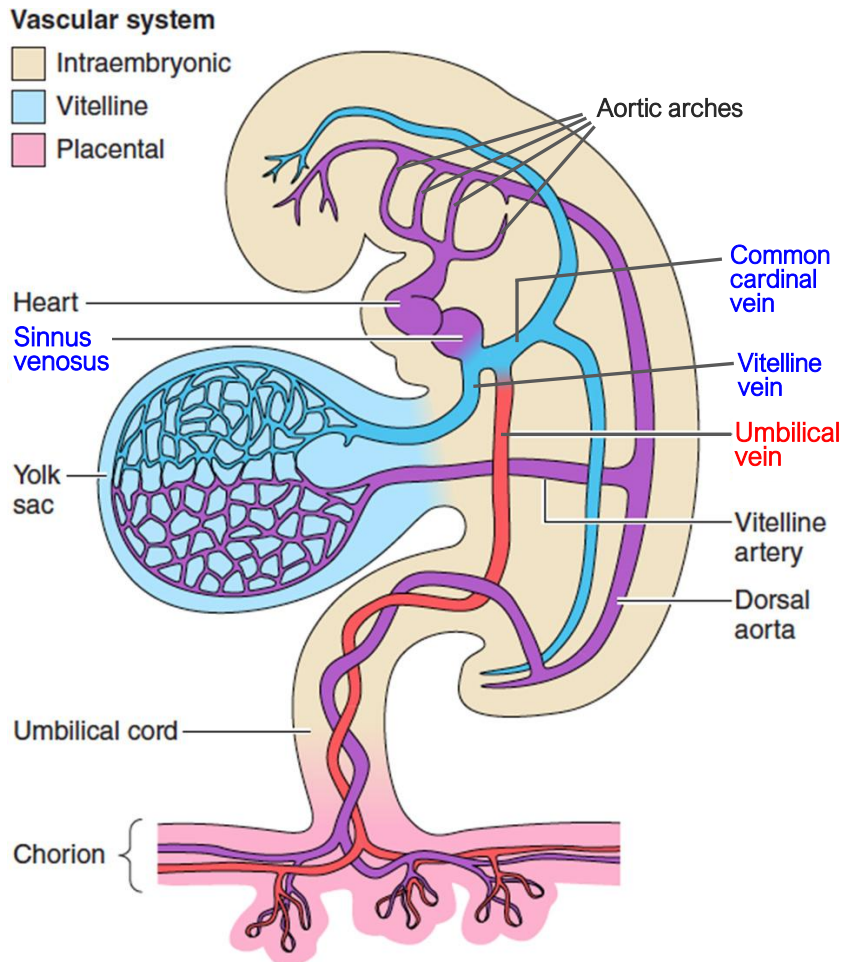
- Conus cordis and arterial trunk move ventral to the atrium to form the later outflow tract. The atrium now lies superior to the ventricle
- This looping places the atria above the ventricles and aligns the basic architecture of the adult heart

Blood flow through the primitive heart tube



- Unidirectional flow
- Blood enters the primordial heart tube at the caudal venous end via the sinus venosus and exits cranially through the truncus arteriosus

Early embryonic circulation

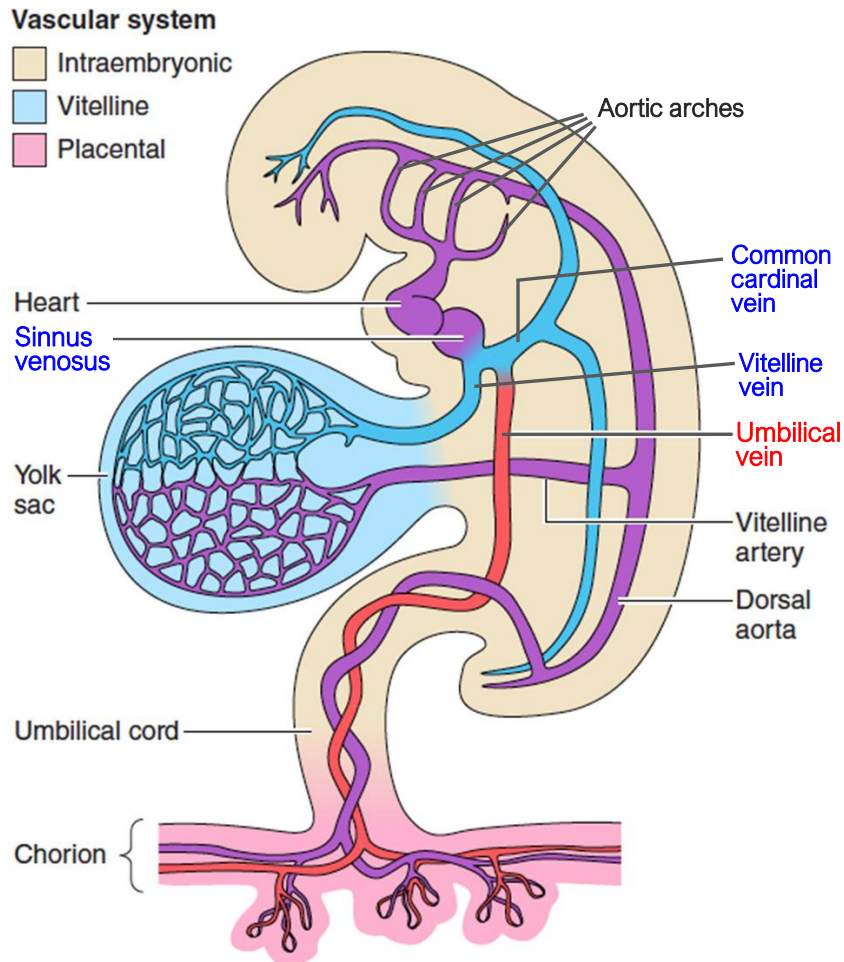


1. Intraembryonic (Systemic) circuit

2. Extraembryonic circuit

- Vitelline (Omphalomesenteric) circuit
- Umbilical (Allantoic) circuit

Intraembryonic (Systemic) circuit

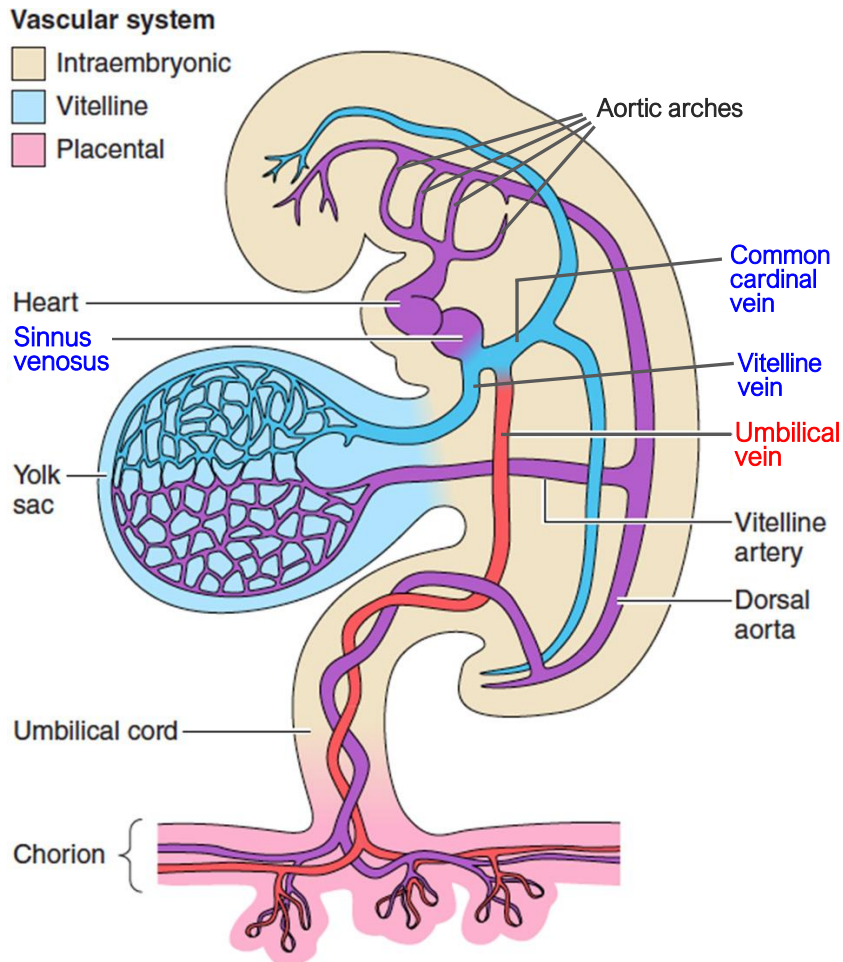


Function: Delivers oxygen and nutrients directly to the developing organs of the embryo proper

Arterial supply: Blood pumped through the dorsal aorta passes over the aortic arches and down into the embryo

Venous return: The cardinal veins (anterior, posterior, and common) drain the head, neck, and body wall back into the embryonic heart

Extraembryonic circuit: Vitelline (Omphalomesenteric) circuit



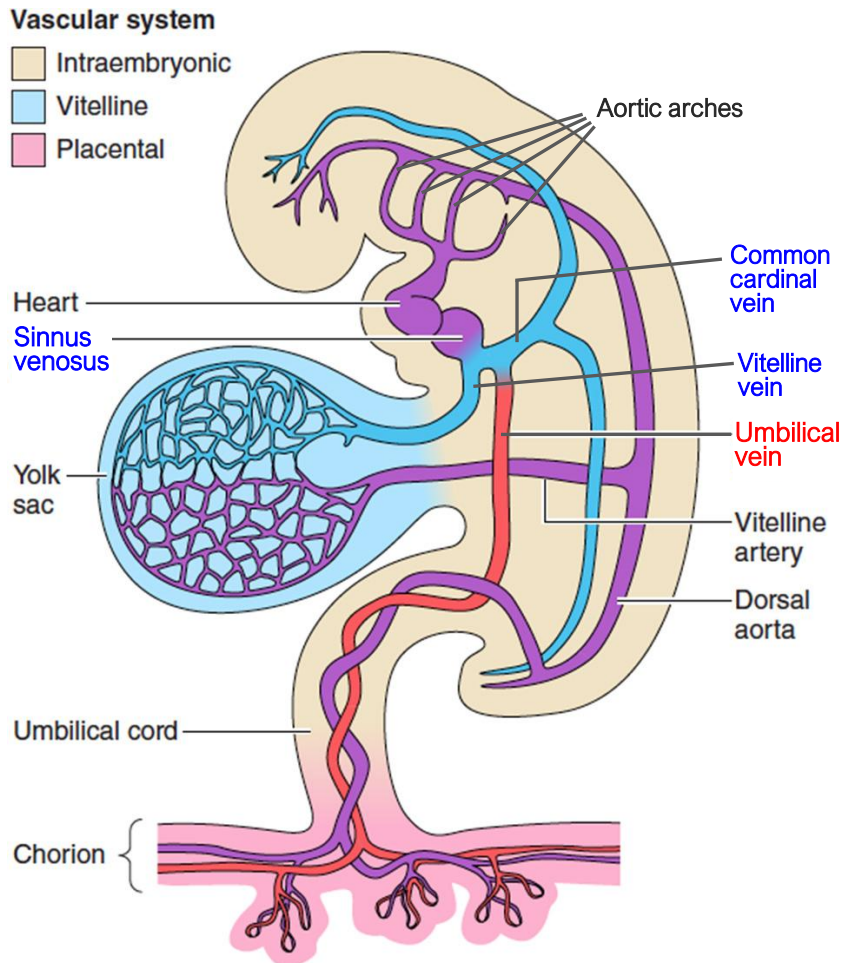
Function: Connects the embryo to the yolk sac

Arterial supply: Vitelline arteries branch from the dorsal aorta to carry blood to the yolk sac

Venous return: Vitelline veins return poorly oxygenated blood, nutrient-rich blood from the yolk sac to the venous end (sinus venosus) of the embryonic heart

Significance: This is the site of primitive hematopoiesis, where the embryo's very first red blood cells are generated before the liver and bone marrow take over

Extraembryonic circuit: Umbilical (Allantoic) Circuit



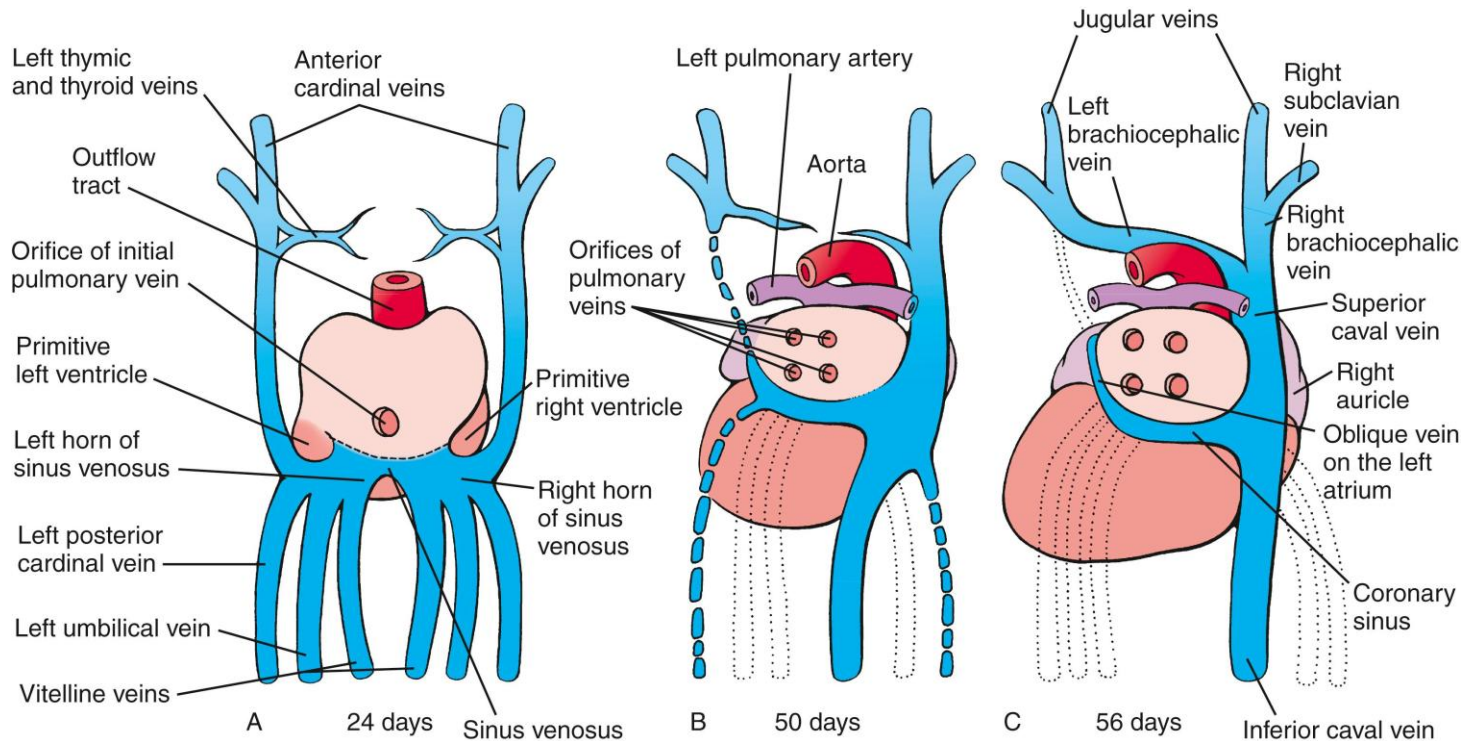
<https://www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2018.00373/full>

Function: Links the embryo to the fetomaternal interface at the **placenta**

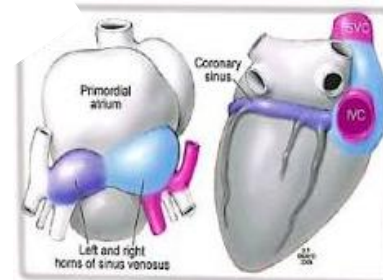
Arterial supply: **Umbilical arteries** carry deoxygenated, waste-rich blood from the caudal dorsal aorta out through the connecting stalk to the placenta

Venous return: **Umbilical veins** carry freshly oxygenated, nutrient-dense blood from the maternal placenta back into the embryo, routing through the developing liver

Remodeling of the inflow end of the heart

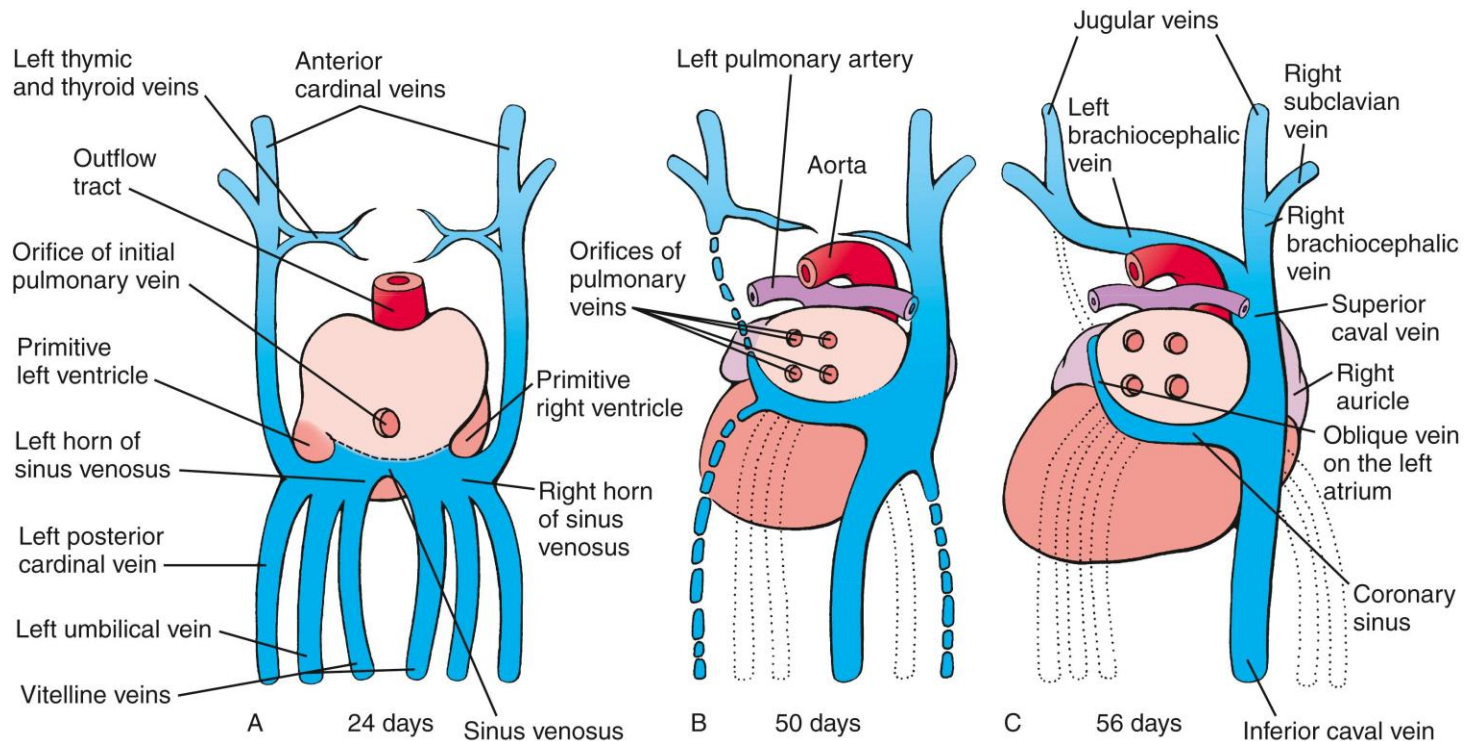


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- Left sinus horn is reduced and pulled to the left
- Left sinus horn loses connection with left anterior cardinal vein and becomes the coronary sinus, draining blood only from the heart wall

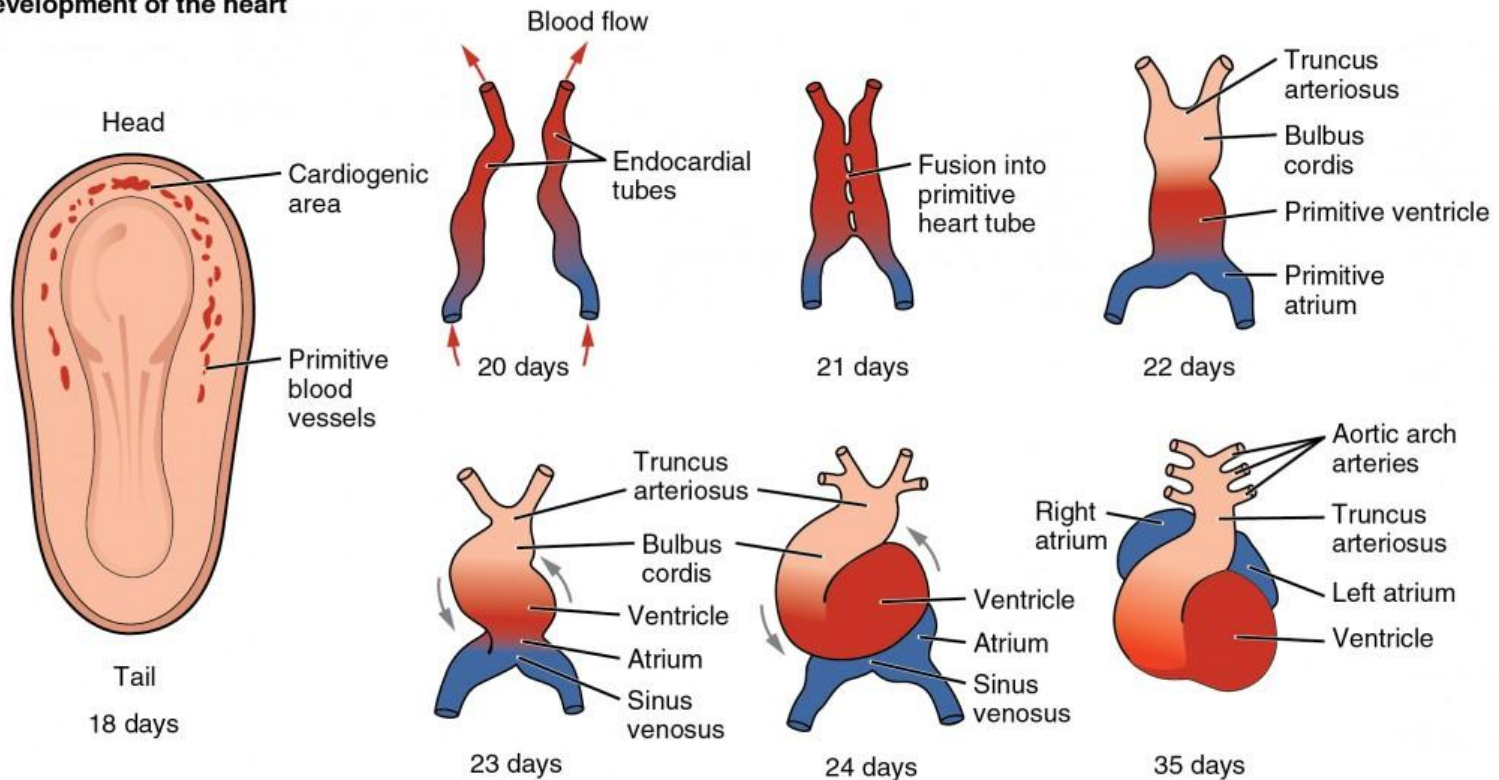
Remodeling of the inflow end of the heart



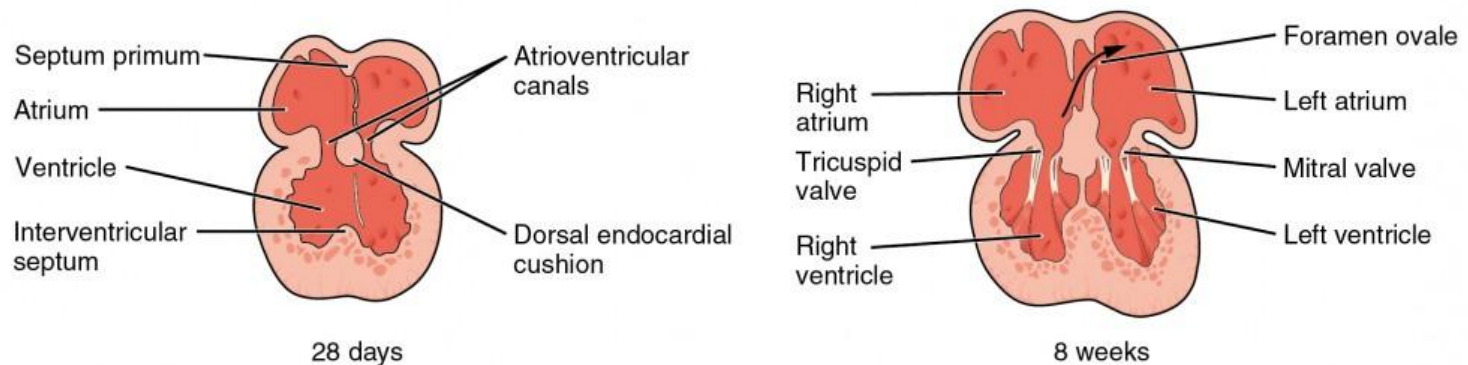
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- Left anterior cardinal vein connects to the right anterior cardinal vein, which form the left brachiocephalic vein
- Remnant of right vitelline vein becomes the terminal segment of IVC

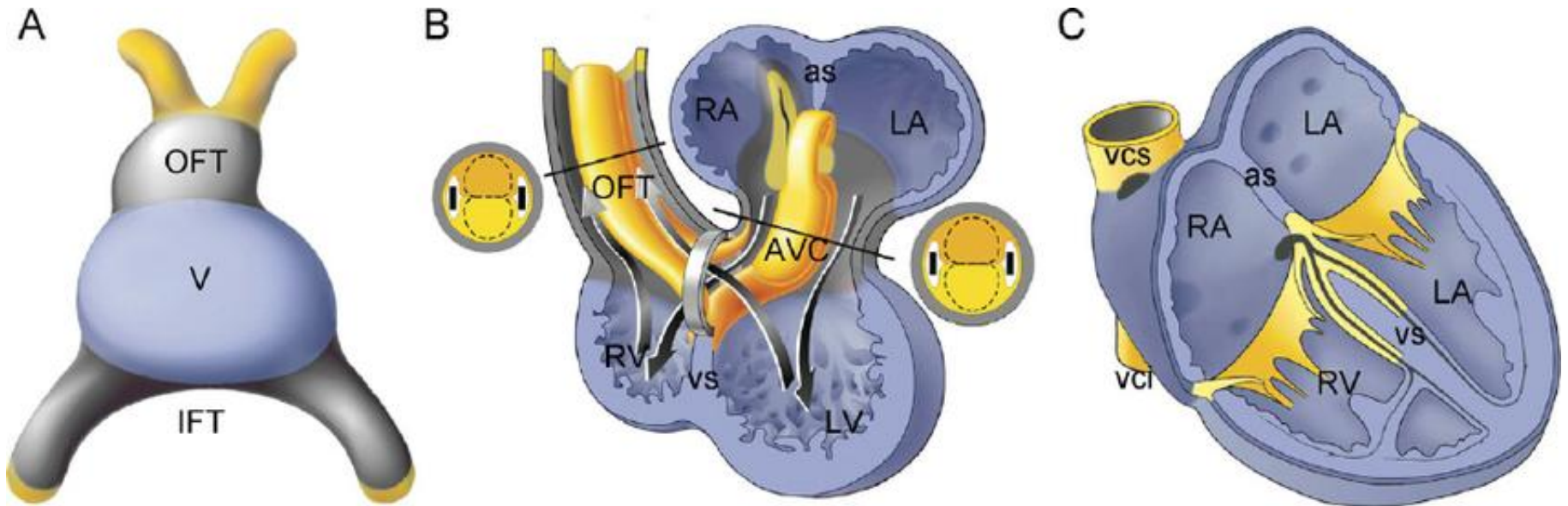
Development of the heart



Partitioning of the heart into four chambers



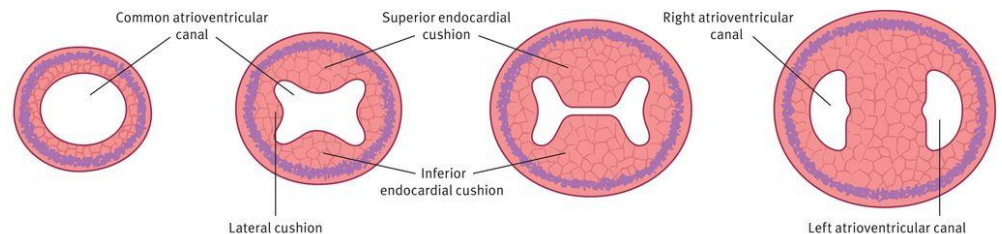
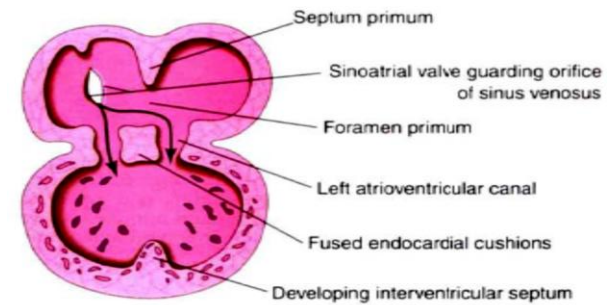
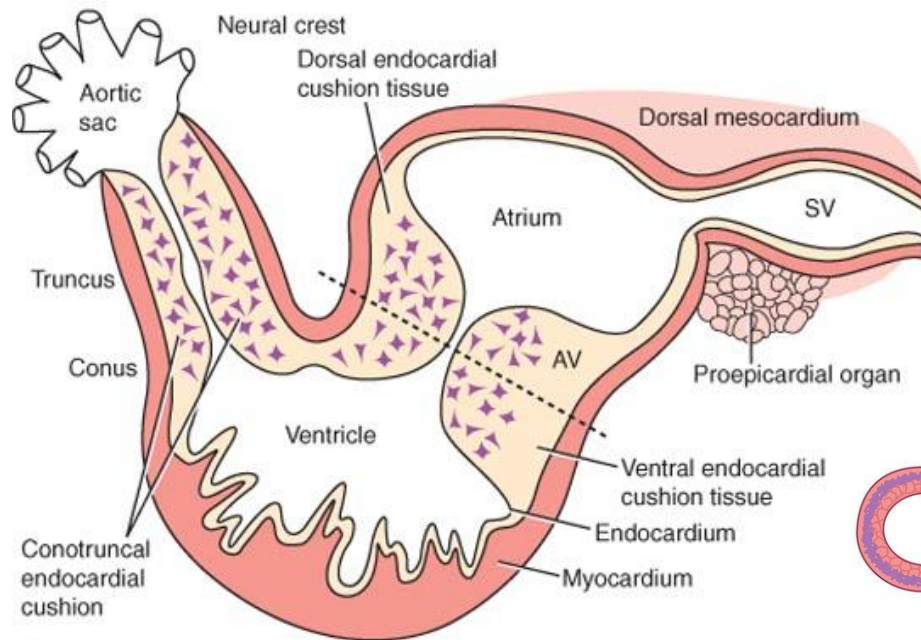
Partitioning of the heart into four chambers



<https://academic.oup.com/cardiovasres/article/74/2/244/284365>

- Partitioning of atrioventricular canal
- Septation of the atrium
- Septation of the ventricle
- Separation of the outflow tract

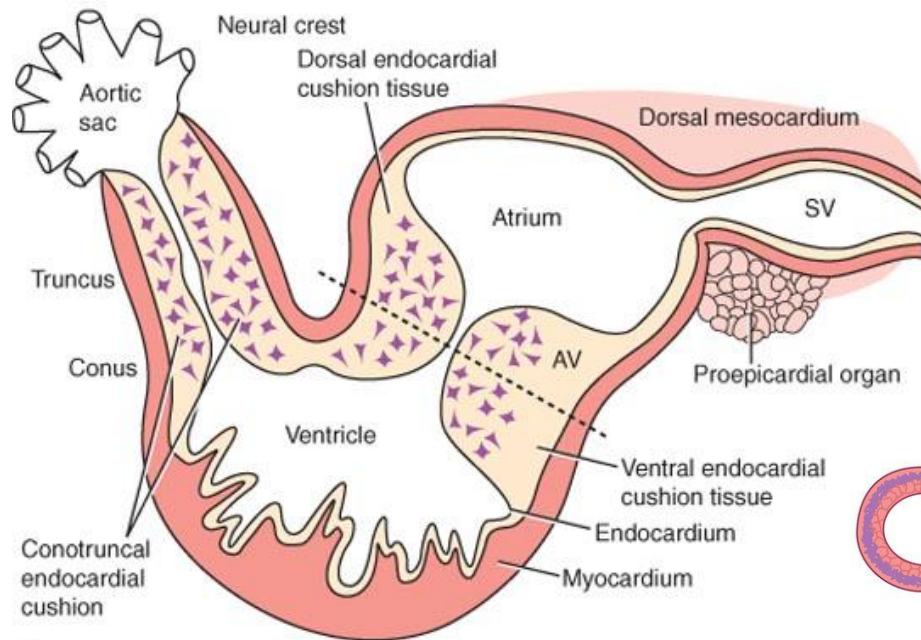
Partitioning of the atrioventricular canal



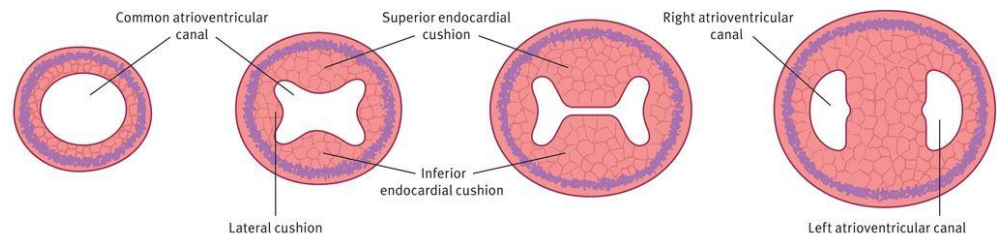
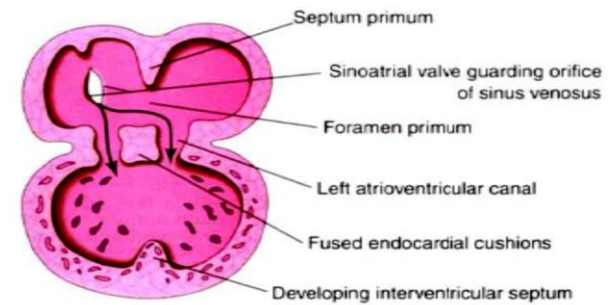
<https://obgynkey.com/development-of-the-heart-and-the-fetal-circulation/>

- Cushion Formation: Mesenchyme on superior (dorsal) and inferior (ventral) sides form endocardial cushions grow toward each other and fuse in the middle
- Channel Division: This fusion splits the single AV canal into a right AV orifice and a left AV orifice, separating the primitive atrium from the primitive ventricle on each side

Partitioning of the atrioventricular canal



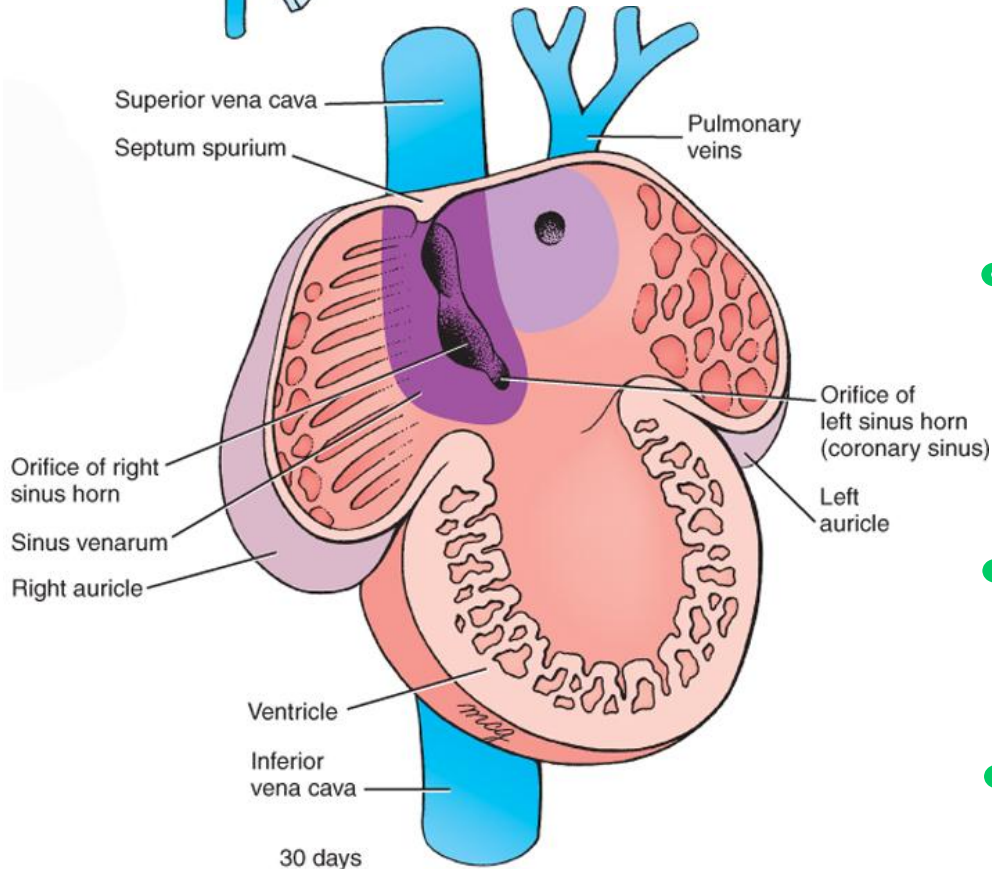
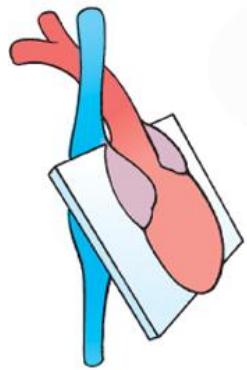
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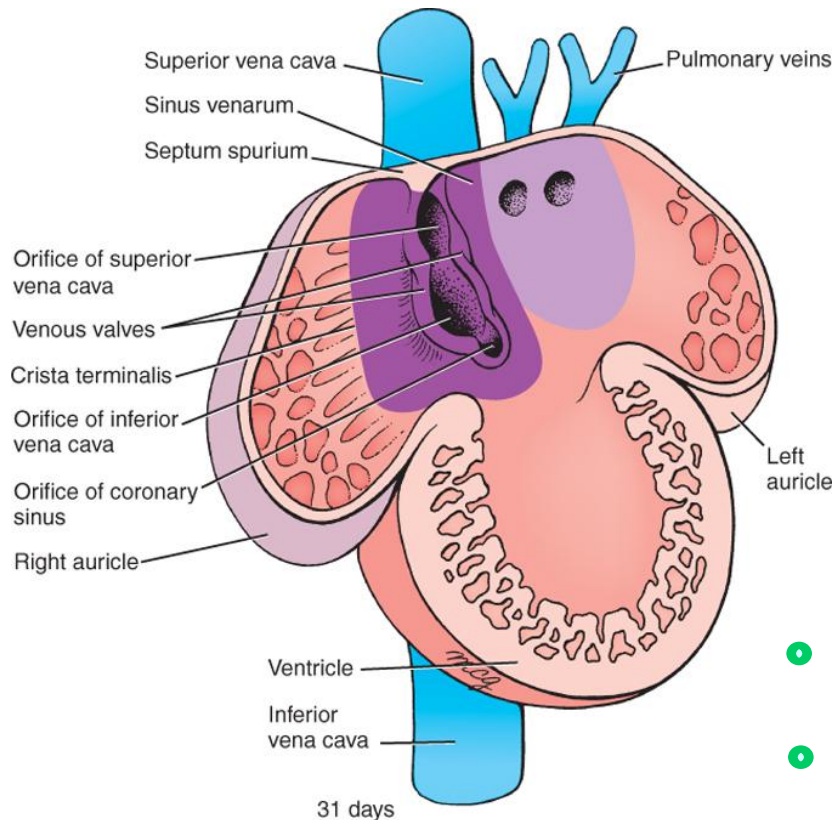
- Valve Development: The fused cushions and lateral tissues later remodel and hollow out to form the definitive tricuspid valve (on the right) and mitral valve (on the left)

Differentiation of atrium I



- 5th week
- Form trabeculated portion at ventral and lateral part of both Lt and Rt primitive atrium and auricles
- Rt atrium: right sinus horn incorporate into dorsal wall of right atrium as smooth-walled sinus venarum
- Lt atrium: single pulmonary vein develop in Lt primitive atrium
- Pulmonary vein branches twice to produce 2 right and left pulmonary veins

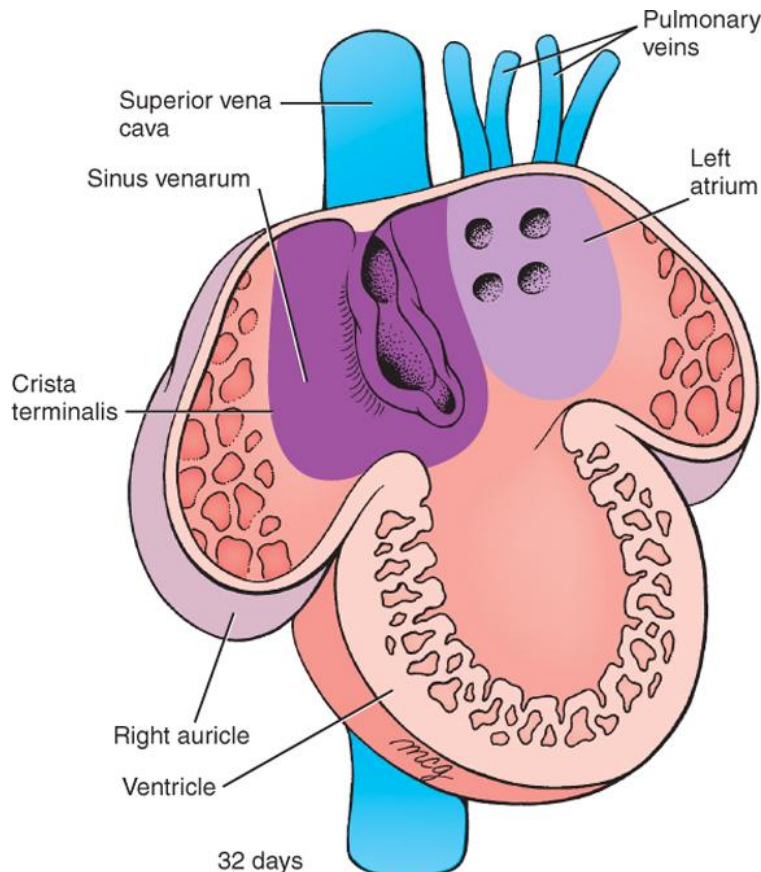
Differentiation of atrium II



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- Late 5th week
 - Pulmonary vein system undergo intussception (folding in of an outer layer) into dorsal wall of primitive atrium to form the definitive left atrium
 - Right horn sinus venosus :SVC, IVC orifices
 - Left horn sinus venosus: coronary sinus orifices
-
- Left and right venous valves join cranially to form septum spurium (part of s. secundum)
 - Right venous valve: valve of IVC, valve of coronary sinus

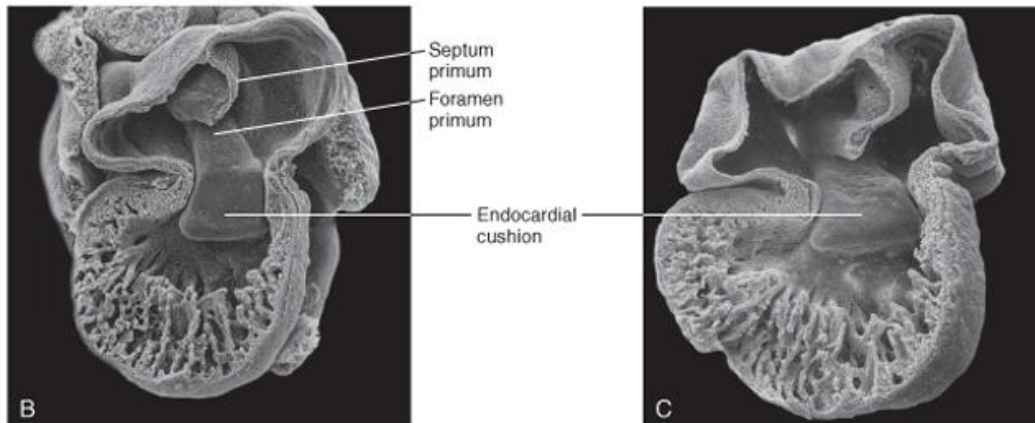
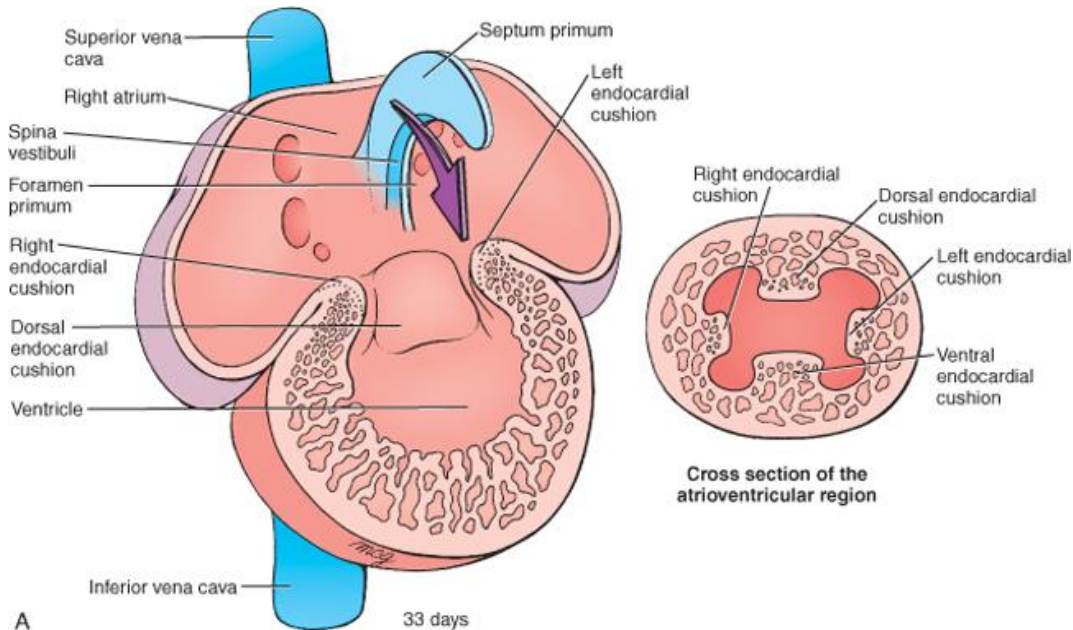
Differentiation of atrium III



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- **Crista terminalis:**
 - Ridge tissue separate trabeculated right atrium from smooth walled sinus venarum
 - Conducting system carry impulse from SA to AV node
- **4 pulmonary veins** are incorporated into the dorsal wall of left side primitive atrium
- **Complete formation of smooth walled part of future left atrium**

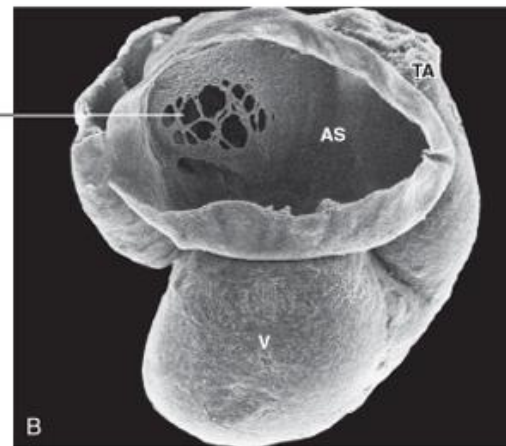
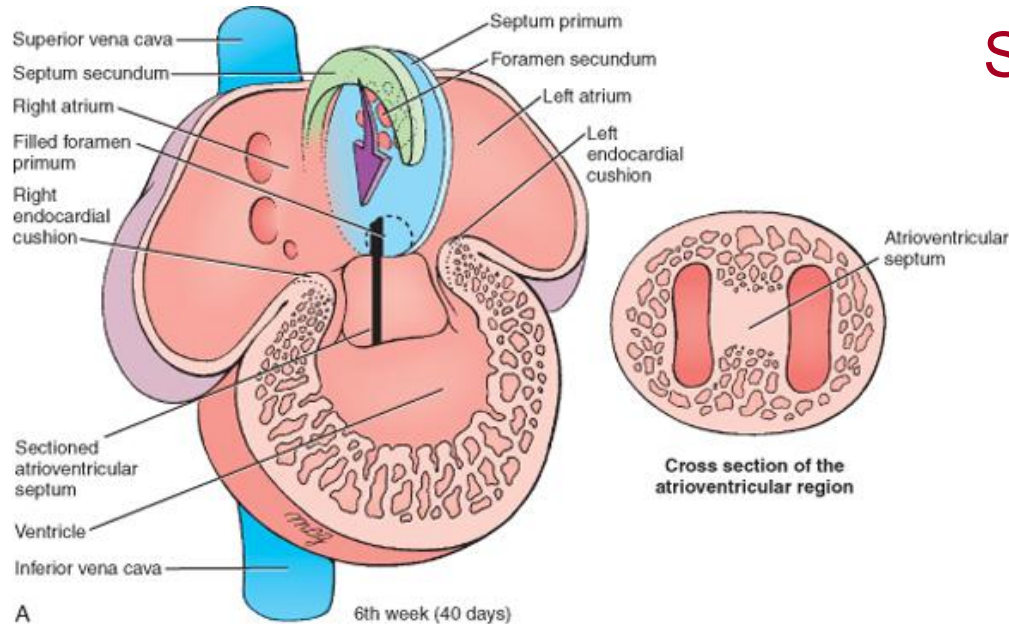
Septation of atria I



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- 5th week
- Septum primum grows from the dorsal wall of the primitive atrium as a crescent shaped and leaves a gap called foramen primum
- Foramen primum closes as the septum primum continues to grow to endocardial cushion

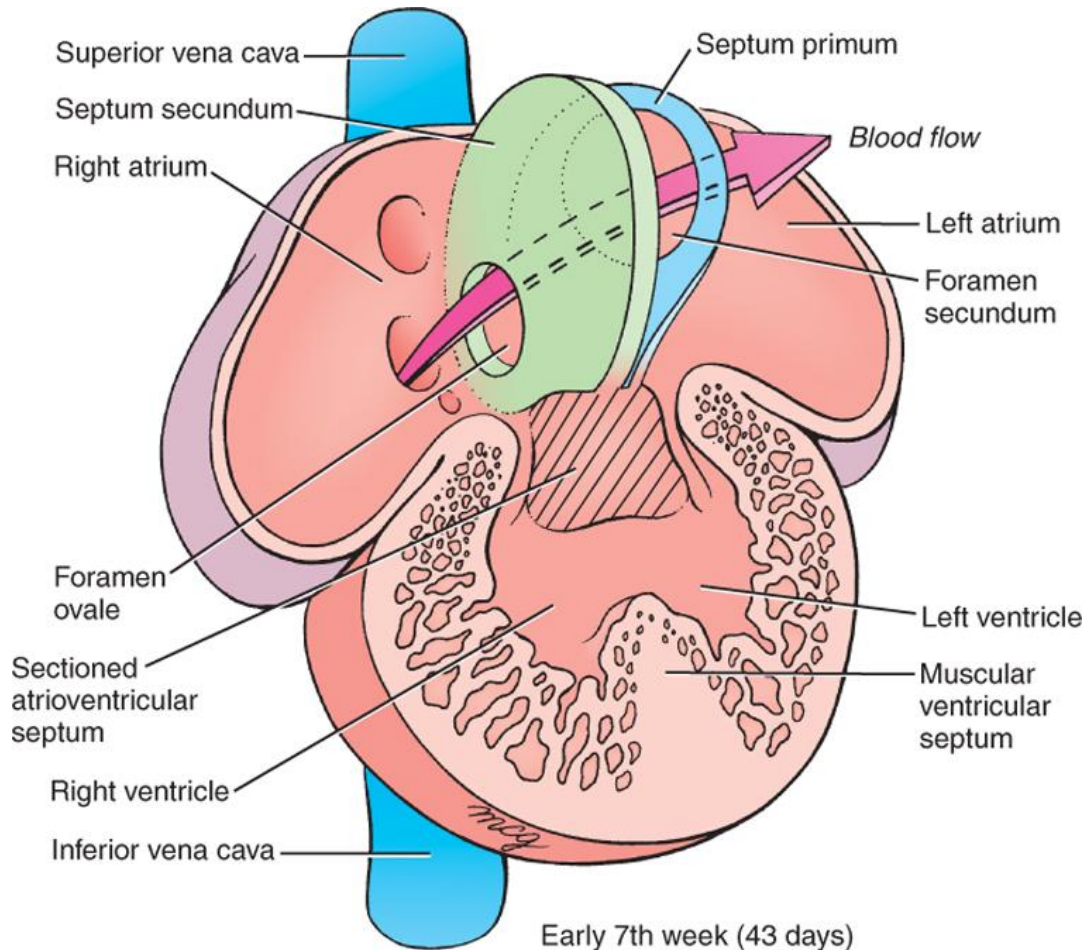
Septation of atria II



Mid 6th week (38 days)

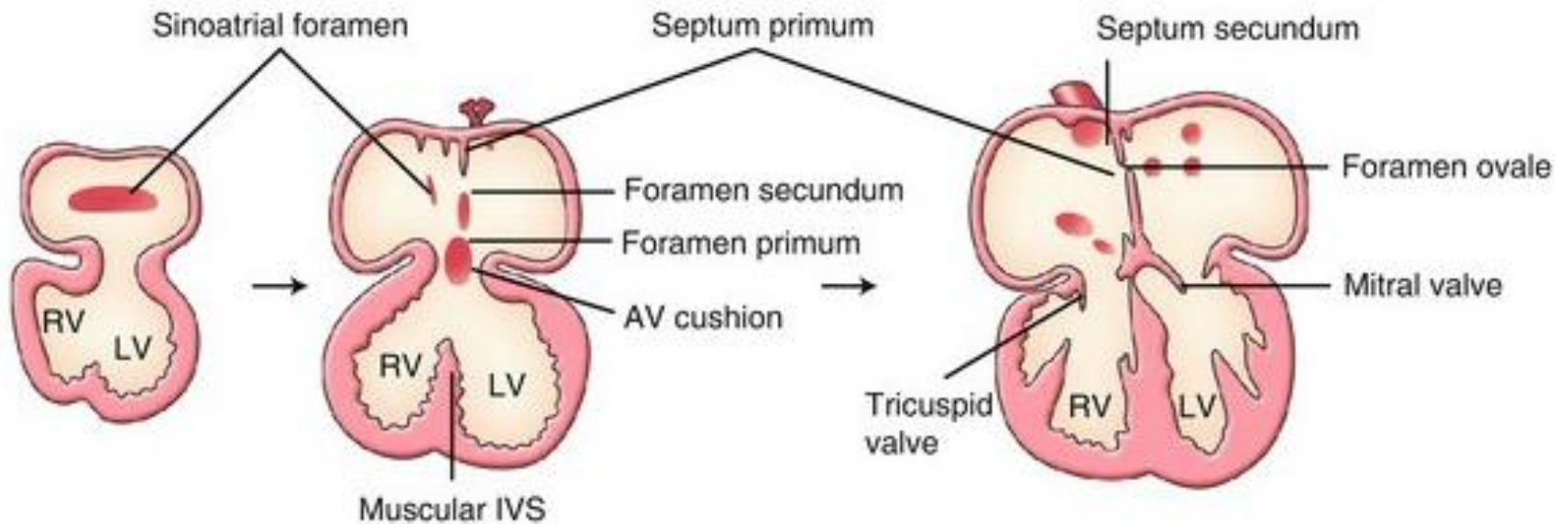
- 5th week
- Before septum primum close, foramen primum is obliterated that coalesce to form foramen secundum
- Septum secundum form at the right side of the septum primum

Septation of atria III



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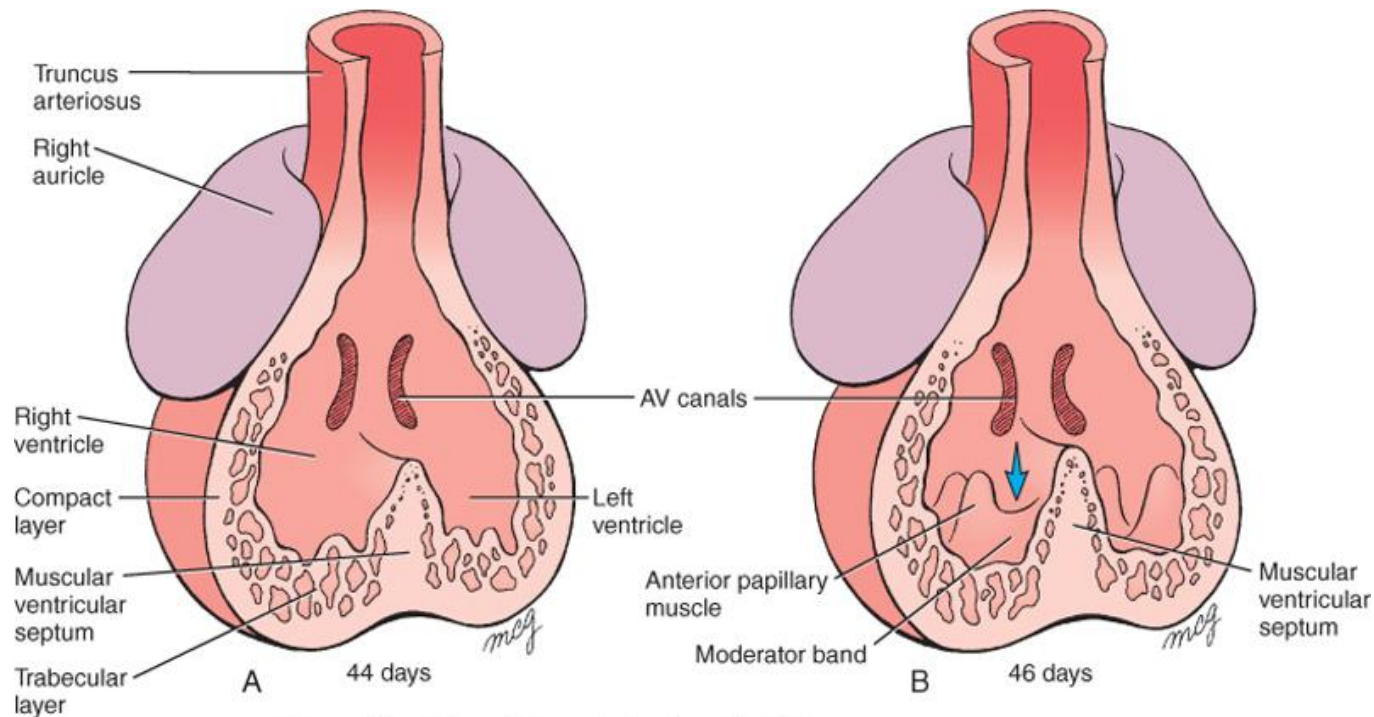
- Septum secundum covers up the foramen primum, but it leaves the foramen ovale
- Oxygenated blood can pass From right to left atria in the developing fetus before birth
- After birth, it closes because the septum primum fuses with the septum secundum, fossa ovalis



https://media.springernature.com/lw685/springer-static/image/chp%3A10.1007%2F978-1-4614-8875-0_2/MediaObjects/302553_1_En_2_Fig6_HTML.jpg

At this point the right and left atria have formed, having the foramen ovale through which oxygenated blood is passing, now have divided with valves in place to prevent the blood only from going from atria to ventricle and not back up, by the formation of the AV septum.

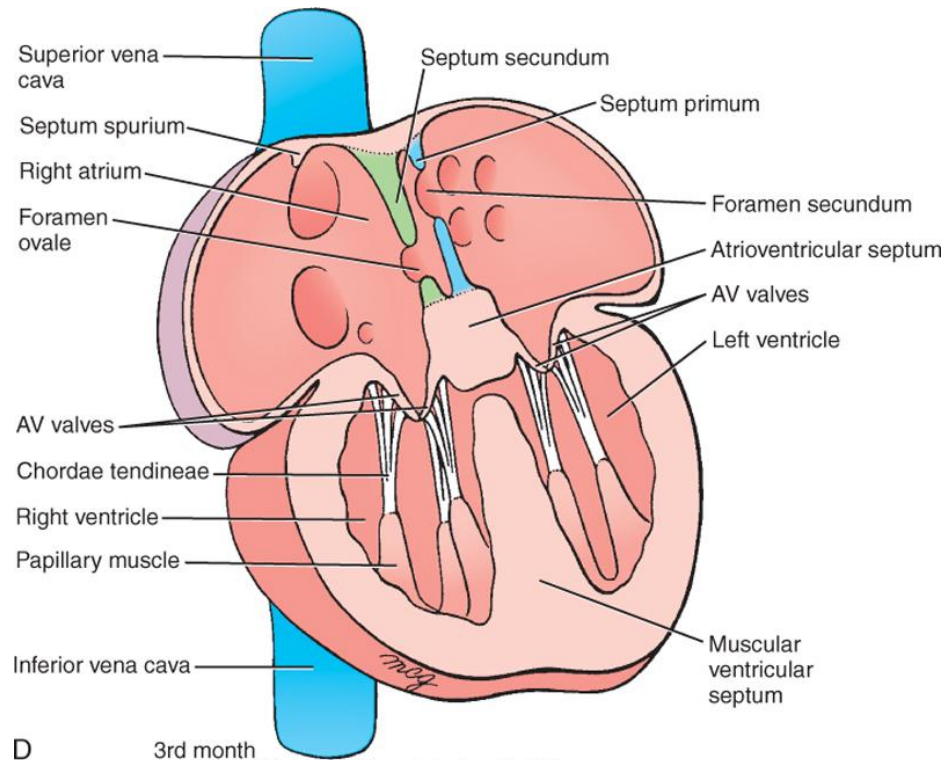
Septation of ventricles I



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- Thickening of myocardium and trabeculae form on inner wall of both ventricles
- Anterior papillary muscle: ventral portion of muscular interventricular septum
- Moderator band: right wall of muscular interventricular septum to anterior papillary muscle

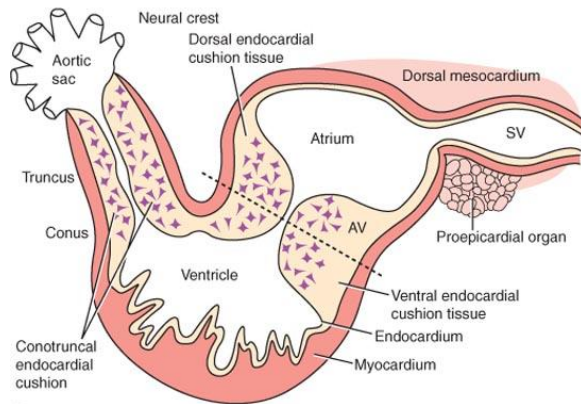
Septation of ventricles II



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- Muscular ventricular septum: groove between bulbus cordis - primitive ventricle, grows up forward the endocardial cushion but stop before it complete separate the ventricles.
- It will remain incomplete until the membranous septum is form later on in development, connecting the muscular septum to endocardial cushions

Septation of cardiac outflow I



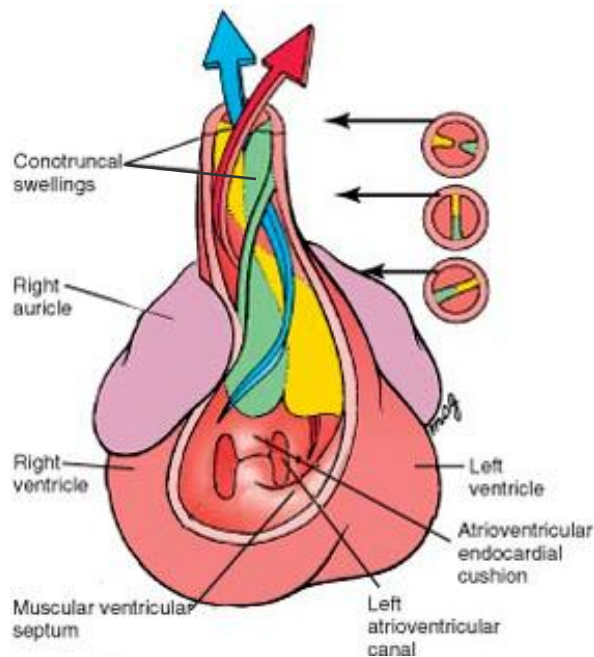
- 5th week, right and left conotruncal swellings grow out from the walls of common outflow tract

- Develop in spiral configuration

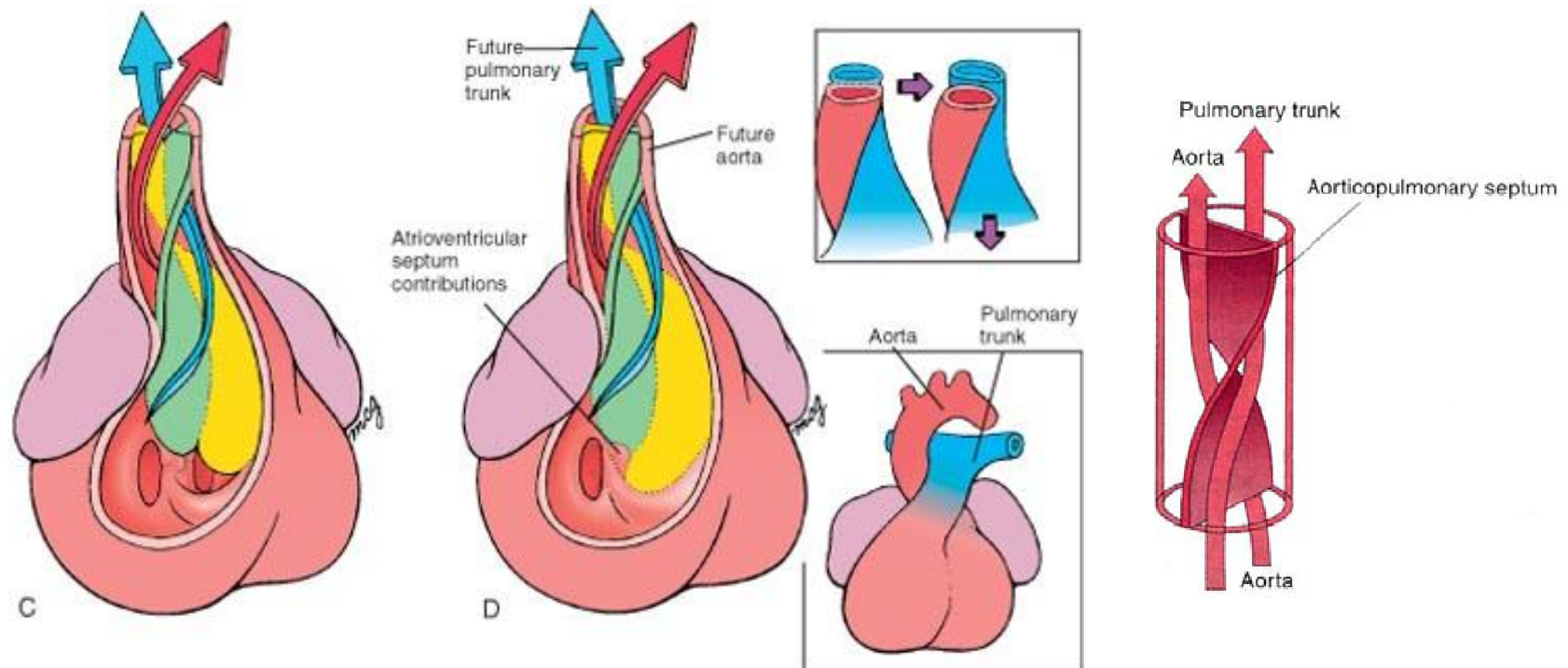
- Fuse in cranial-caudal direction to form conotruncal septum

- Aorta: send blood out of left ventricle

- Pulmonary trunk: outflow of right ventricle to lungs



Septation of cardiac outflow II

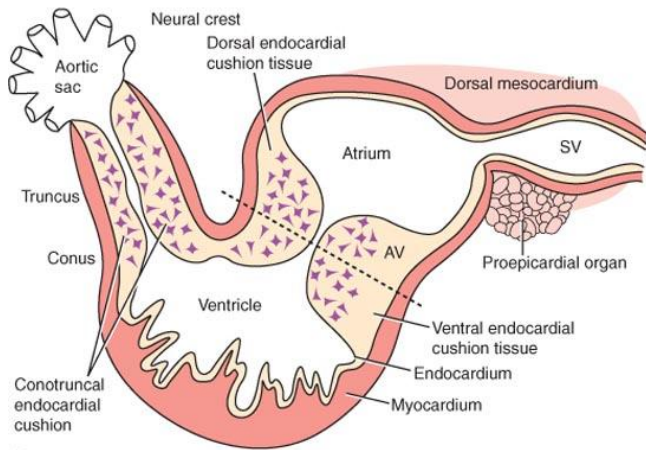


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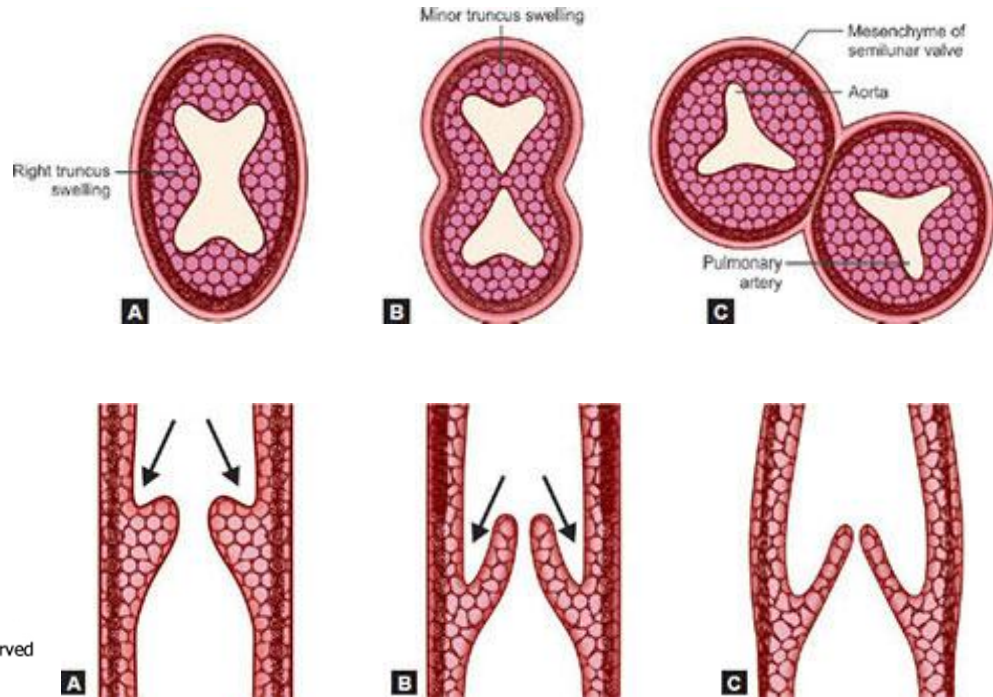
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- 9th week, caudal end of conotruncal septum reach muscular interventricular septum and atrioventricular septum
- Complete the interventricular septum

Formation of semilunar valve



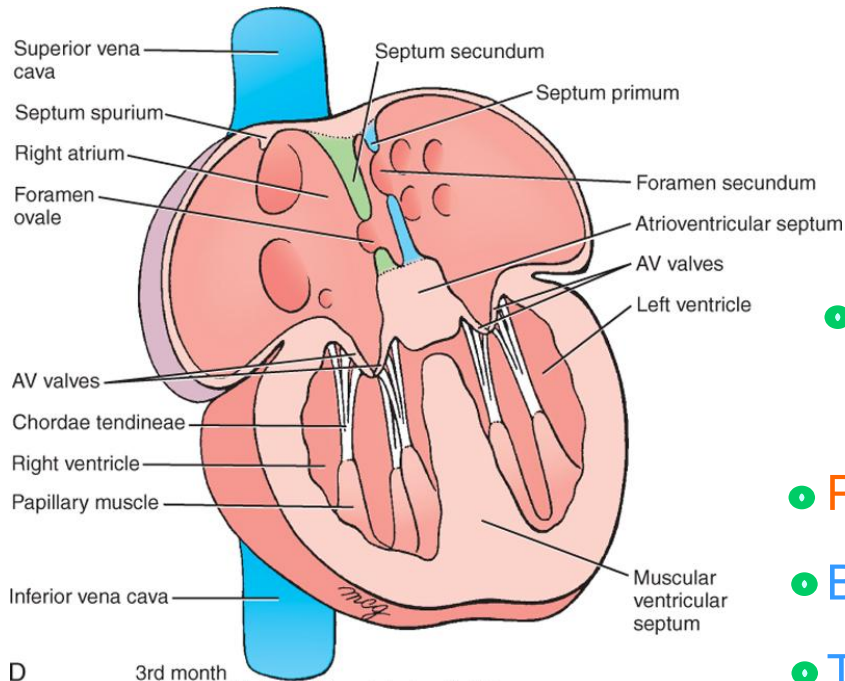
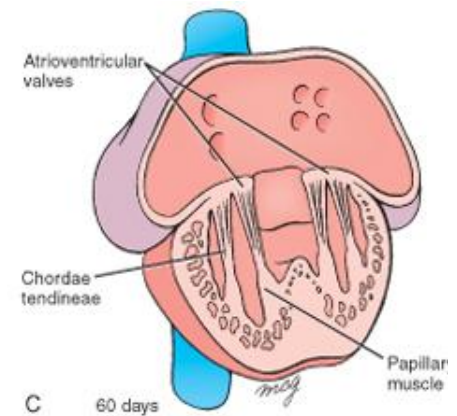
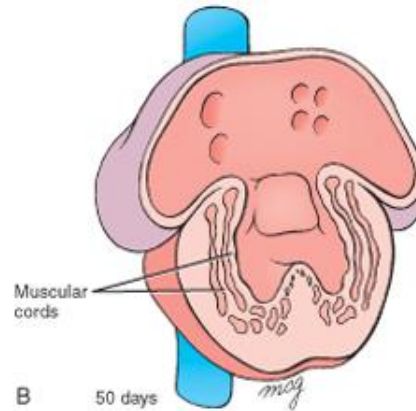
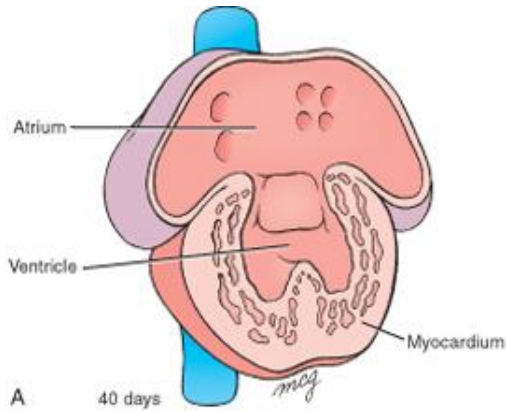
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<https://www.jaypeedigital.com/eReader/chapter/9789350902677/ch1>

- During formation of conotruncal septum (aorta - pulmonary trunk)
- Two small cushion tissues form in the opposite quadrants of outflow tract
- New cushion tissue is excavated and remodeling to form 2 cavities
- The cushion tissue slowly hollow out at their upper surface, forming the semilunar valves

Formation of atrioventricular valve



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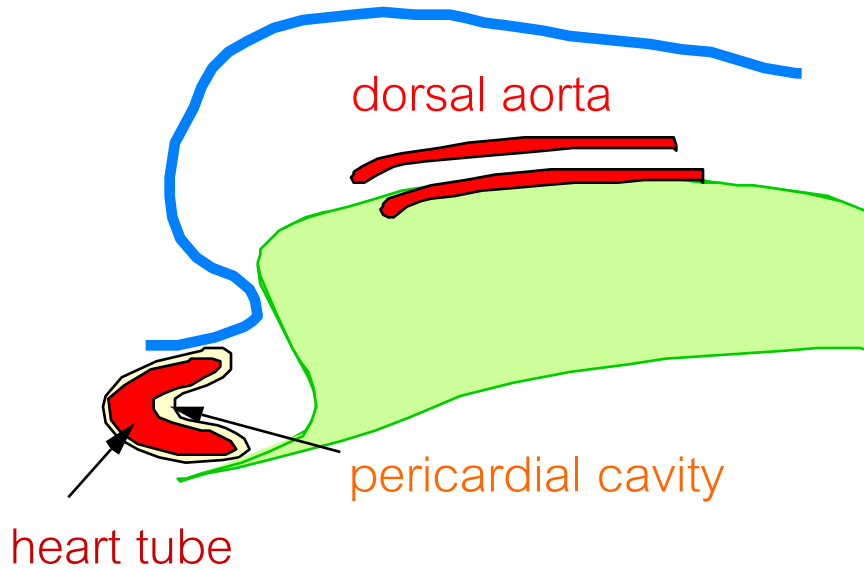
- Begin in 5th - 8th wk.
- Cusp: proliferate & differentiate from endocardial cushion
- Chordae tendineae: remodeling and erosion of ventricular myocardial wall
- Papillary muscles: small hillocks of myocardium
- Bicuspid valve: anterior & posterior cusp
- Tricuspid valve: septal cusp develop during 3rd month



Development of the aortic arches

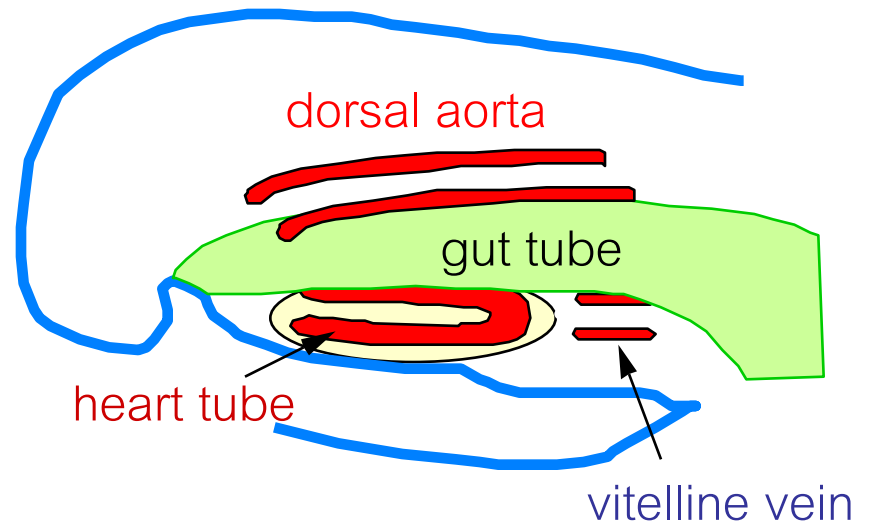
Assoc.Prof.Dr.Pornsawan Duangsuwan

Early development of heart

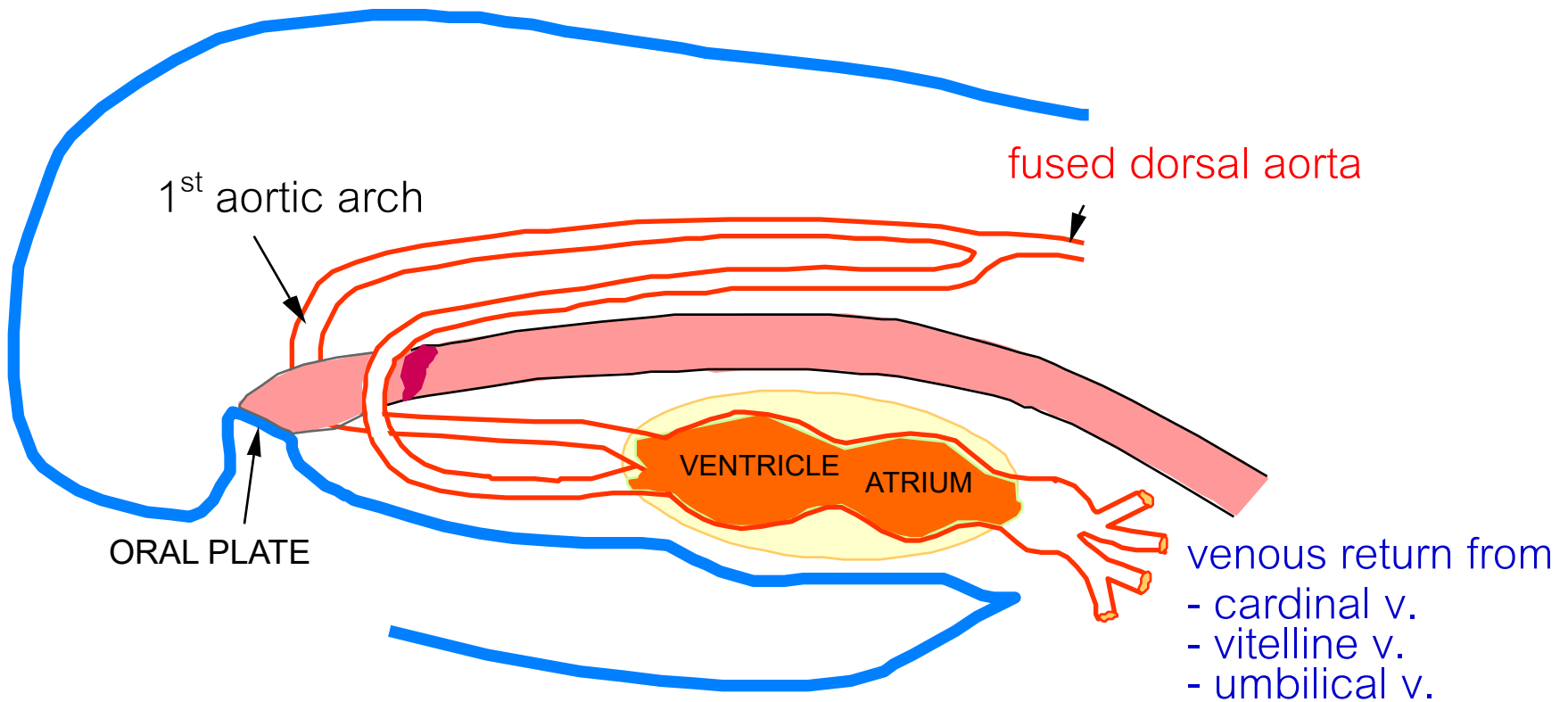


Head fold, heart tube pulled down to
beneath the pharynx

U-shaped heart tube and the forming
blood vessels are initially unconnected

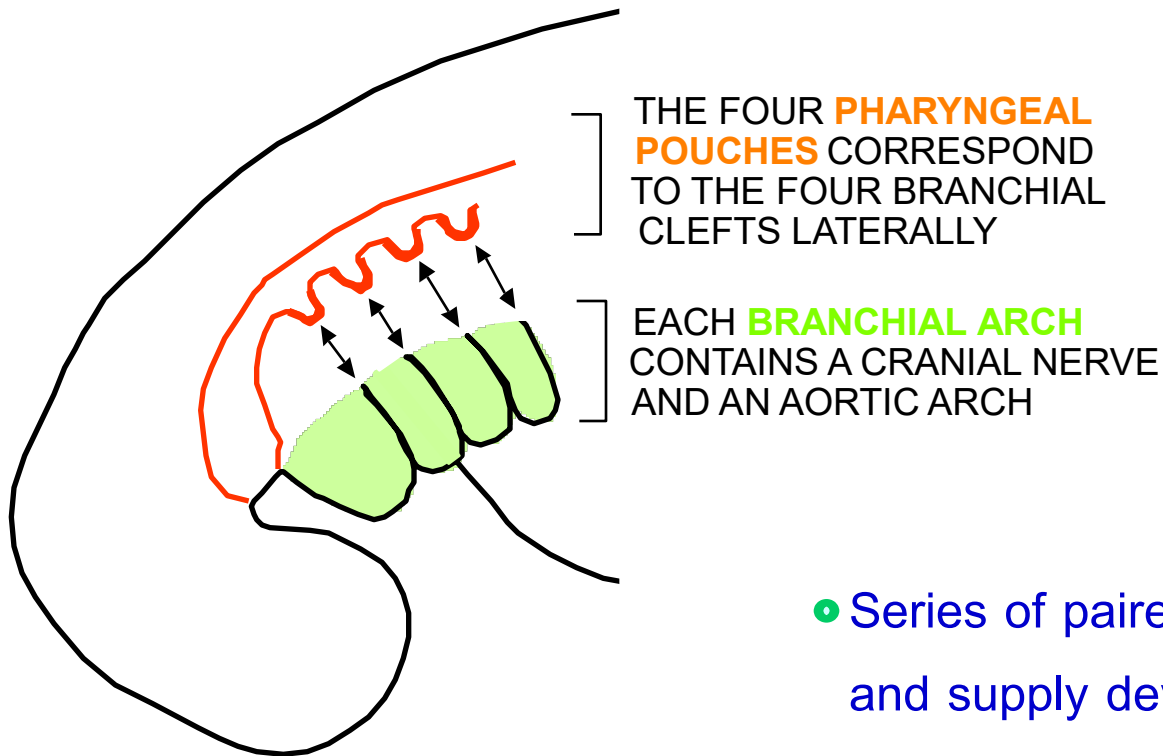


First aortic arch



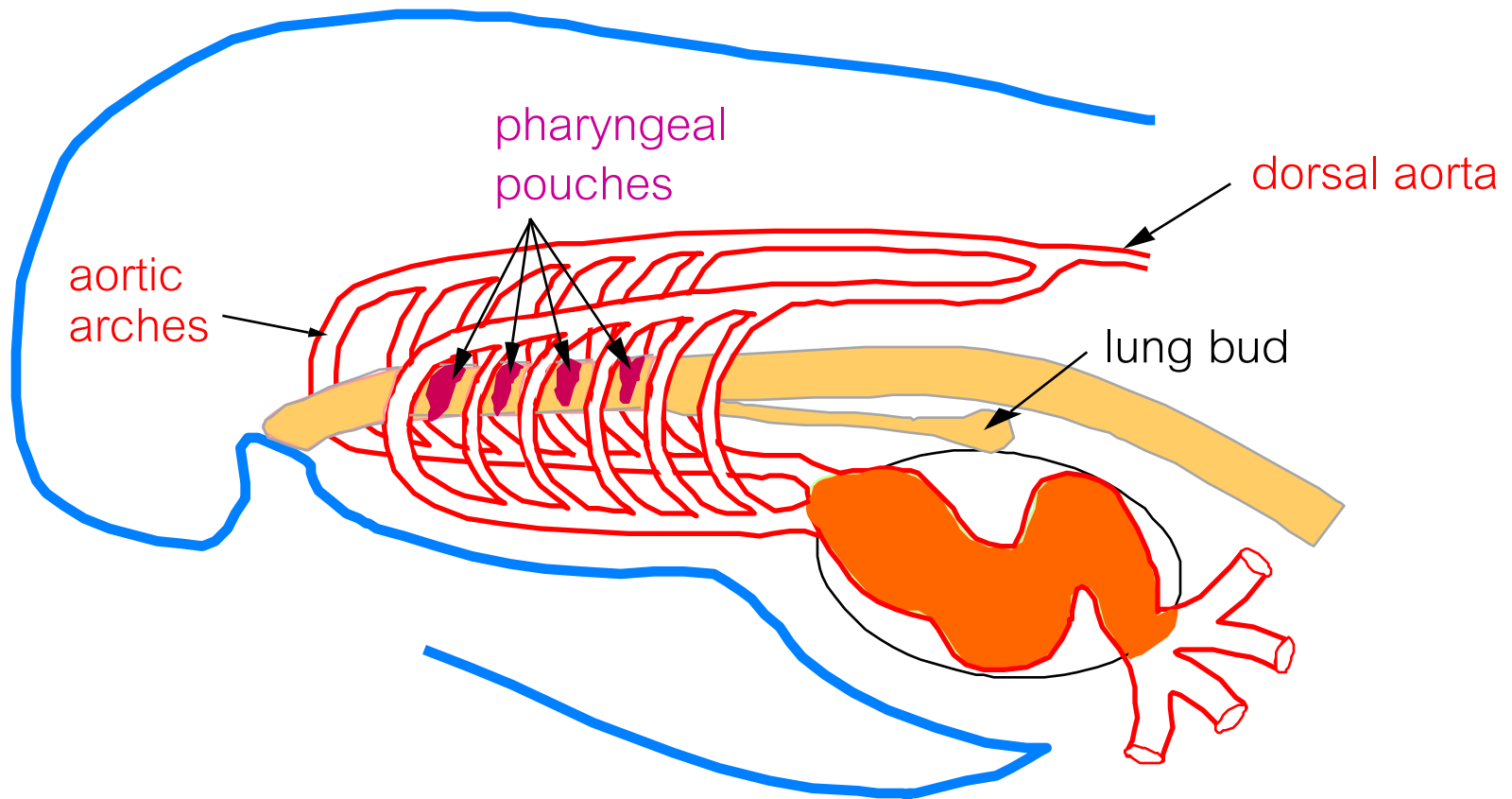
- Dorsal aorta form independently and then grow to meet the ventral output from the heart in the aortic arches
- Paired dorsal aorta fuse to form single aorta, just caudal to pharyngeal arches

Aortic arches



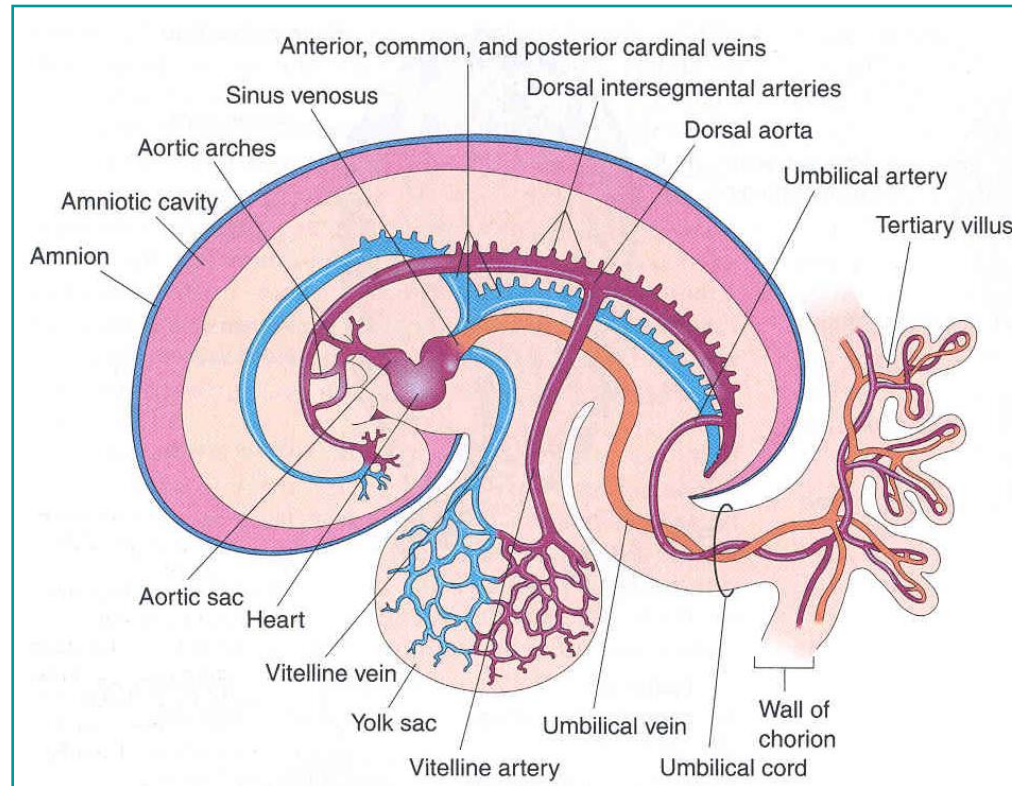
- Series of paired arterial channels encircling and supply developing pharyngeal arches
- Develop in the 4th week
- Arise from aortic sac
- Terminate in right and left dorsal aortae

Aortic arches



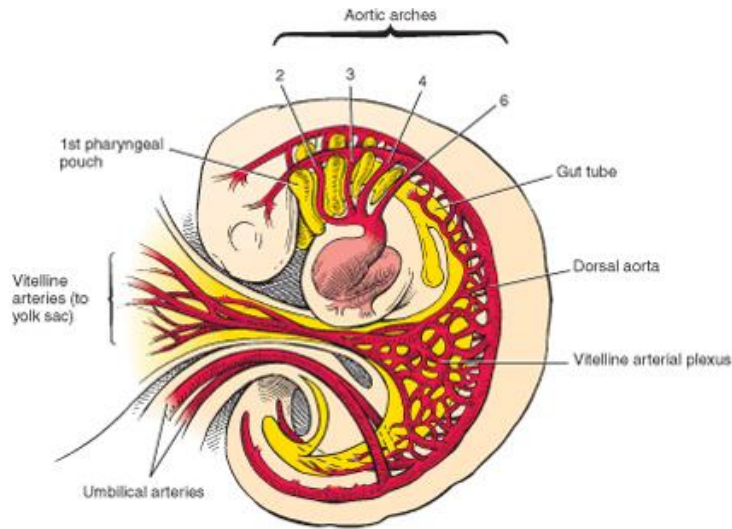
- Six pairs, but the 5th pair is poorly developed and disappears soon after formation
all are not present at the same time
- Dorsal aorta fuse to form single median vv. from 4th thoracic – 4th lumbar segment

Dorsal aorta



- Ventral aortic branch : supply gut tube and derivatives
- Lateral aortic branch : supply kidney, suprarenal gland and gonad
- Posterolateral aortic branch : intersegmental a.

Ventral branches of descending aorta

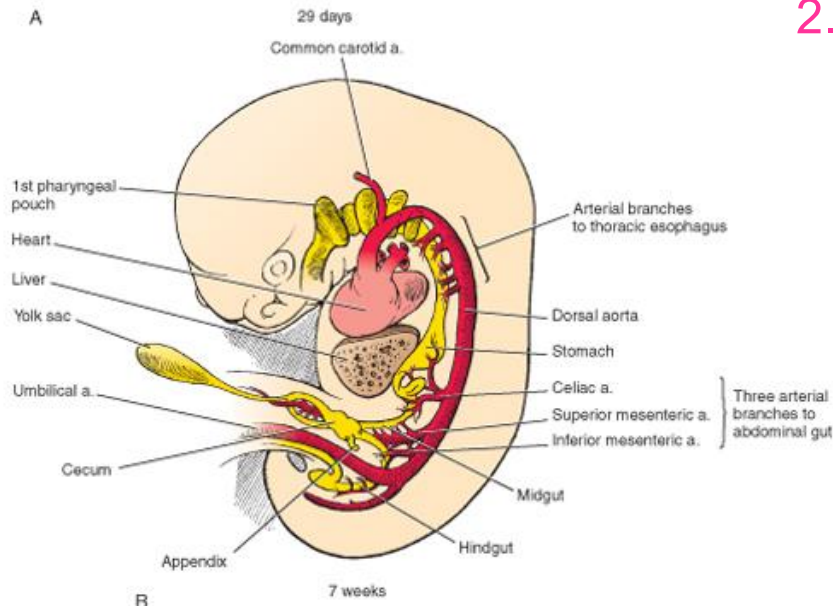


1. Vitelline a. : supply gut tube and derivatives

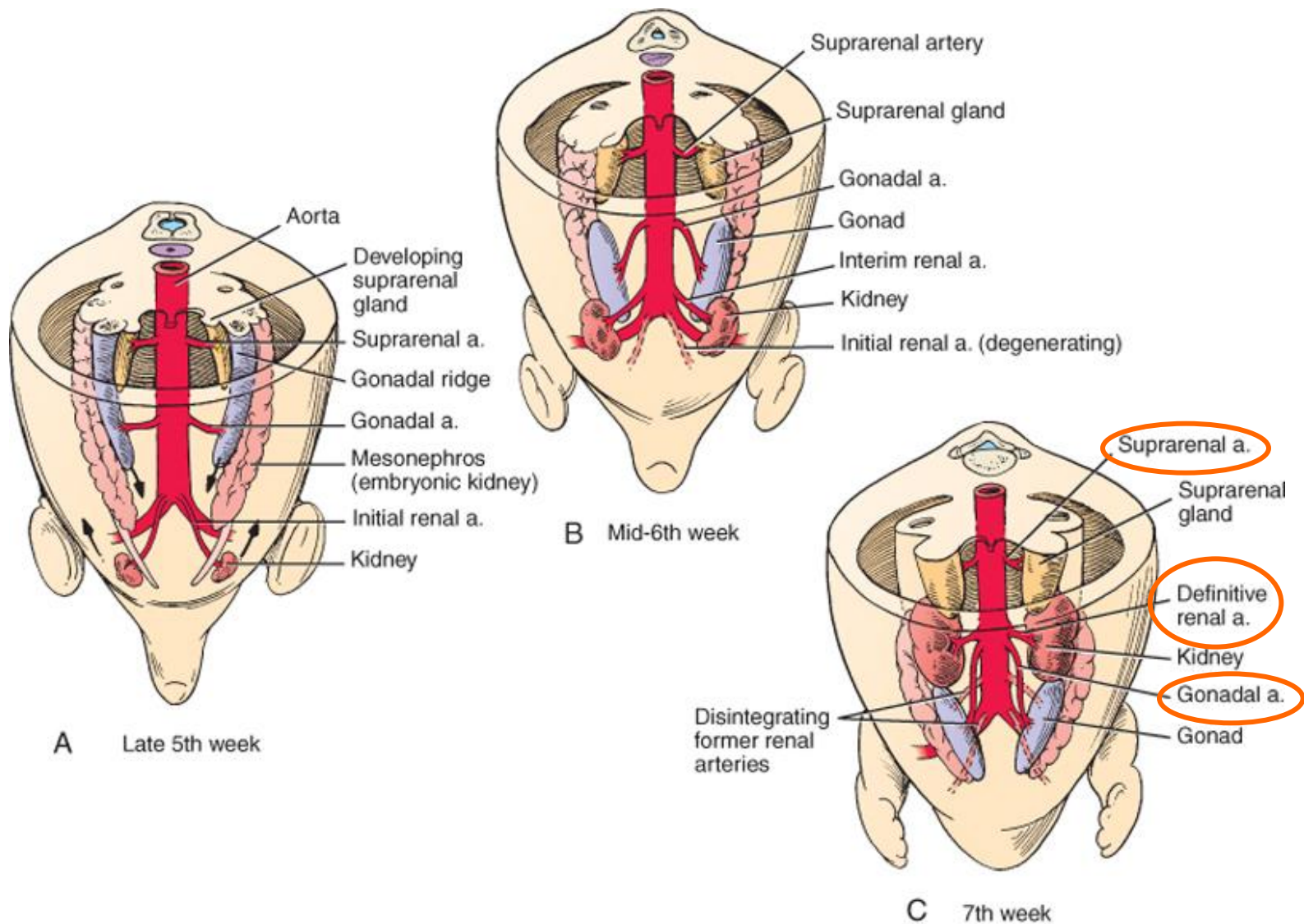
- Celiac trunk: fore gut
- Superior mesenteric a.: mid gut
- Inferior mesenteric a.: hind gut

2. Umbilical a. :

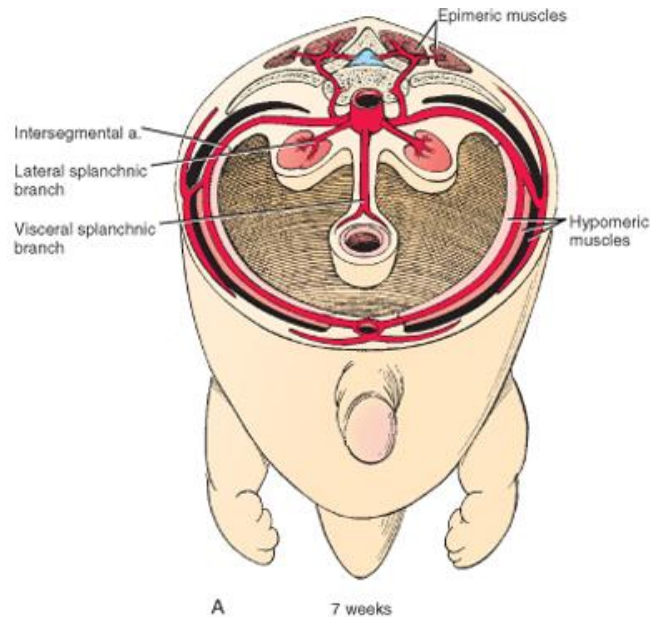
- Carry poorly O₂ blood to placenta
- Proximal: superior vesical a.
- Distal: medial umbilical ligament



Lateral branches of descending aorta



Postero-lateral branches of descending aorta



- Somatic branches
- Thirty or more pairs arise at serial segmental levels, run between somites
- Supply body wall, limbs, brain, spinal cord

Derivatives:

Cervical : join to form vertebral a.

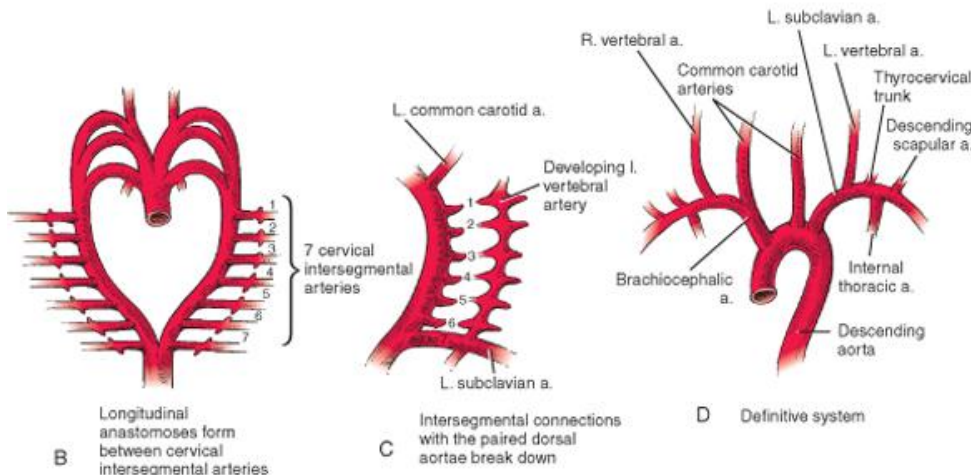
: 7th pair --->subclavian a.

Thoracic : posterior intercostal a.

Abdominal : lumbar a.

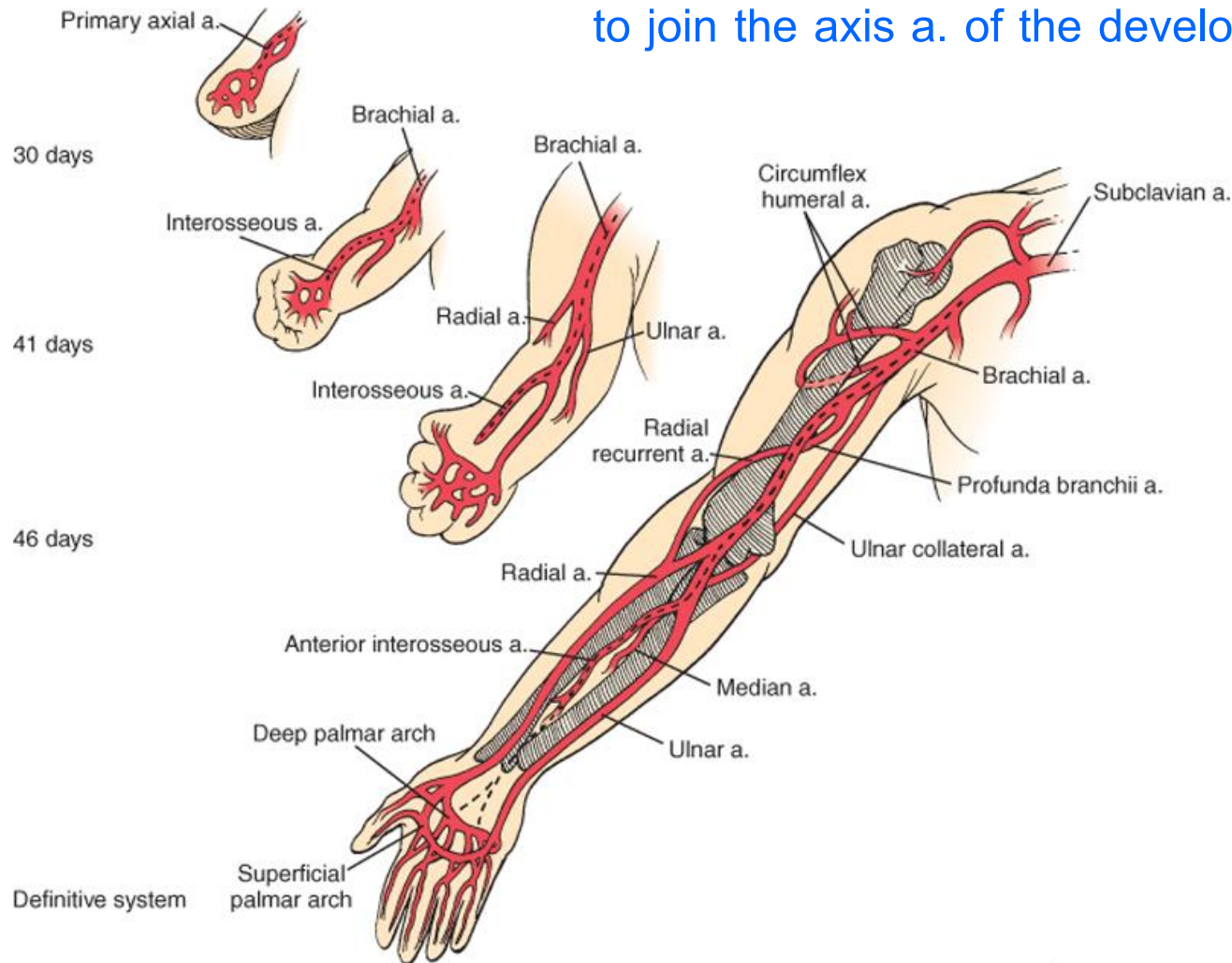
: 5th pair ---->common iliac a.

Sacral : lateral sacral a.

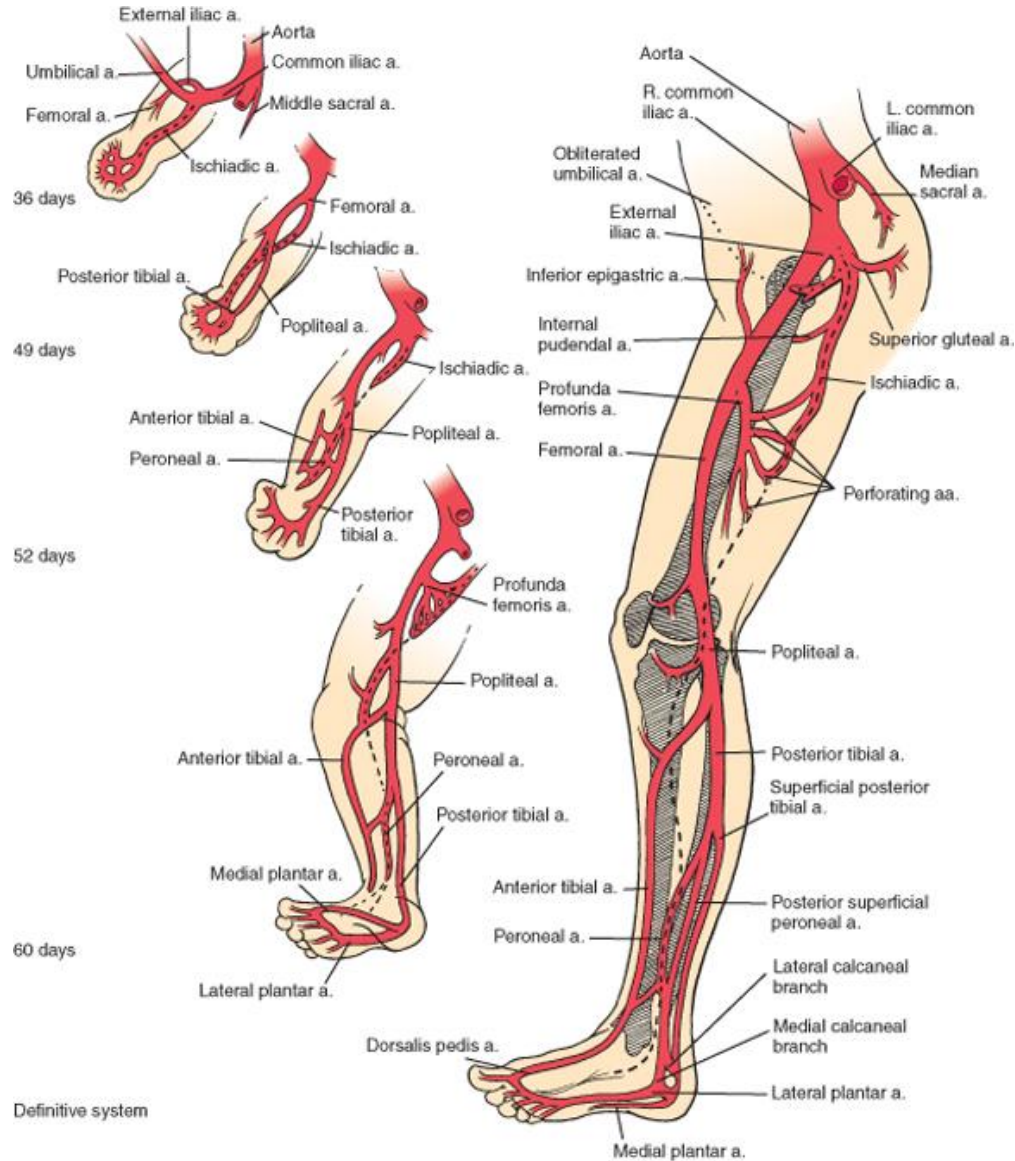


Arterial system of the upper limb

- 7th cervical intersegmental a. grow into limb buds to join the axis a. of the developing upper limb

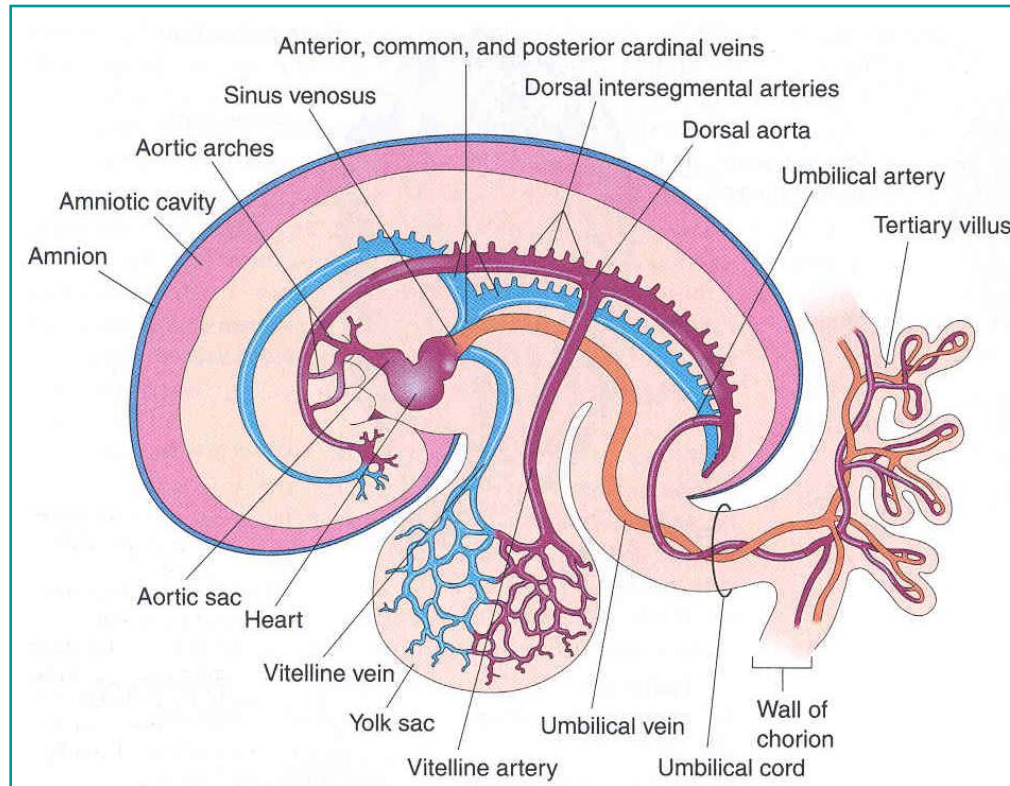


Arterial system of the lower limb



- 5th lumbar intersegmental a. join the axis a. forming in the lower limb
- Remnants: ishiadic a. (popliteal a., peroneal a.)

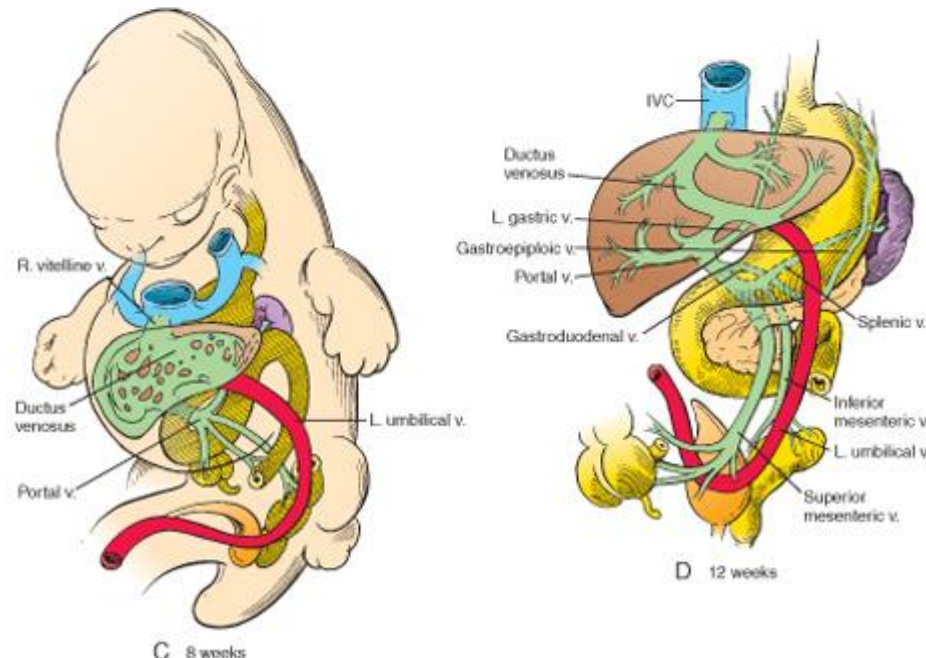
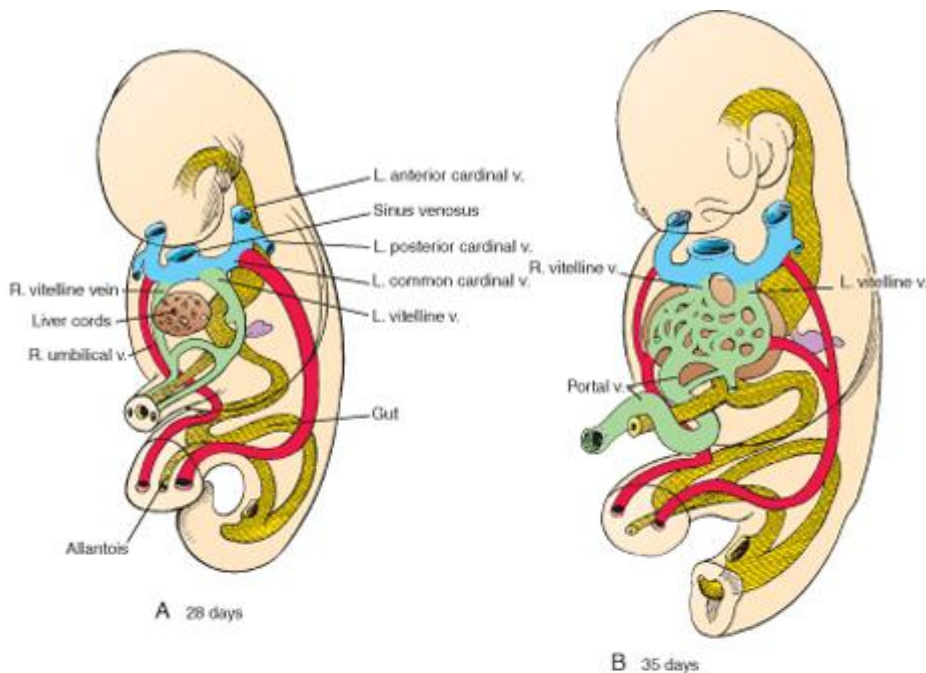
Embryonic venous system



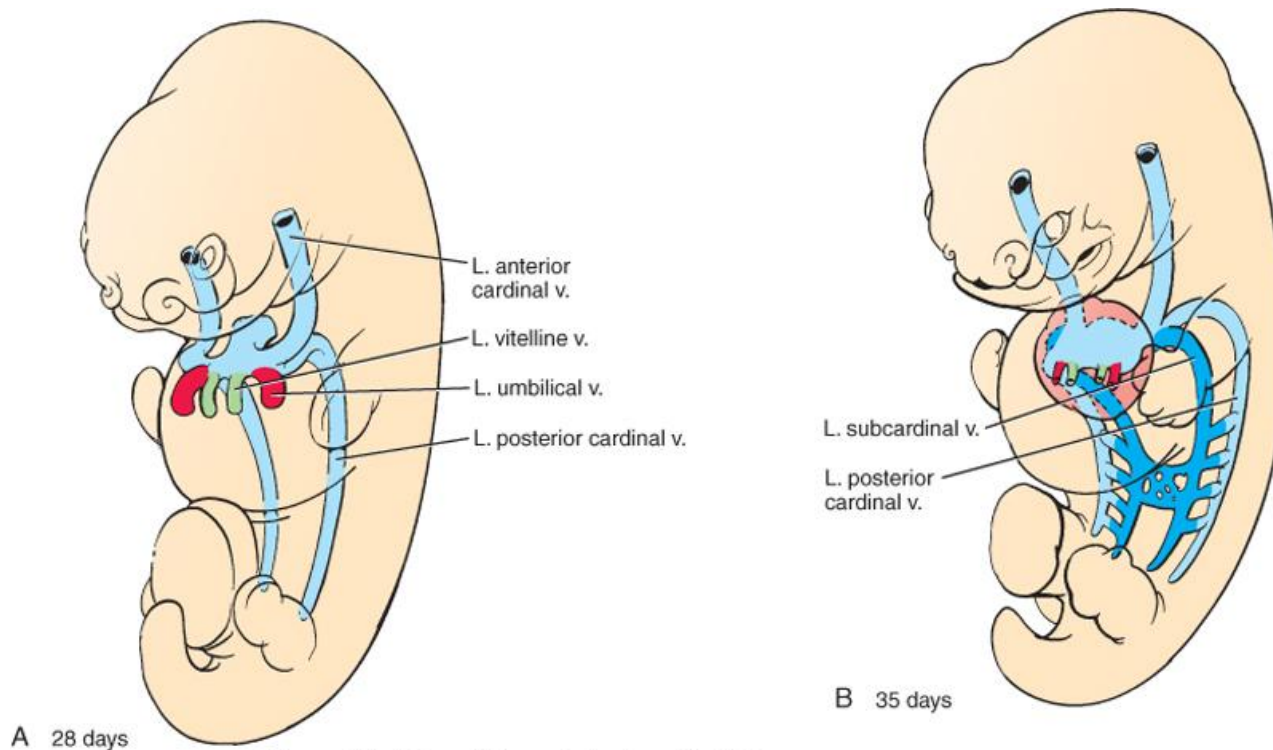
- Vitelline system: give rise liver sinusoid, portal system, portion of IVC
- Umbilical veins
- Cardinal system

Vitelline and umbilical veins

- Rt. & Lt. vitelline v. form a portal system to drain blood from foregut, midgut and part of anorectal canal
- Lt. vitelline v.: disappear
- Rt. umbilical v.: disappear
- Lt. umbilical v.: anastomose with ductus venosus ---> shut O₂ placental blood into IVC and right atrium
- Lt. umbilical v. --> lig. teres hepatis
- ductus venosus --> lig. venosum



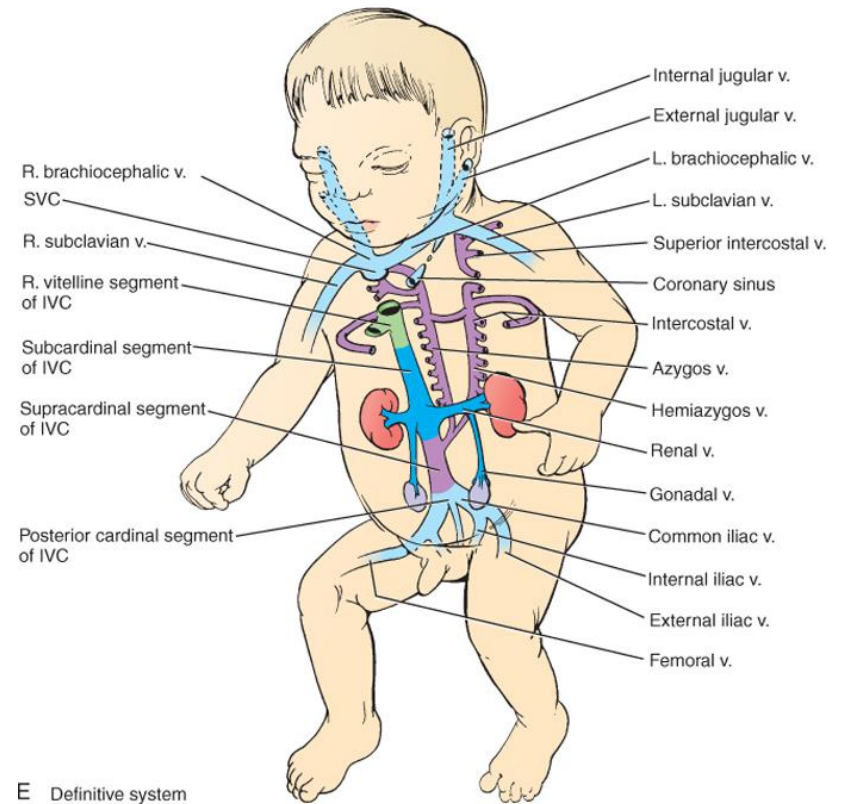
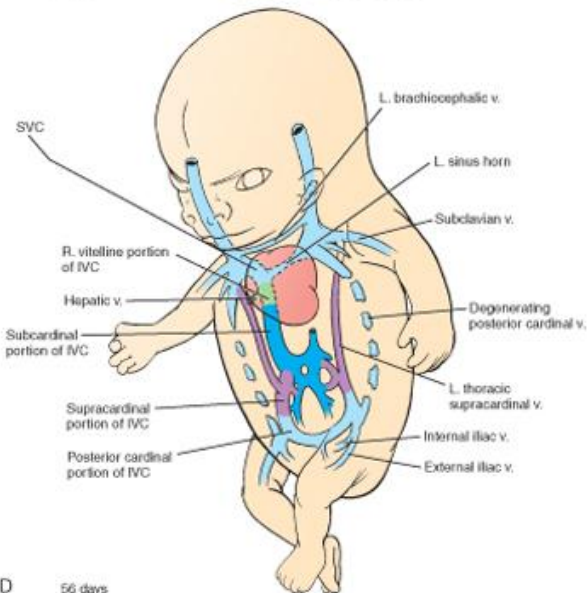
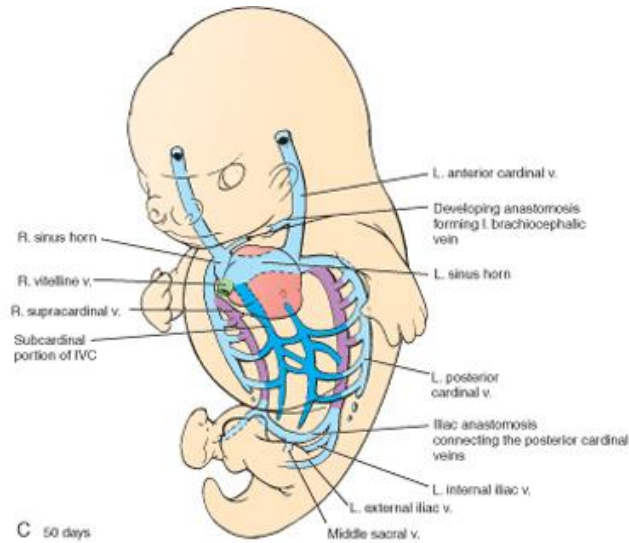
Systemic venous system



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- Anterior cardinal v. drain blood from both side of head and neck
- Posterior cardinal v.: drain blood from trunk
- Posterior cardinal v. are replaced by set of subcardinal and supracardinal v.

Systemic venous system

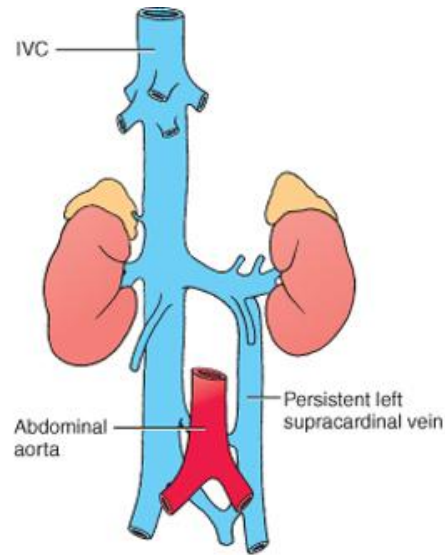


E Definitive system

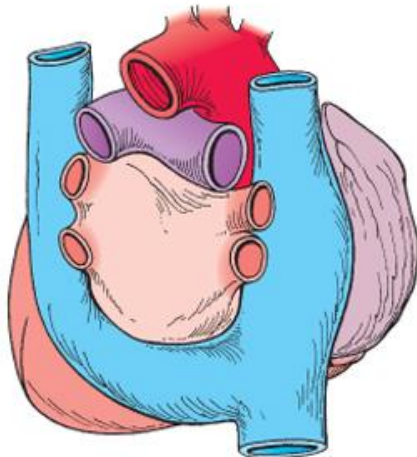
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- Following remodeling of the subcardinal system, the supracardinal v. sprout
- Remodeling of abdominal venous system occurs through obliteration of left supracardinal v.

Left superior vena cava (LSVC)



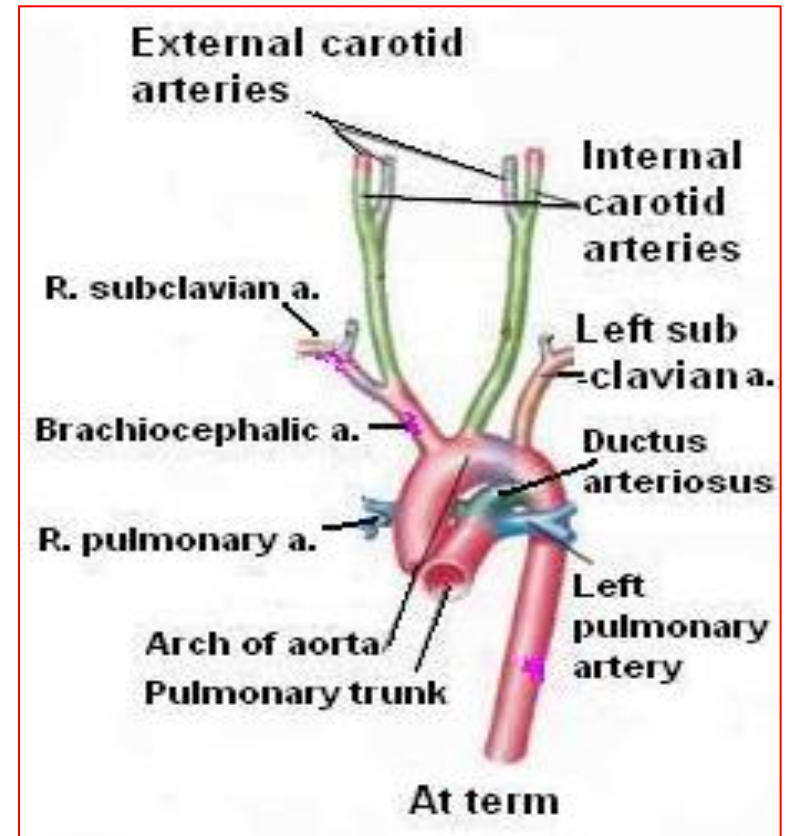
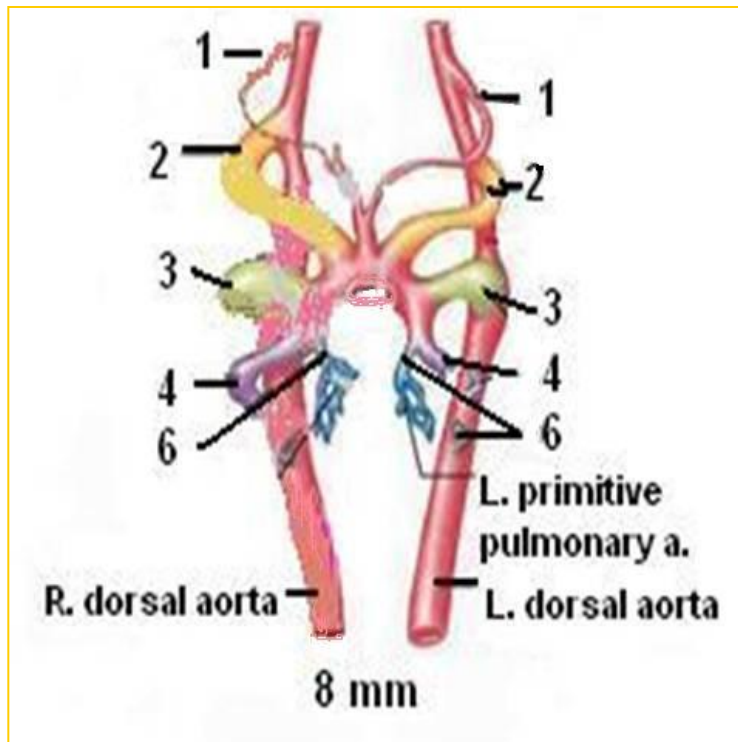
A Double inferior vena cava



B Double superior vena cava

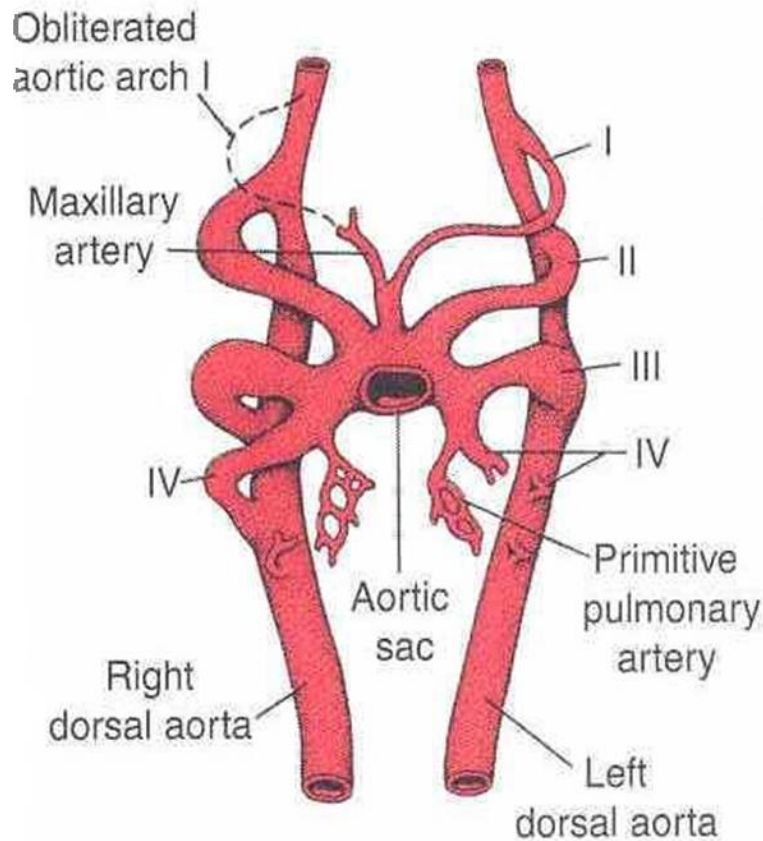
- LSVc occurs in 0.3% to 0.5% of the normal population
- In 65% of cases, left brachiocephalic vein is also missing
- 4% of patients with CHD have an LSVc
- Usually drains to the coronary sinus

Aortic arch development



- During week 6 to 8, the primitive aortic arch pattern is transformed into adult arterial arrangement of carotid, subclavian, and pulmonary arteries

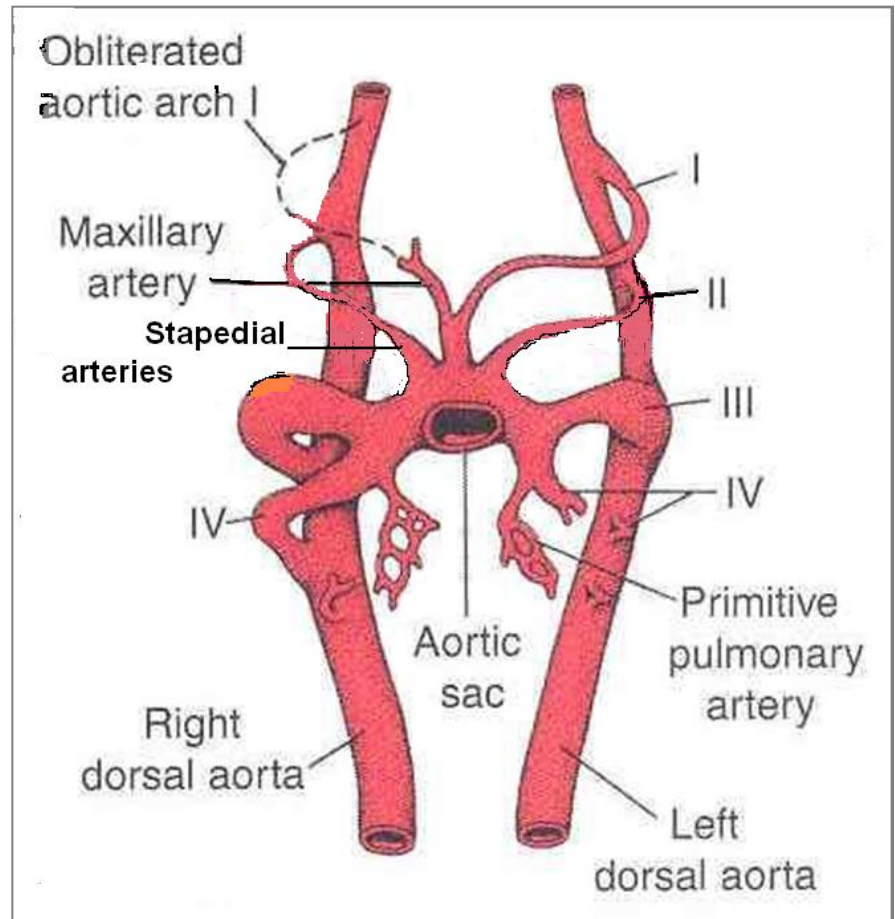
Derivatives of 1st pair



- Complete by Day 24
- Regress as 2nd arches form by Day 26
- Remaining parts form maxillary a.
- Supply ear, teeth, muscles of eyes and face

Derivatives of 2nd pair

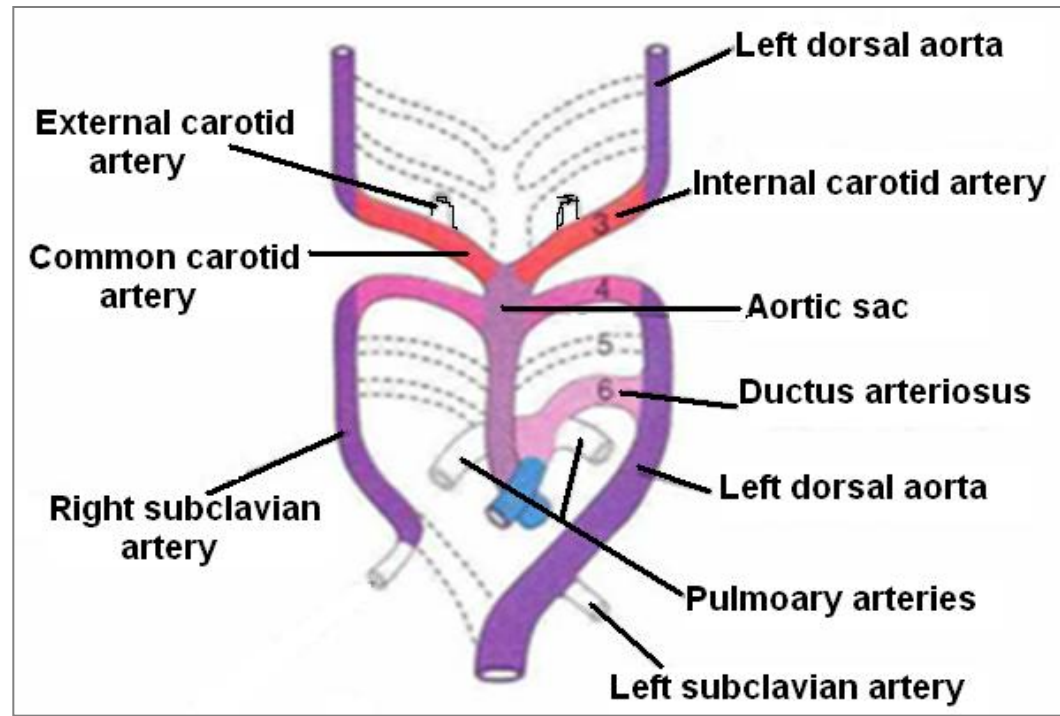
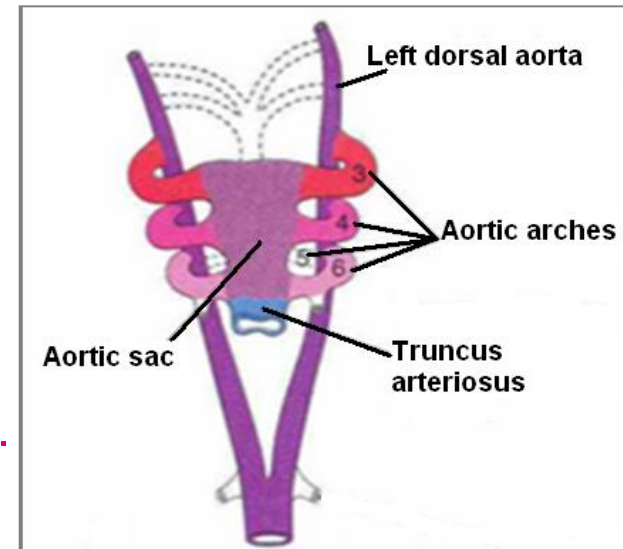
- Complete by Day 26
- Regress as 6th arches form by Day 29
- Remaining parts form stapedial a.
- Supply stapes bone in developing ear



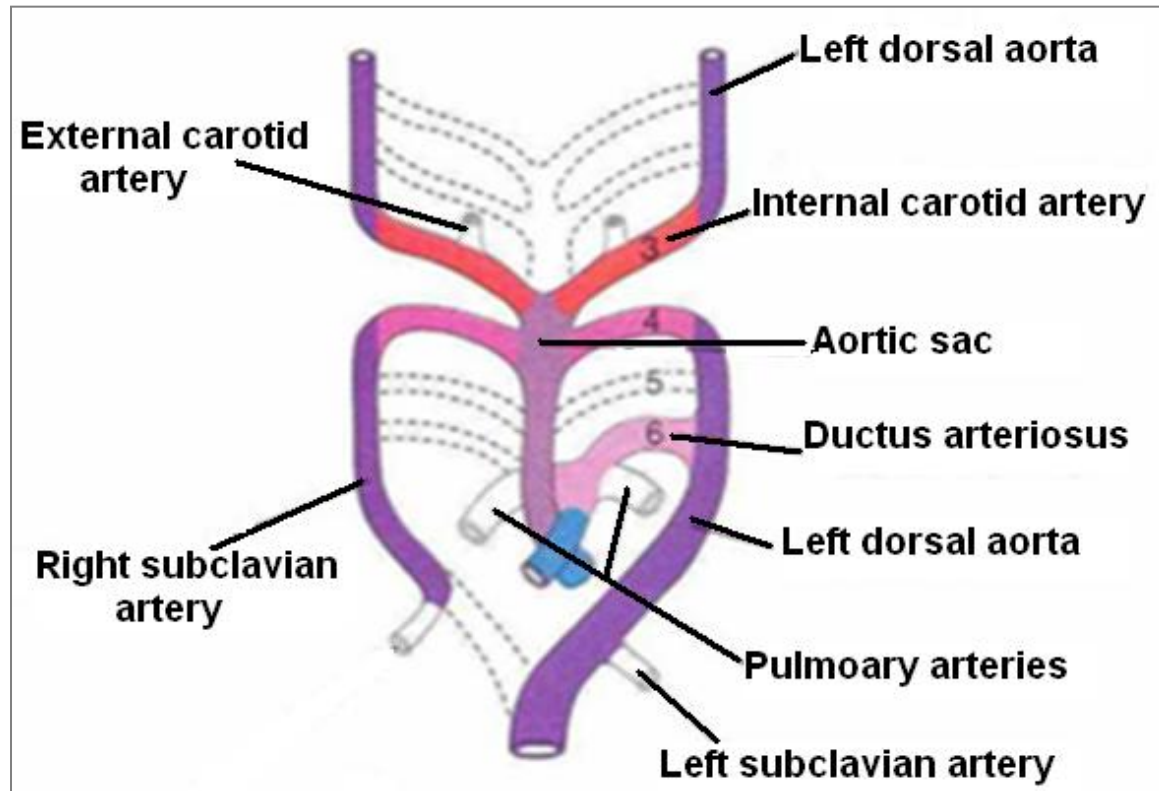
Derivatives of 3rd pair

- Complete by Day 28
- Proximal part form common carotid a.
 - : supply structures in head
- Distal parts join with dorsal aorta to form int. carotid a.
 - : supply ears, orbits, brain and meninges

External carotid a. sprout from common carotid a.



Derivatives of 5rd pair



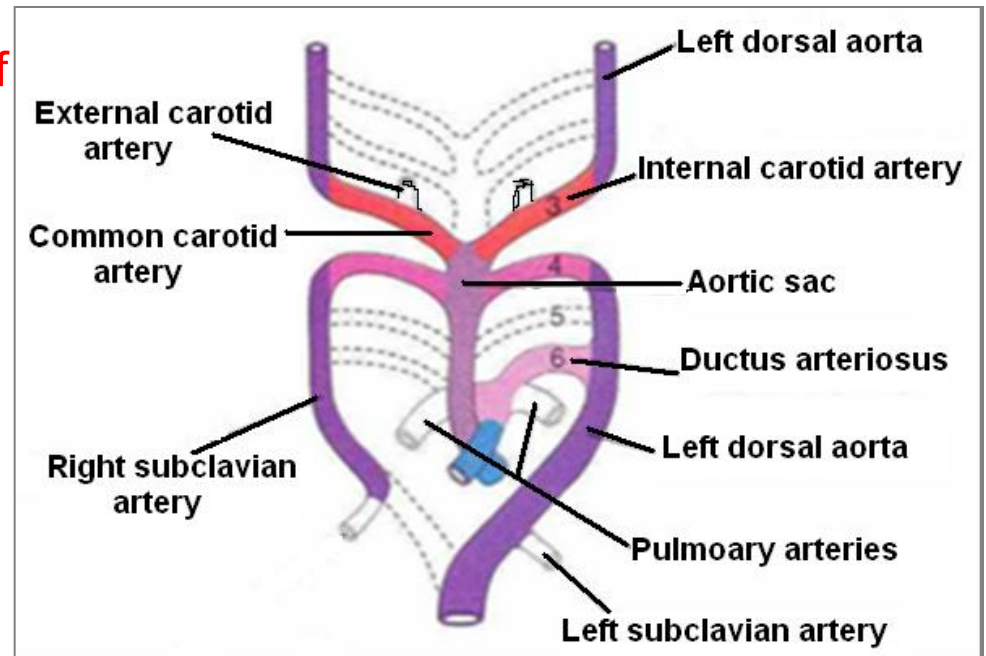
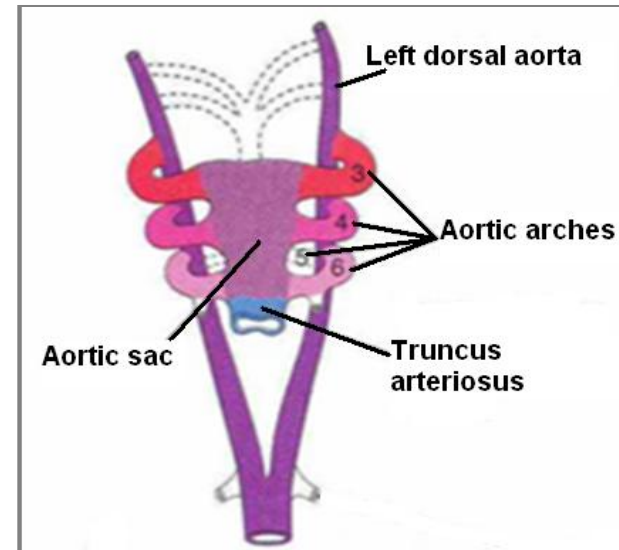
- Disappears completely with NO vascular derivatives
- In other 50% of embryos, these arteries do not develop

The fate of 1st - 3rd pairs of aortic arches are bilateral.

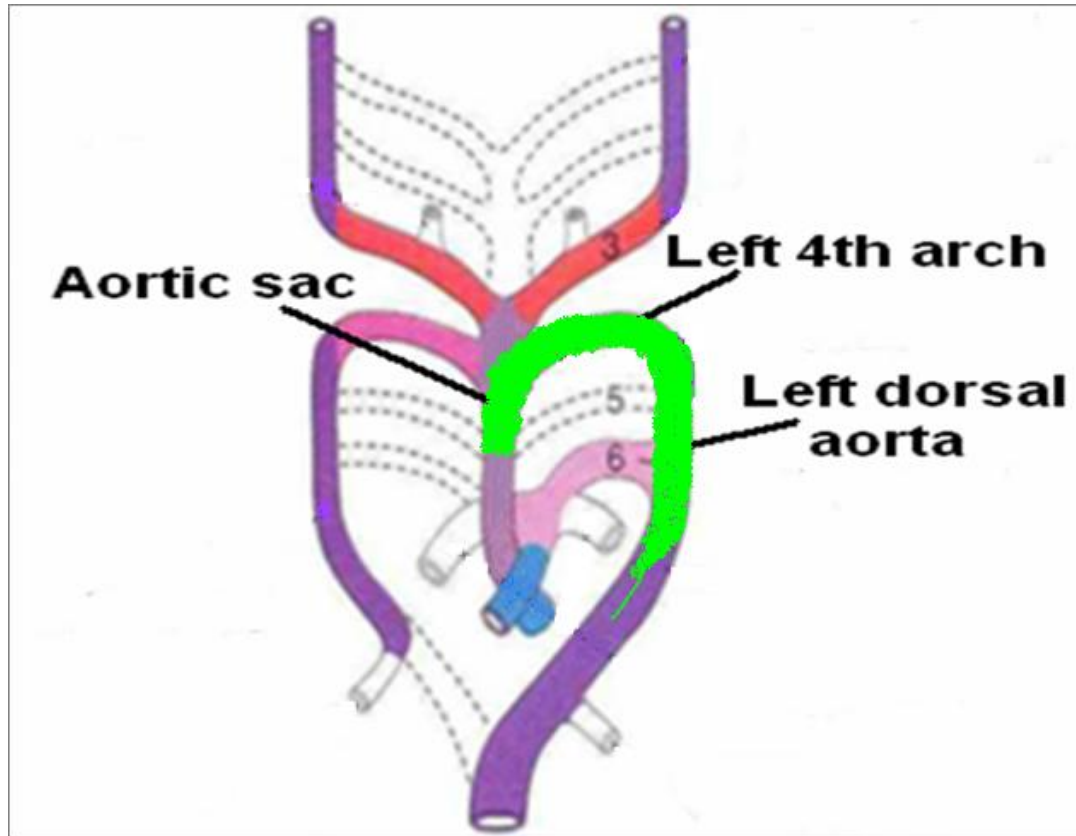
The fate of 4th & 6th pairs of aortic arches differs on the right and left side.

Derivatives of 4rd pair

- Form by day 28
- LEFT 4th aortic arch
 - : Form part of arch of aorta
- RIGHT 4th aortic arch
 - : Becomes the proximal part of right subclavian a.

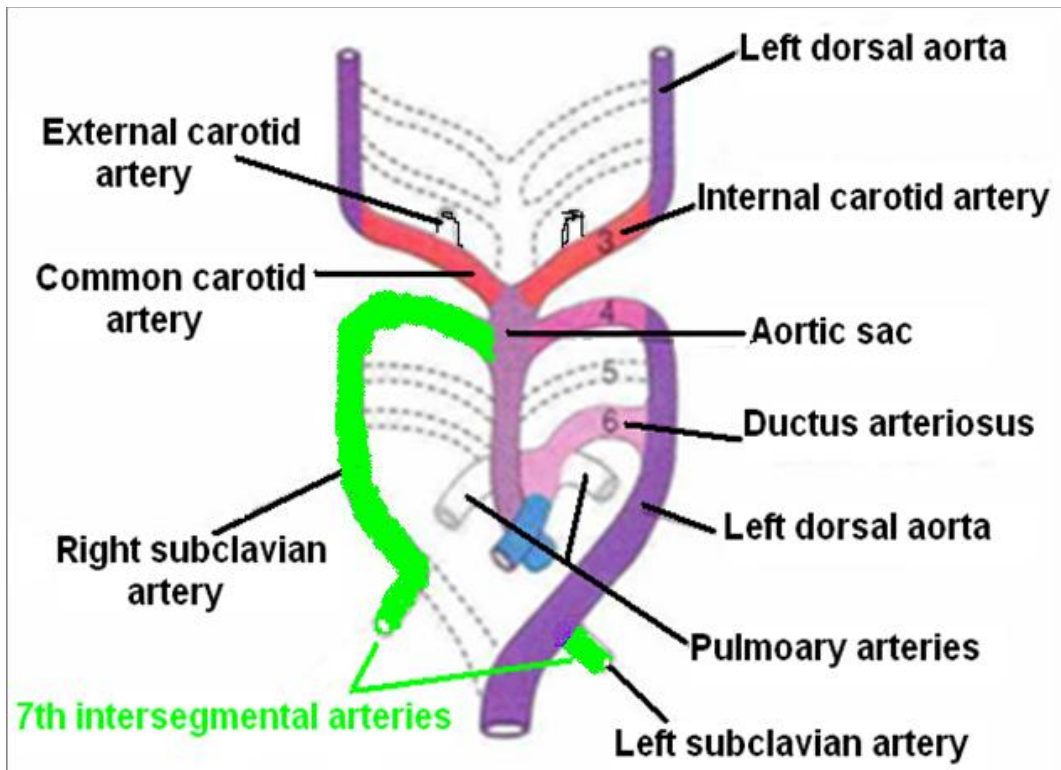


Arch of aorta



- Proximal segment: derived from aortic sac
- Middle segment: derived from left 4th aortic arch
- Distal segment: derived from left dorsal aorta

Subclavian arteries



RIGHT subclavian a.

- Right 4th aortic arch
- Right dorsal aorta
- Right 7th intersegmental a.

LEFT subclavian a.

- Left 7th intersegmental a.

Derivatives of 6rd pair

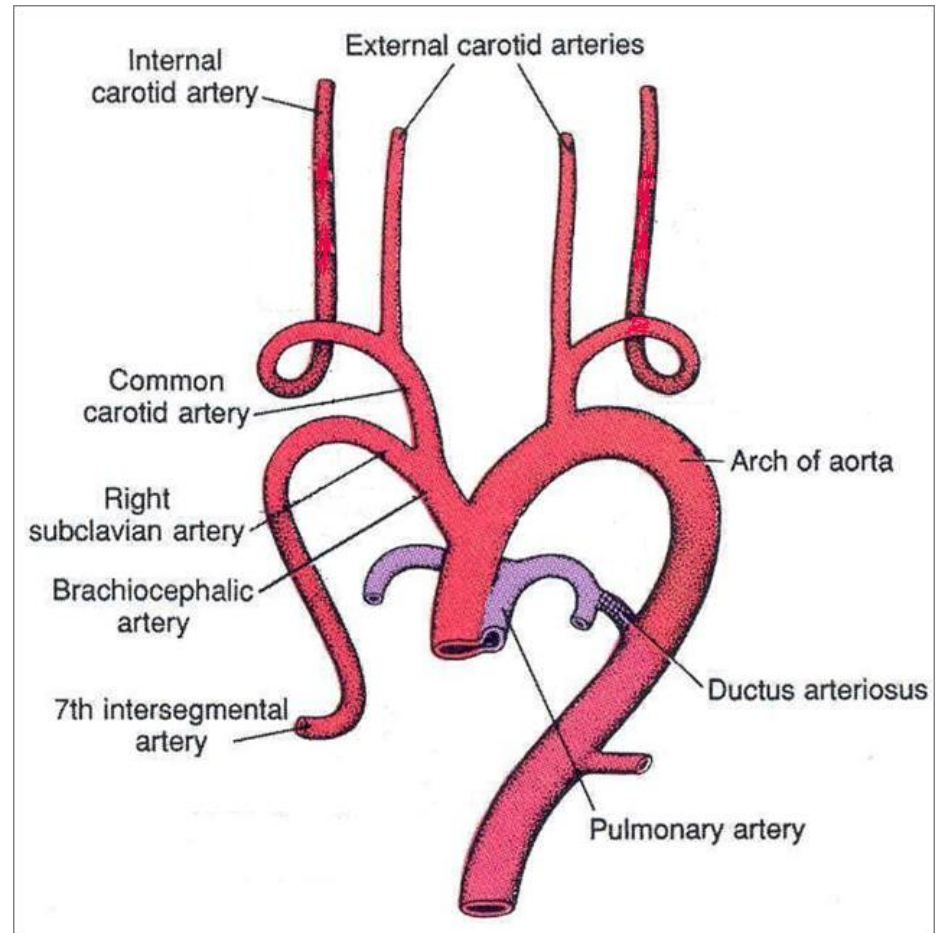
- Form by day 29

LEFT 6th aortic arch

- Proximal part: proximal part of left pulmonary a.
- Distal part: ductus arteriosus (shunt between Lt. pul. a. & dorsal aorta)

RIGHT 6th aortic arch

- Proximal part: proximal part of right pulmonary a.
- Distal part: degenerate

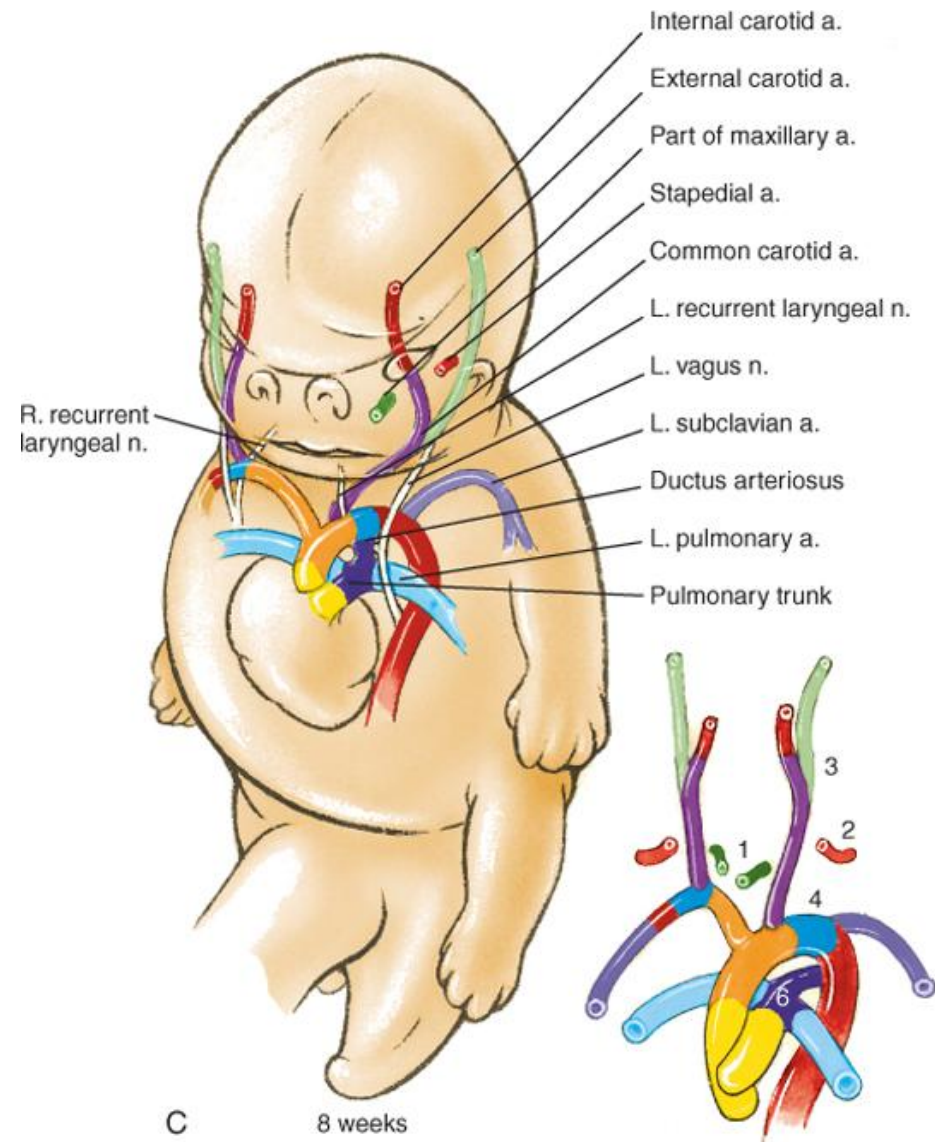


Change in the original aortic arch system

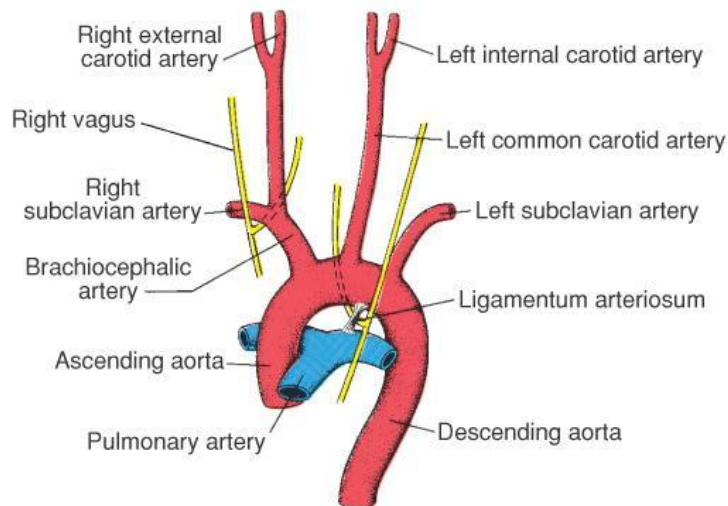
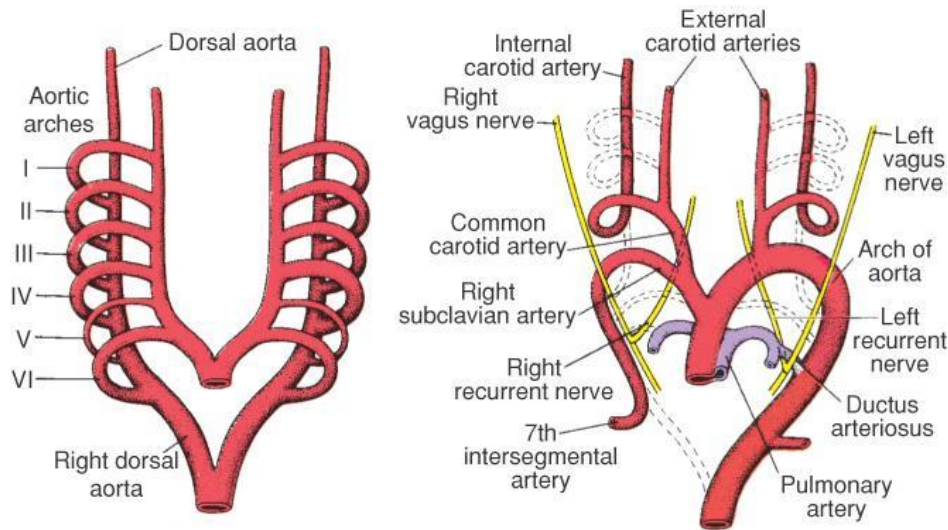
Obliteration of:

1. Most of the 1st & 2nd arches
2. 5th arches completely
3. Distal part of the right 6th arch
4. The segment of both aortae lying between the 3rd & 4th arch

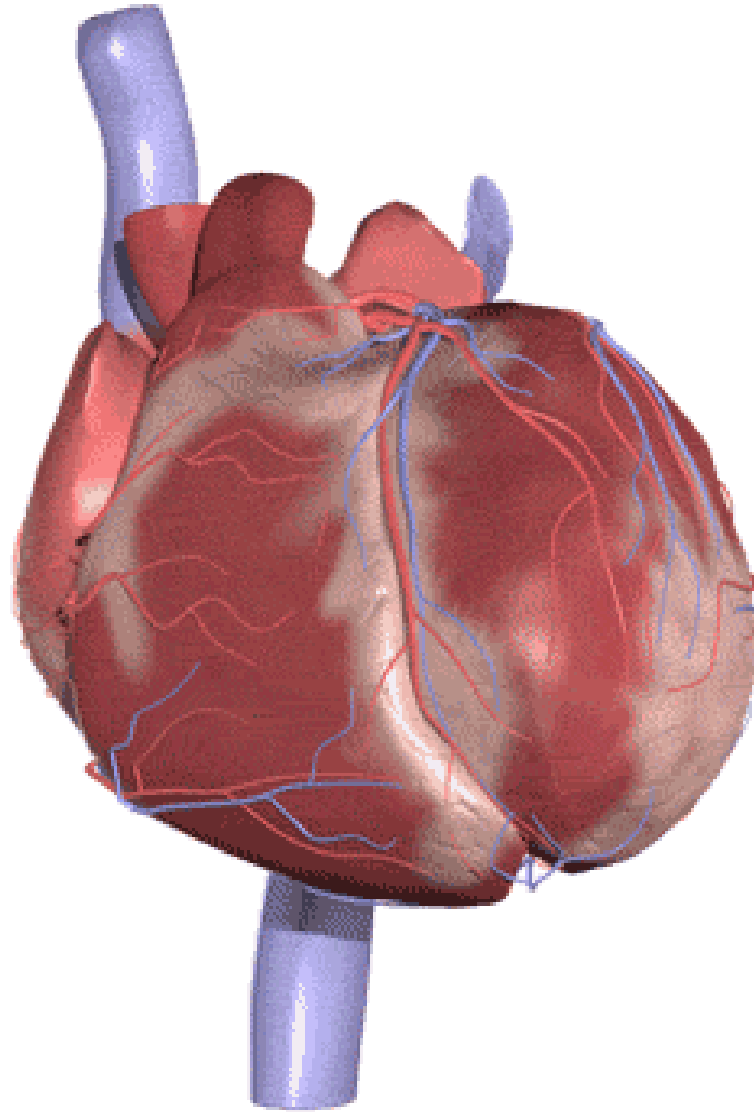
The segment of right aorta lying between the 7th intersegmental artery & the fused dorsal aortae



Relation of recurrent laryngeal nerves to aortic arches



- Recurrent laryngeal n. originally arise below the level of 6th arch and cross under right and left 6th arches
- LEFT recurrent laryngeal n.
 - : remain loop under ductus arteriosus
- RIGHT recurrent laryngeal n.
 - : Right 6th arch disappear
 - : 5th arch do not develop
 - : remain loop under 4th arch (right subclavian a.)



THE END